

Minimal Winding in the Self-Similar Collapse of Point Vortices

Chase Hendrick

Independent Researcher

chase@hendrickresearch.com

ORCID 0009-0002-9754-6087

Abstract

In a self-similar collapse of point vortices every vortex moves on a logarithmic spiral, and the angle P through which the configuration turns while the square of its size decreases by the factor e measures how tightly the spiral winds; for the Euler equation $P = |\omega_0|t_c$, the initial angular velocity times the collapse time. For three Euler vortices with circulations $(1, \mu, -\mu/(1+\mu))$, $0 < \mu \leq 1$, the collapsing configurations form two arcs, one for each orientation of the triangle, and P has exactly one critical point, a minimum, on each. The squared minima are roots of an explicit cubic, and the smaller one increases from $\sqrt{3}/2$, approached as $\mu \rightarrow 0$, to $\sqrt{2}$ at $\mu = 1$. Hence $P > \sqrt{3}/2$ for every self-similar collapse of three point vortices, and the constant is sharp: every vortex travels more than twice its initial distance from the collision point. In the α -models, where a vortex induces the velocity $\Gamma r^{-\alpha-1}/(2\pi)$, every self-similar collapse of three vortices has $P > \sqrt{3+\alpha}/(2+\alpha)$ for $\alpha > -2$, again sharply. For two concentric regular n -gons, with or without a vortex at the center, P has a closed form, and its minimum is explicit and exceeds $\sqrt{3}/2$. A strong vortex with weak, tight opposite-signed pairs has $P > \sqrt{3}/2$ once the pairs are weak enough. For more vortices the bound fails. Computer-assisted proofs show that four, five and six Euler vortices collapse self-similarly with $P < \sqrt{3}/2$, that P has strict local minima $0.7978967838\dots$, $0.7448144569\dots$ and $0.7136801485\dots$ on these collapses, that for $\alpha = 1$ and $\alpha = 2$ four vortices go below the three-vortex bounds, and that eleven vortices in the $\alpha = 2$ model and sixty in the SQG model collapse without rotating. Numerically, sixty-one Euler vortices reach $P = 0.498\dots$

Keywords: point vortices; vortex collapse; self-similar motion; logarithmic spiral; surface quasi-geostrophic equation; computer-assisted proof.

MSC 2020: 76B47, 37N10, 76U60, 65G20.

1 Introduction

In the Euler equation, three point vortices collapse self-similarly only if $1/\Gamma_1 + 1/\Gamma_2 + 1/\Gamma_3 = 0$ and their angular impulse about the center of vorticity vanishes (Lemma 2 below). Under these conditions every configuration moves self-similarly: the triangle keeps its shape and rotates while it shrinks to a point in finite time, expands, or rotates rigidly [14, Sect. 4.5.2], [28]. Such collapse was found by Gröbli [16] and found again by Aref [1] and by Novikov and Sedov [33], who also constructed collapsing configurations of four and five vortices; see [4, p. 17, Fig. 3] for the history and [32] for the general theory. Synge [39, Sect. 4, Theorem 8] identified $\Gamma_1\Gamma_2 + \Gamma_2\Gamma_3 + \Gamma_3\Gamma_1 = 0$ as an exceptional case: the configurations whose shape lies on a certain conic in his trilinear coordinates then come in one-parameter families of similar triangles, so that the shape does not determine the size. Conte and de Seze [9] solved the motion of three vortices of arbitrary circulations exactly and showed that when $\sum_{j < k} \Gamma_j \Gamma_k = 0$ and the angular impulse vanishes, each vortex runs a logarithmic

spiral about the center of vorticity while the triangle keeps its shape, and the configuration either collapses in finite time or expands [9, Sect. 4]. Kimura [25] studied similarity solutions of point-vortex systems, Aref [3] derived formulas for the rate of collapse or expansion and the angular frequency of rotation, and Gotoda [15] gave explicit formulas for the self-similar motions of three vortices. Borisov and Lebedev [6] obtained conditions for the collapse and the scattering of three vortices within the Lie–Poisson formulation of the problem, and Krishnamurthy, Aref and Stremmer [28] recast the motion of three vortices in terms of the circumcircle and the interior angles of the vortex triangle and derived equations of motion for these quantities. Collapse also occurs in configurations of higher symmetry. Aref [2] reduced the motion of two concentric regular n -gons of vortices to an integrable Hamiltonian system with two degrees of freedom, and Koiller et al. [26] found collapsing configurations of two such rings, whose vortices move on logarithmic spirals. Demina and Kudryashov [11, Sect. 3] gave explicitly a family of such configurations that collapse or scatter, with an additional vortex, possibly of zero circulation, at the center, together with the complex constant that determines their rates of collapse and rotation.

Self-similar collapse of three vortices also occurs in the generalized Euler or α -models of two-dimensional flow, in which the streamfunction is obtained from the vorticity by a fractional inverse Laplacian and which include the surface quasi-geostrophic (SQG) equation. Badin and Barry [5] derived conditions for collapse in these models, and the self-similar motions of three vortices were studied by Yasunaga, Ootobe and Iwayama [42], Reinaud, Dritschel and Scott [37], Chen and Liu [7] and Grotto and Pappalettera [17], among others; we return to this literature in Section 9.

In a self-similar collapse at time t_c the configuration has rotated by time t through the angle $-\omega_0 t_c \ln(1 - t/t_c)$, where ω_0 is the initial angular velocity. The angular velocity at time t is $\omega_0 t_c / (t_c - t)$, so the product of the angular velocity and the remaining time does not depend on which instant is taken as initial. Each vortex moves on a logarithmic spiral about the collision point, whose exponent is fixed by this product [33, p. 298], [9, Sect. 4], [3, Eq. (29c)], [12, App. A, Eq. (A.19)], and the dimensionless number $P = |\omega_0| t_c$, which is invariant under rescaling of lengths, times and circulations, is the angle through which the configuration turns while the square of its size decreases by the factor e ; in the α -models below this angle is $P = \beta |\omega_0| t_c$ with $\beta = 1 + \alpha/2$ (Section 2), and we keep this geometric definition of P . Equivalently, the path of each vortex makes the constant angle $\arctan 2P$ with the direction to the collision point, and a vortex that starts at distance r_0 from that point travels the distance $r_0 \sqrt{1 + 4P^2}$ before the collapse. Gröbli gave the coefficient of the spiral, whose absolute value is $2P$, in closed form for every triple of circulations that can collapse [16, § 10] (Section 2); Kimura [25] and Aref [3] computed the rate of collapse and the rate of rotation, and the dimensionless combination P of the two rates fixes the spiral [3, Eq. (29c)]. But we have not found P minimized in the literature, Gröbli’s § 10 included, which describes the possible shapes of the triangle but not the extremum of his coefficient. Within a collapsing family it is natural to ask which configuration winds least, and whether any configuration can collapse without turning at all.

For three vortices in the Euler equation we answer this question for every circulation ratio (Section 3). After normalization the circulations are $(1, \mu, -\mu/(1 + \mu))$ with $0 < \mu \leq 1$. The collapsing configurations form two arcs, one for each orientation of the triangle, and P has a unique minimum on each (Theorem 1). For $\mu < 1$ the two minima differ, and the smaller one increases strictly with μ , from $\sqrt{3}/2$ as $\mu \rightarrow 0$ to $\sqrt{2}$ at $\mu = 1$, the value for two equal circulations. Thus $P > \sqrt{3}/2$ for every self-similar collapse of three vortices, and the bound is sharp (Corollary 1); the bound also has a short direct proof from Lemma 3 and an elementary inequality. For $\mu = 1/2$ the two minima are conjugate algebraic numbers of degree six, given explicitly in Proposition 1. In Section 4 we extend the bound to the α -models: every self-similar collapse of three vortices has $P > \sqrt{3 + \alpha}/(2 + \alpha)$ for every $\alpha > -2$, whether the velocity induced by a vortex decays

with distance ($\alpha > -1$) or not, and the constant is sharp (Theorem 2). The proof rests on two elementary identities: the circulations of a collapsing triangle are fixed, up to a factor, by its side lengths, and P is a symmetric function of the side lengths alone. At $\alpha = 0$ the theorem is Corollary 1. In Section 5 we treat two concentric regular n -gons, for which the minimization reduces to an elementary inequality once P is written in a suitable form. With a vortex of any circulation at the center the minimum is again explicit; it exceeds $\sqrt{3}/2$ for every n and tends to $\sqrt{3}/2$ as the central circulation grows (Proposition 3). In both Euler families the infimum is approached as a strong vortex carries weak, tight pairs of opposite sign, and Theorem 3 in Section 6 shows that this structure alone forces the constant: for a strong vortex with any number of such pairs, $P \geq \sqrt{3}/2 - o(1)$ as the circulations of the pairs tend to zero, and P comes close to $\sqrt{3}/2$ only if every pair makes the angle 60° with the direction away from the strong vortex and all the pairs are at nearly the same distance from it. The theorem is asymptotic; Proposition 4 gives $P > \sqrt{3}/2$, by a margin quadratic in the pair circulation, once the pairs are weaker than a threshold that is not explicit.

In Section 7 we turn to more vortices, where the bound fails. With computer-assisted proofs in interval arithmetic we show that four, five and six Euler vortices can collapse self-similarly with $P < \sqrt{3}/2$, that P has strict local minima $P_4 = 0.7978967838\dots$, $P_5 = 0.7448144569\dots$ and $P_6 = 0.7136801485\dots$ on these collapses, and that four vortices in the SQG and $\alpha = 2$ models go below the three-vortex bounds (Theorem 4); and that eleven vortices in the $\alpha = 2$ model and sixty vortices in the SQG model can each collapse self-similarly without rotating, every vortex moving on a straight line into the collision point (Theorem 5), which three vortices cannot do in any α -model with $\alpha > -2$. Numerically, P goes below $1/2$ for sixty-one Euler vortices. Section 8 describes the numerical checks and the computer-assisted proofs, Section 9 discusses the results and the literature, and Section 10 lists open problems.

2 Self-similar collapse

In the α -models the point vortices, at positions $z_j \in \mathbb{C}$ with real circulations Γ_j , move by

$$\frac{dz_j}{dt} = \frac{i}{2\pi} \sum_{k \neq j} \Gamma_k \frac{z_j - z_k}{|z_j - z_k|^{2\beta}}, \quad \beta = 1 + \frac{\alpha}{2}, \quad (1)$$

so that a vortex of circulation Γ induces the speed $|\Gamma|r^{-\alpha-1}/(2\pi)$ at distance r . The case $\alpha = 0$ is the Euler equation, which we write as

$$\dot{z}_j = \frac{1}{2\pi i} \sum_{k \neq j} \frac{\Gamma_k}{z_j - z_k}, \quad (2)$$

and $\alpha = 1$ ($\beta = 3/2$) is SQG. A constant factor in front of the sum, which differs between papers, only rescales time and does not change the number P below. In the conventions of Badin and Barry [5], Yasunaga, Ootobe and Iwayama [42] and Grotto and Pappalattera [17] the same system has the parameter $2 - \alpha$. The fluid models cover $-1 \leq \alpha < 2$ in our convention, and the constant factor of their Green's function vanishes at $\alpha = 2$ [42, Eq. (2)]; for $\alpha < -1$ and $\alpha \geq 2$ our statements are statements about the point-vortex system (1) itself.

A motion is *self-similar* if $z_j(t) = a(t) + b(t)z_j(0)$ for every j , with complex $a(t)$ and $b(t) \neq 0$, and it is a *self-similar collapse* if moreover $b(t) \rightarrow 0$ in finite time. Suppose $\sum_j \Gamma_j \neq 0$, let $z_c = \sum_j \Gamma_j z_j / \sum_j \Gamma_j$ be the center of vorticity, which is conserved, and suppose that at $t = 0$ there is $\kappa_0 \in \mathbb{C}$ with $\dot{z}_j = \kappa_0(z_j - z_c)$ for all j . The right-hand side of (1) is equivariant under

rotations, and replacing every $z_j - z_c$ by $b(z_j - z_c)$ multiplies it by $b|b|^{-2\beta}$, so the solution is $z_j(t) = z_c + \lambda(t)e^{i\phi(t)}(z_j(0) - z_c)$ with $\lambda^{2\beta-1}\dot{\lambda} = \text{Re } \kappa_0$ and $\lambda^{2\beta}\dot{\phi} = \text{Im } \kappa_0$. Hence

$$\lambda^{2\beta} = 1 + 2\beta \text{Re } \kappa_0 t, \quad \phi = \frac{\text{Im } \kappa_0}{2 \text{Re } \kappa_0} \ln \lambda^2, \quad (3)$$

and $\dot{z}_j = \kappa(t)(z_j - z_c)$ with $\kappa(t) = \kappa_0 \lambda^{-2\beta}$. If $\text{Re } \kappa_0 < 0$ the vortices collide at z_c at the time $t_c = -1/(2\beta \text{Re } \kappa_0)$, and the number

$$P = \frac{|\text{Im } \kappa|}{-2 \text{Re } \kappa} \quad (4)$$

does not depend on t . By (3) it is the angle through which the configuration turns while the square of its size decreases by the factor e : each vortex moves on a logarithmic spiral about the collision point, its path makes the constant angle $\arctan 2P$ with the direction to that point, and a vortex that starts at distance r_0 from it travels the distance $r_0\sqrt{1+4P^2}$ before the collapse. The initial angular velocity $\omega_0 = \text{Im } \kappa_0$ satisfies $|\omega_0|t_c = P/\beta$, and by the time t the configuration has turned through the angle $-\omega_0 t_c \ln(1 - t/t_c)$. For the Euler equation, $\beta = 1$, $t_c = -1/(2 \text{Re } \kappa_0)$ and $P = |\omega_0|t_c$, and $\phi = -P \ln \lambda^2$ when $\omega_0 > 0$, which is the description of the spiral given in the introduction.

For $\alpha = 0$ the first closed form of P is Gröbli's [16, § 10, Eqs. (8), (9), (12), pp. 56–57], for every triple of circulations with $1/\Gamma_1 + 1/\Gamma_2 + 1/\Gamma_3 = 0$: he writes the squared sides as constant multiples of the time measured from the collision and the rotation as $d\vartheta = \varkappa dt/(2t)$, so that $P = |\varkappa|/2$, and he gives his coefficient $\varkappa$ (a symbol distinct from our κ) explicitly in a shape constant; we write it out after Lemma 3. The same motion is Kimura's similarity solution [25, Sect. 3.1]: up to the factor 2π in his normalization of the circulations, κ is his constant $A + iB$, (4) reads $P = |B|/(-2A)$, and the spirals and the collision time $t_c = -1/(2A)$ are his Eqs. (3.7) and (3.9). Earlier, Conte and de Seze [9, Sect. 4, pp. 24–25] wrote this motion as $z_j = z_{j,0}(1 - t/t_c)^{1/2 - i\omega t_c}$ and gave $-2\omega + i/t_c$ in closed form in their shape variable; there ω is the initial angular velocity, so $|\omega t_c| = P$. Demina and Kudryashov [11, Eqs. (7)–(12)] derive the same solution, like Kimura for any number of vortices but with the normalization of (2), so that their complex constant c_1 equals κ ; they give the collapse time $1/(2|\text{Re } c_1|)$ [11, Eq. (12)] and the rotation angle $(\text{Im } c_1/(2 \text{Re } c_1)) \ln(1 + 2t \text{Re } c_1)$ [11, Eq. (10)], whose coefficient has absolute value P . Galloway and Šverák [14, Prop. 5.2, Eqs. (5.5), (5.11)] write the collapsing solution as $z_j(t) = (1 - t/T)^{1/2 + is} a_j$ with $z_c = 0$; comparing the two descriptions gives $T = t_c$ and $s = -\omega_0 t_c$, so $|s| = P$. For the α -models the spiral (3) is in [42, Eqs. (31), (32), (A.9)] and [22, Eqs. (3)–(6)], in [12, App. A, Eq. (A.19)], whose exponent $1/(\alpha + 1)$, in a parametrization where the Euler case is $\alpha = 1$, is our $1/(2\beta)$, and in [17, Prop. 2.1]. For $\beta \neq 1$ the product $|\omega_0|t_c$ is the rotation per unit decrease of $\ln(t_c - t)$, and it differs from P by the factor $1/\beta$; we state bounds for P , which measures the geometry of the spiral, and for $|\omega_0|t_c$ in Corollary 2.

The angular impulse about the center of vorticity satisfies

$$\sum_j \Gamma_j |z_j - z_c|^2 = \frac{1}{\sum_j \Gamma_j} \sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^2. \quad (5)$$

It is conserved for every α , since the terms of its derivative cancel in pairs, and in a self-similar motion it is proportional to λ^2 , so it must vanish for a collapse. Three of the minimizations below (Remark 2 and Propositions 2 and 3) and the direct proof of Corollary 1 reduce to the following elementary inequality.

Lemma 1. *Let $a > |b|$. For $0 < \tau < \pi$, $(a - b \cos \tau)/\sin \tau \geq \sqrt{a^2 - b^2}$, with equality if and only if $\cos \tau = b/a$.*

Proof. The numerator is positive, and $(a - b \cos \tau)^2 - (a^2 - b^2) \sin^2 \tau = (a \cos \tau - b)^2$. $\square$

3 Three vortices in the Euler equation

In this section and in Sections 5 and 6 the vortices obey the Euler law (2).

3.1 Normalization and the collapsing configurations

The Hamiltonian $H = -(4\pi)^{-1} \sum_{j < k} \Gamma_j \Gamma_k \ln |z_j - z_k|^2$ of (2) is conserved, and in a self-similar collapse all mutual distances are $\lambda(t) = |b(t)|$ times their initial values, with $\lambda(t) \rightarrow 0$ in finite time.

Lemma 2. *In a self-similar collapse of three point vortices all circulations are nonzero, $1/\Gamma_1 + 1/\Gamma_2 + 1/\Gamma_3 = 0$, $\sum_j \Gamma_j \neq 0$, and the angular impulse about the center of vorticity vanishes.*

Proof. The harmonic condition and the vanishing of the angular impulse are the classical necessary conditions [3, Sect. II B] (see also [25, Eqs. (3.20)–(3.21), (3.33)], [6, Sect. 3] and, for both conditions and any number of vortices, [11, Eqs. (26)–(28)]); we include the short argument. Along the motion

$$H = H(0) - \frac{1}{4\pi} \left(\sum_{j < k} \Gamma_j \Gamma_k \right) \ln \lambda^2,$$

and $\ln \lambda^2 \rightarrow -\infty$, so $\sum_{j < k} \Gamma_j \Gamma_k = 0$. If one circulation vanished, this would force a second one to vanish; the remaining vortex would then be at rest and the other two would move on circles about it, so no collapse would occur. Hence all $\Gamma_j \neq 0$, and $\sum_{j < k} \Gamma_j \Gamma_k = \Gamma_1 \Gamma_2 \Gamma_3 \sum_j 1/\Gamma_j$ gives the harmonic condition. Moreover $(\sum_j \Gamma_j)^2 = \sum_j \Gamma_j^2 > 0$. Finally, the angular impulse (5) is conserved and proportional to λ^2 , so it vanishes. $\square$

By Lemma 2 two of the three circulations of a self-similarly collapsing configuration have the same sign. Multiplying all circulations by a positive constant rescales time and leaves P unchanged. Multiplying them by -1 reverses time, and so does complex conjugation of the positions; the composition of the two maps solutions of (2) to solutions, and collapsing solutions to collapsing solutions with the same P . After relabeling we may therefore assume, as Gotoda does [15, Sect. 3],

$$\Gamma = \left(1, \mu, -\frac{\mu}{1+\mu} \right), \quad 0 < \mu \leq 1. \quad (6)$$

Then $\sum_j \Gamma_j = R/(1+\mu) > 0$, where $R = 1 + \mu + \mu^2$. For $w = (z_3 - z_1)/(z_2 - z_1)$ the sum $\sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^2$ in (5) equals $\mu(1+\mu)^{-1} |z_2 - z_1|^2 (1 + 2\mu \operatorname{Re} w - (1+\mu)|w|^2)$, which vanishes exactly on the circle $|w - \mu/(1+\mu)| = \sqrt{R}/(1+\mu)$. This circle appears in Kimura [25, Eqs. (3.34)–(3.35)], who notes that it splits into two arcs of collapse and two of expansion, and in Aref [3, Eq. (20a)]; we use the parametrization of it given by Gotoda [15, Sect. 3]:

$$z_1 = \frac{\mu(1 + \sqrt{R}e^{-i\theta})}{(1+\mu)^2}, \quad z_2 = \frac{\mu - \sqrt{R}e^{-i\theta}}{(1+\mu)^2}, \quad z_3 = 1, \quad \theta \in [0, 2\pi). \quad (7)$$

Here $\sum_j \Gamma_j z_j = 0$, so $z_c = 0$, and

$$z_1 - z_2 = \frac{\sqrt{R}e^{-i\theta}}{1+\mu}, \quad z_3 - z_1 = \frac{\sqrt{R}(\sqrt{R} - \mu e^{-i\theta})}{(1+\mu)^2}, \quad z_3 - z_2 = \frac{\sqrt{R}(\sqrt{R} + \mu e^{-i\theta})}{(1+\mu)^2}. \quad (8)$$

The shape ratio is $w = \mu/(1+\mu) - (\sqrt{R}/(1+\mu))e^{i\theta}$, which traverses the zero-impulse circle exactly once, so every zero-impulse configuration is obtained, up to translation, rotation and dilation, for exactly one θ . Since $\operatorname{Im} w = -(\sqrt{R}/(1+\mu))\sin \theta$, the triangle $z_1 z_2 z_3$ is negatively oriented for $\sin \theta > 0$ and positively oriented for $\sin \theta < 0$. Put $C = \sqrt{R} \cos \theta$ and

$$N(C) = 2(1 + \mu^2)R + (1 - \mu)(2 + \mu + 2\mu^2)C - 2\mu C^2, \quad M(C) = 1 - \mu + 2C.$$

Lemma 3. For every θ the three quotients $\dot{z}_j/(z_j - z_c)$ are equal to

$$\kappa = \frac{i(1+\mu)^3}{2\pi\sqrt{R}} \cdot \frac{\sqrt{R} + (1-\mu)e^{i\theta}}{(\sqrt{R} - \mu e^{i\theta})(\sqrt{R} + e^{i\theta})},$$

and, with $D = (R + \mu^2 - 2\mu C)(R + 1 + 2C) > 0$,

$$\operatorname{Re} \kappa = -\frac{(1+\mu)^3 \mu M(C) \sin \theta}{2\pi\sqrt{R} D}, \quad \operatorname{Im} \kappa = \frac{(1+\mu)^3 N(C)}{2\pi R D}.$$

Moreover $N(C) > 0$ for every θ .

Proof. Since $z_c = 0$ and $z_3 = 1$, $\kappa = \dot{z}_3$, and by (2), $\bar{z}_3 = (2\pi i)^{-1}((z_3 - z_1)^{-1} + \mu(z_3 - z_2)^{-1})$. Inserting (8) gives the stated κ . The same computation for $j = 1$ and $j = 2$ gives the same value, so all three vortices move with the same κ , in agreement with [25], [29, Sect. 3.1]. The real and imaginary parts follow on multiplying numerator and denominator by the complex conjugate of the denominator, since $|\sqrt{R} - \mu e^{i\theta}|^2 |\sqrt{R} + e^{i\theta}|^2 = D$. The two factors of D are at least $(\sqrt{R} - \mu)^2$ and $(\sqrt{R} - 1)^2$, which are positive because $R > \mu^2$ and $R > 1$. Finally, N is concave in C , and $N(\sqrt{R})N(-\sqrt{R}) = 3\mu^2(1+\mu)^2R$ and $N(\sqrt{R}) + N(-\sqrt{R}) = 4R(1 - \mu + \mu^2)$ are positive, so $N > 0$ on $|C| \leq \sqrt{R}$. $\square$

Gotoda [15, Sect. 3] gives the rates in this parametrization. In arXiv:2002.09624v1 the imaginary part in his Eq. (3.3) differs from Lemma 3 when $\mu \neq 1$; replacing $(\Gamma_1^2 + \Gamma_2^2)(\Gamma_2\lambda_1 + \Gamma_1\lambda_2)$ there by $(\Gamma_1 + \Gamma_2)(\Gamma_1^2\lambda_1 + \Gamma_2^2\lambda_2)$, in his notation, removes the difference, so we take it to be a misprint.

Gröbli's closed form of P [16, § 10, pp. 55–59] is older and holds for every triple of circulations (m_1, m_2, m_3) with $1/m_1 + 1/m_2 + 1/m_3 = 0$. We write his shape constant and shape parameters a and μ_i as a_G and μ_i^G , since a and μ have other meanings here. By his Eqs. (1), (5) and (8) the squared sides $|z_2 - z_3|^2$, $|z_3 - z_1|^2$ and $|z_1 - z_2|^2$ are proportional to

$$\mu_i^G = a_G - \frac{m_j - m_k}{m_i}, \quad (i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2),$$

where a_G is arbitrary, subject to the μ_i^G having one sign and forming a triangle, and by his Eqs. (9), (11) and (12), with the time measured from the collision, the rotation is $d\vartheta = \varkappa dt/(2t)$, where

$$\varkappa = \pm \frac{2a_G^2 + \frac{(m_2 - m_3)(m_3 - m_1)(m_1 - m_2)}{m_1 m_2 m_3} a_G - 3}{\sqrt{2\mu_2^G \mu_3^G + 2\mu_3^G \mu_1^G + 2\mu_1^G \mu_2^G - (\mu_1^G)^2 - (\mu_2^G)^2 - (\mu_3^G)^2}},$$

the sign going with the orientation of the triangle. Since the squared sides are proportional to t , $P = |\varkappa|/2$ by (3). By our calculation, Lemma 6 below with $\beta = 1$ gives the same value for a collapse: with r_i^2 proportional to μ_i^G , say $r_i^2 = \lambda \mu_i^G$ with λ real and of the sign of the μ_i^G , its sum S equals $-\lambda$ times the numerator of $\varkappa$, and $16A^2$ equals λ^2 times the radicand, by Heron's formula, so $|S|/(8A) = |\varkappa|/2$ for either sign of λ . For the family (6) Gröbli's $\varkappa$ is thus a closed form of P in a variable other than θ ; Remark 2 writes it out for $\mu = 1$. Section 10 of the dissertation also notes that for given circulations there are infinitely many shapes, that the triangle is right-angled for certain values of a_G and never isosceles, and works one example [16, p. 59, Fig. 6]; it does not discuss the extremum of $\varkappa$.

By Lemma 3 every zero-impulse configuration rotates in the positive sense, and it collapses exactly when $M(C) \sin \theta > 0$. Let $\theta_0 \in (0, \pi)$ be defined by $\cos \theta_0 = (\mu - 1)/(2\sqrt{R})$. The configurations with $\theta = 0$ and $\theta = \pi$ are collinear, and those with $\theta = \pm\theta_0$ are equilateral triangles; these four are relative equilibria. The collapsing configurations form the two arcs

$$\mathcal{A}_+ = (0, \theta_0), \quad \mathcal{A}_- = (\pi, 2\pi - \theta_0),$$

the triangle being negatively oriented on $\mathcal{A}_+$ and positively oriented on $\mathcal{A}_-$, and on them (4) gives

$$P(\theta) = \frac{N(C)}{2\mu\sqrt{R}M(C)\sin\theta}, \quad C = \sqrt{R}\cos\theta. \quad (9)$$

Since $z_j(-\theta) = \overline{z_j(\theta)}$, the reflection $\theta \mapsto -\theta$ maps $\mathcal{A}_+$ and $\mathcal{A}_-$ onto the expanding arcs $(2\pi - \theta_0, 2\pi)$ and (θ_0, π) .

3.2 The minimal winding

Let $u = \mu + 1 + 1/\mu \geq 3$ and $Q(\mu, y) = \mu^6 \tilde{Q}(u, y)$, where

$$\begin{aligned} \tilde{Q}(u, y) = & 1728(u+1)^2 y^3 - 144(u+1)(8u^3 - 9u - 9)y^2 \\ & - 4(16u^6 - 288u^4 - 288u^3 - 81u^2 - 162u - 81)y + 3(4u^3 - 3u - 3)^2. \end{aligned} \quad (10)$$

Then Q is a polynomial in μ and y with integer coefficients, irreducible over $\mathbb{Q}$. Its discriminant with respect to y is

$$28311552 \mu^4 (\mu - 1)^2 (\mu + 1)^4 (\mu + 2)^2 (2\mu + 1)^2 R^6 \Delta_1^3,$$

where $\Delta_1 = 4\mu^6 + 12\mu^5 + 51\mu^4 + 82\mu^3 + 51\mu^2 + 12\mu + 4$; the sum of the roots of $Q(\mu, \cdot)$ is $(8u^3 - 9u - 9)/(12(u+1)) > 0$, and their product is $-((4u^3 - 3u - 3)/(24(u+1)))^2 < 0$. Hence for $0 < \mu < 1$ the cubic $Q(\mu, \cdot)$ has three distinct real roots, one negative and two positive, which we denote $y_1(\mu) < y_2(\mu)$.

Theorem 1. *Let $0 < \mu \leq 1$.*

(a) *On each of $\mathcal{A}_+$ and $\mathcal{A}_-$ the function P has exactly one critical point, and it is the minimum of P on that arc. Denote the two minima by $P_+(\mu)$ and $P_-(\mu)$.*

(b) *$P_+(\mu)^2$ and $P_-(\mu)^2$ are roots of $Q(\mu, \cdot)$. For $\mu < 1$, $P_-(\mu)^2 = y_1(\mu)$ and $P_+(\mu)^2 = y_2(\mu)$, so $P_-(\mu) < P_+(\mu)$; for $\mu = 1$, $P_+ = P_- = \sqrt{2}$.*

(c) *The least value of P over the collapsing configurations, $P_-(\mu)$, is strictly increasing on $(0, 1]$, and $P_-(\mu) \rightarrow \sqrt{3}/2$ as $\mu \rightarrow 0^+$.*

Proof. (a) A direct computation from (9) gives

$$\frac{dP}{d\theta} = -\frac{G(C)}{2\mu M(C)^2 \sin^2 \theta},$$

where

$$G(C) = 4(1 - \mu)C^3 + 4(2\mu^2 - \mu + 2)C^2 + 2(1 - \mu)^3 C - (2\mu^4 + 7\mu^3 + 6\mu^2 + 7\mu + 2). \quad (11)$$

As θ runs over $\mathcal{A}_+$, C decreases monotonically from $\sqrt{R}$ to $C_0 = (\mu - 1)/2$; over $\mathcal{A}_-$ it increases from $-\sqrt{R}$ to C_0 . The critical points on the two arcs therefore correspond to the roots of G in $(C_0, \sqrt{R})$ and in $(-\sqrt{R}, C_0)$. Now $G(\pm\sqrt{R}) = \pm\sqrt{R}N(\pm\sqrt{R})M(\pm\sqrt{R})/R$, which is positive

because $M(\sqrt{R}) > 0 > M(-\sqrt{R})$, and $G(C_0) = -3(1+\mu)^4/2 < 0$. So G has a root in each interval. For $\mu < 1$, G is a cubic with positive leading coefficient, so it also has a root below $-\sqrt{R}$, and each interval contains exactly one root. For $\mu = 1$, $G = 12C^2 - 24$, with one root in each of $(-\sqrt{3}, 0)$ and $(0, \sqrt{3})$. The function P is differentiable on each arc, and at the ends of each arc $M(C) \sin \theta \rightarrow 0$ while $N > 0$, so $P \rightarrow +\infty$; hence the single critical point is the minimum.

(b) At a critical point $G(C) = 0$, and since $R \sin^2 \theta = R - C^2$, the number $y = P^2$ satisfies $4\mu^2 y(R - C^2)M(C)^2 = N(C)^2$. The resultant of these two polynomials with respect to C is $-48\mu^4(1+\mu)^8 Q(\mu, y)$, so $P_{\pm}(\mu)^2$ are roots of $Q(\mu, \cdot)$ for $\mu < 1$. For $\mu = 1$, $P_{\pm} = \sqrt{2}$ by Remark 2 below, and $Q(1, y) = 6912(y-2)^2(4y+1)$, so $P_{\pm}(1)^2 = 2$ is a root of $Q(1, \cdot)$. For $0 < \mu < 1$ both are positive roots, hence each lies in $\{y_1(\mu), y_2(\mu)\}$. The three roots of G are simple, so $P_+(\mu)$ and $P_-(\mu)$ depend continuously on μ , and so do $y_1(\mu) < y_2(\mu)$. By connectedness each of P_+^2 and P_-^2 coincides with the same y_i on all of $(0, 1)$. At $\mu = 1/2$ the proof of Proposition 1, which uses only (a) and the first part of (b), gives certified enclosures $P_-^2 \in [0.99, 1.30]$ and $P_+^2 \in [4.70, 5.02]$, each containing exactly one root of $Q(1/2, \cdot)$; so $P_-^2 = y_1$ and $P_+^2 = y_2$.

(c) For $0 < \mu < 1$ the roots $y_i(\mu)$ are simple, hence differentiable, and $y'_i = -(\partial Q / \partial \mu) / (\partial Q / \partial y)$. A zero of y'_1 at some μ would be a common root of $Q(\mu, \cdot)$ and $(\partial Q / \partial \mu)(\mu, \cdot)$, but their resultant with respect to y ,

$$7044820107264 \mu^7 (\mu - 1)^3 (\mu + 1)^7 (\mu + 2)^3 (2\mu + 1)^3 R^6 \Delta_2 \Delta_1^3,$$

where $\Delta_2 = 4\mu^6 + 12\mu^5 + 21\mu^4 + 22\mu^3 + 21\mu^2 + 12\mu + 4$, has no zero in $(0, 1)$. So y_1 is strictly monotone on $(0, 1)$. The leading coefficient of $Q(\mu, \cdot)$ does not vanish at $\mu = 1$, so the roots stay bounded as $\mu \rightarrow 1^-$, and y_1 tends to a nonnegative root of $Q(1, \cdot) = 6912(y-2)^2(4y+1)$, which is 2. Since $y_1(1/2) = 1.1335 \dots < 2$, y_1 is increasing, and therefore $y_1(\mu) < 2 = P_-(1)^2$ for $\mu < 1$. As $\mu \rightarrow 0^+$ it therefore decreases to a limit $L \geq 0$, which is a root of $Q(0, y) = -16(4y-3)$, so $L = 3/4$. Finally, $y_2 = \sigma - y_1 - y_0 > \sigma - 2$, where $y_0 < 0$ is the negative root and $\sigma = (8u^3 - 9u - 9)/(12(u+1))$ the sum of the roots; since $\sigma \rightarrow \infty$ as $\mu \rightarrow 0^+$, $P_+(\mu)$ grows without bound. $\square$

Corollary 1. *Every self-similar collapse of three point vortices has $|\omega_0|t_c > \sqrt{3}/2$, and for every $\delta > 0$ there is one with $|\omega_0|t_c < \sqrt{3}/2 + \delta$. Equivalently, each vortex travels more than twice its initial distance from the collision point, and the factor 2 is sharp.*

Proof. By Lemma 2, the normalization (6) and Theorem 1, $P \geq P_-(\mu) > \sqrt{3}/2$, and $P_-(\mu) \rightarrow \sqrt{3}/2$ as $\mu \rightarrow 0^+$. The path length is $r_0 \sqrt{1 + 4P^2} > 2r_0$, and the angle $\arctan 2P$ between each path and the direction to the collision point exceeds $\arctan \sqrt{3} = \pi/3$; and no vortex starts at the collision point, since z_1, z_2 and z_3 in (7) never vanish. $\square$

A direct proof of Corollary 1. The bound needs only Lemmas 1 and 3, not Theorem 1. By Lemma 2 and (6) it suffices to take $0 < \mu \leq 1$ and $\theta \in \mathcal{A}_+ \cup \mathcal{A}_-$, where $M(C) \sin \theta > 0$. Put $k = (1 - \mu)(2 + \mu)(1 + 2\mu)$, $T = 2R + (1 - \mu)C$ and $V = 3(1 + \mu)^2(R - C^2) = 3(1 + \mu)^2 R \sin^2 \theta > 0$. Comparing coefficients of μ in the first identity and of C in the other two shows

$$k^2 + 27\mu^2(1 + \mu)^2 = 4R^3, \quad T^2 = RM(C)^2 + V, \quad 9(1 + \mu)^2 N(C) = 2R(T^2 + V) + kM(C)T;$$

in the third, the coefficients of C^2 , C and 1 on each side are $-18\mu(1 + \mu)^2$, $9(1 - \mu)(1 + \mu)^2(2 + \mu + 2\mu^2)$ and $18(1 + \mu)^2(1 + \mu^2)R$. By the first identity there is $\psi \in (0, \pi)$ with $\cos \psi = k/(2R^{3/2})$ and $\sin \psi = 3\sqrt{3}\mu(1 + \mu)/(2R^{3/2})$. By the second, $T \neq 0$, and $x = \sqrt{R}M(C)/T$ and $y = \sqrt{V}/|T|$ satisfy $x^2 + y^2 = 1$ and $0 < |x| < 1$. The third, divided by $2RT^2$, reads $9(1 + \mu)^2 N(C)/(2RT^2) = 2 - x^2 + x \cos \psi$, and $2|x|y \sin \psi = 9\mu(1 + \mu)^2 M(C) \sin \theta / (\sqrt{R}T^2)$, so (9) becomes

$$P = \frac{2 - x^2 + x \cos \psi}{2|x|y \sin \psi}.$$

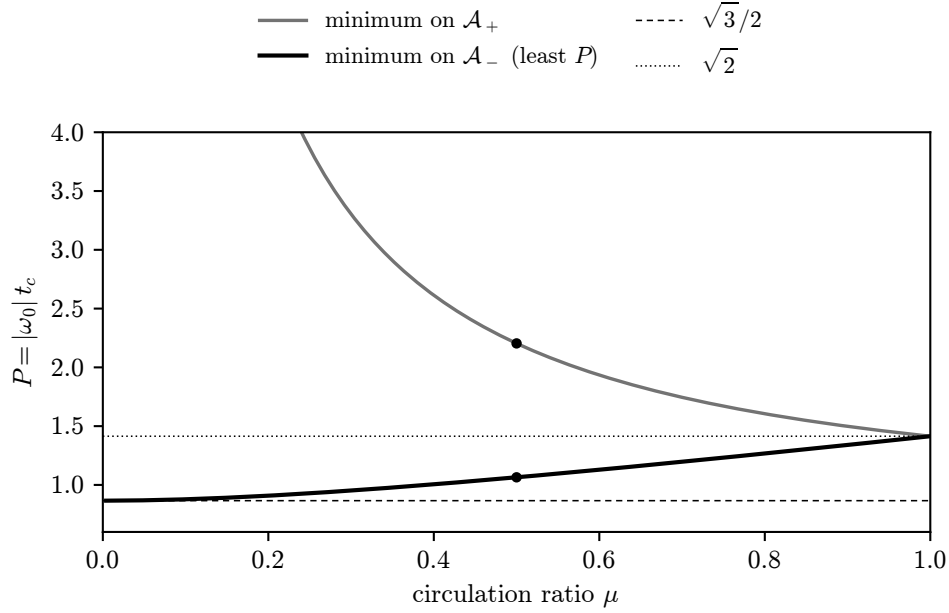


Figure 1: The minima $P_-(\mu)$ and $P_+(\mu)$ of $P = |\omega_0|t_c$ on the two collapsing arcs, as functions of the circulation ratio μ . The dots mark $\mu = 1/2$ (Proposition 1). As $\mu \rightarrow 0$, P_- tends to $\sqrt{3}/2$ and P_+ grows without bound. Both axes are dimensionless; values follow the cubic of Theorem 1. The figure is reproduced by `plot_minimal_winding.py`.

Lemma 1 with $a = 2 - x^2$ and $b = -x$, where $a - |b| = (1 - |x|)(2 + |x|) > 0$, gives $2|x|yP \geq \sqrt{(2 - x^2)^2 - x^2}$, and $(2 - x^2)^2 - x^2 = 3x^2y^2 + 4y^4$ because $x^2 + y^2 = 1$. Hence

$$P^2 \geq \frac{3}{4} + \frac{y^2}{x^2} = \frac{3}{4} + \frac{3(1 + \mu)^2 \sin^2 \theta}{M(C)^2} > \frac{3}{4}.$$

Equivalently, with $N = N(C)$ and $M = M(C)$, the three identities combine into

$$4R^3(N^2 - 3\mu^2(R - C^2)M^2) = (kN + 3\mu^2MT)^2 + 48\mu^2(1 + \mu)^2R^2(R - C^2)^2.$$

For sharpness, take θ with $\sin \theta = -\sqrt{3}\mu/2$ and $C = -\sqrt{R}\sqrt{1 - 3\mu^2/4}$, which lies in $\mathcal{A}_-$ because $C < -1/2 \leq (\mu - 1)/2 = \sqrt{R}\cos \theta_0$. As $\mu \rightarrow 0$, $C + \sqrt{R} = 3\mu^2/8 + O(\mu^3)$; by the proof of Lemma 3, $N(-\sqrt{R}) = 3\mu^2(1 + \mu)^2R/N(\sqrt{R}) = 3\mu^2/4 + O(\mu^3)$; and $N'(C) = 2 + O(\mu)$ near $C = -1$. So $N(C) = 3\mu^2/2 + O(\mu^3)$, $M(C) = -1 + O(\mu)$, and (9) gives $P = \sqrt{3}/2 + O(\mu)$. The remaining statements follow as in the first proof. $\square$

Every motion of three point vortices in the plane that ends in a total collision is self-similar [19, Theorem 1], [20, Prop. 2.8], [21, Sect. 3], [29, Sect. 3.1], [14, Prop. 5.2], [13, Theorem 1.1], so the bound of Corollary 1 holds for every collapse of three point vortices. The bound fails for more vortices: the collapsing configuration of seven vortices in [11, Table 1, Fig. 1a], with circulation 6383/2250 at the origin, 14/15 at ± 2 , $-62/45$ at $\pm 2e^{i\varphi}$ where $\cos 2\varphi = 13/18$, and 1 at $\pm 4/3$, has the constant $\Omega = 12433/9000 - (31\sqrt{155}/450)i$, where $\Omega = 2\pi i \bar{\kappa}$ [11, Eq. (11)], so $P = |\operatorname{Re} \Omega|/(2|\operatorname{Im} \Omega|) = 12433/(1240\sqrt{155}) = 0.8053 \dots < \sqrt{3}/2$ (see also Section 7).

The two minima are shown in Figure 1, and Figure 2 shows two minimizing configurations and the paths of their vortices.

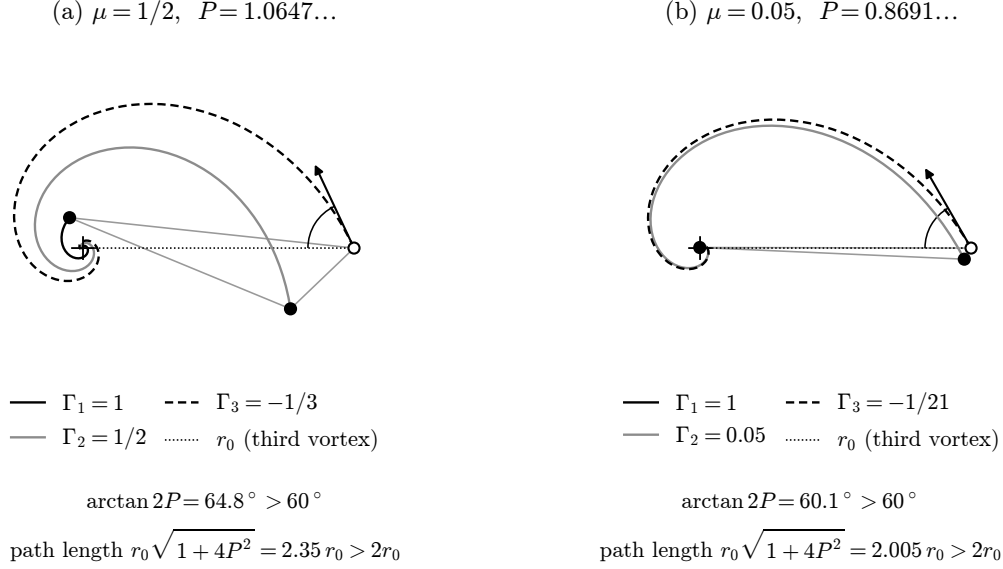


Figure 2: Minimizing configurations on $\mathcal{A}_-$ (dots, filled for positive circulation) and the logarithmic spiral paths of their vortices into the collision point (+), drawn until the distances have shrunk by the factor 100, for (a) $\mu = 1/2$ and (b) $\mu = 0.05$, where the two weak vortices form a tight pair beside the strong one, as in Theorem 3, and P is close to $\sqrt{3}/2$. Each path meets the line to the collision point at the constant angle $\arctan 2P > 60^\circ$ and has length $r_0 \sqrt{1 + 4P^2} > 2r_0$ (Corollary 1), marked here for the third vortex. The collision point is the origin and the third vortex initially lies at 1 in the complex plane, fixing the length scale. The figure is reproduced by `plot_minimal_winding.py`.

3.3 The ratio $\mu = 1/2$

For $\Gamma = (1, 1/2, -1/3)$ we have $R = 7/4$ and $\cos \theta_0 = -\sqrt{7}/14$, and (9) becomes

$$P(\theta) = \frac{14 \sin^2 \theta + 6\sqrt{7} \cos \theta + 21}{2(14 \cos \theta + \sqrt{7}) \sin \theta}.$$

Proposition 1. *Let $X = 245351/5201^{3/2}$ and $\sigma = 7\sqrt{5201}/162$. For $\mu = 1/2$,*

$$P_- = \sqrt{\frac{605}{324} + \sigma \cos\left(\frac{1}{3} \arccos X - \frac{2\pi}{3}\right)} = 1.0647059762712043\dots,$$

$$P_+ = \sqrt{\frac{605}{324} + \sigma \cos\left(\frac{1}{3} \arccos X\right)} = 2.2038550160361327\dots,$$

attained on $\mathcal{A}_-$ at $\cos \theta = c_1 = -0.9243893679\dots$ and on $\mathcal{A}_+$ at $\cos \theta = c_0 = 0.6739838839\dots$, where

$$c_m = -\frac{8\sqrt{7}}{21} + \frac{5\sqrt{70}}{21} \cos\left(\frac{1}{3} \arccos\left(-\frac{124\sqrt{10}}{3125}\right) - \frac{2\pi m}{3}\right).$$

The two minima are roots of the polynomial $8748\xi^6 - 49005\xi^4 + 27794\xi^2 + 18723$, which is irreducible over $\mathbb{Q}$ and whose real roots are exactly these two numbers and their negatives.

Proof. Here $16Q(1/2, \xi^2) = 8748\xi^6 - 49005\xi^4 + 27794\xi^2 + 18723$, and by Theorem 1(b) the squares of both minima are roots of this polynomial. As a cubic in $q = \xi^2$ it is irreducible over $\mathbb{Q}$ and

has positive discriminant, so its roots are real and are given by the trigonometric solution of the cubic, $q_m = 605/324 + \sigma \cos(\frac{1}{3} \arccos X - 2\pi m/3)$, $m = 0, 1, 2$, with $q_0 \approx 4.856977$, $q_1 \approx 1.133599$ and $q_2 \approx -0.388724$. The critical points are the roots of G in the two intervals of the proof of Theorem 1(a). Here $G(C) = 2C^3 + 8C^2 + C/4 - 8$, and with $C = \sqrt{R} \cos \theta = (\sqrt{7}/2)c$, that is $c = \cos \theta$, the equation $G = 0$ becomes, after division by $7\sqrt{7}/4$, $c^3 + (8\sqrt{7}/7)c^2 + c/14 - 32\sqrt{7}/49 = 0$. The substitution $c = s - 8\sqrt{7}/21$ reduces it to $s^3 - (125/42)s + 124\sqrt{7}/1323 = 0$. By the proof of Theorem 1(a) its three roots are real, and the trigonometric solution of the cubic gives them as $s = (5\sqrt{70}/21) \cos(\frac{1}{3} \arccos(-124\sqrt{10}/3125) - 2\pi m/3)$, $m = 0, 1, 2$; so the three roots in c are $c_0 \approx 0.673984$, $c_1 \approx -0.924389$ and $c_2 \approx -2.773310$. Exact sign changes of G place the critical points in $0.67 < \cos \theta < 0.68$ on $\mathcal{A}_+$ and $-0.93 < \cos \theta < -0.92$ on $\mathcal{A}_-$, so they are at c_0 and c_1 , and interval arithmetic applied to $P^2 = N^2/(4\mu^2(R - C^2)M^2)$ on these intervals gives $P_+^2 \in [4.70, 5.02]$ and $P_-^2 \in [0.99, 1.30]$. Each enclosure contains exactly one of q_0, q_1, q_2 , so $P_+^2 = q_0$ and $P_-^2 = q_1$. $\square$

Remark 1. The two minima are conjugate algebraic numbers of degree six. Since the cubic in q is irreducible over $\mathbb{Q}$ and has three real roots, none of its roots is expressible by real radicals (casus irreducibilis), and therefore neither minimum is. The same holds for every $\mu = a/b \in (0, 1)$ in lowest terms with $2 \leq b \leq 30$: for each of these 277 values, $Q(\mu, \xi^2)$ is irreducible over $\mathbb{Q}$, so the minima have degree six and the cubic $Q(\mu, \cdot)$ is irreducible, and its discriminant is positive on $(0, 1)$, so it has three real roots.

Remark 2. For $\mu = 1$, that is $\Gamma = (1, 1, -1/2)$, the zero-impulse circle is $|w - 1/2| = \sqrt{3}/2$. With $w = 1/2 + (\sqrt{3}/2)e^{i\chi}$, $z_1 = 0$ and $z_2 = 1$, the same computation gives $P = (3 - \cos 2\chi)/(2 \sin 2\chi)$ on the collapsing arcs $0 < \chi < \pi/2$ and $\pi < \chi < 3\pi/2$, which are exchanged by interchanging the two equal vortices ($w \mapsto 1 - w$). Since P has period π in χ , Lemma 1 applies on both arcs and gives $P \geq \sqrt{2}$, with equality if and only if $\cos 2\chi = 1/3$, in agreement with Theorem 1. This parametrization is Kimura's [25, Sect. 4], who gives the two rates for $\Gamma = (2, 2, -1)$ in his Eq. (4.4); their ratio $B/(-2A)$ is this expression for P .

In the variable $v = \tan \chi$, which is positive on both arcs, $\cos 2\chi = (1 - v^2)/(1 + v^2)$ and $\sin 2\chi = 2v/(1 + v^2)$ give $P = v + 1/(2v)$, and $P - \sqrt{2} = (\sqrt{2}v - 1)^2/(2v)$ is a second proof of the bound: equality holds if and only if $\tan \chi = 1/\sqrt{2}$, which is $\cos 2\chi = 1/3$. Since P is unchanged under $v \mapsto 1/(2v)$, the configuration $\chi = \pi/4$ ($v = 1$) and the fastest collapse discussed in Section 9 ($v = 1/2$, $\cos 2\chi = 3/5$) both have $P = 3/2$. At the minimum on the first arc, $w = (1 + \sqrt{2})/2 + i/2$, so $\arg w = \arctan(\sqrt{2} - 1) = \pi/8$ and $\arg(w - 1) = \arctan(\sqrt{2} + 1) = 3\pi/8$: the minimizing triangle has the interior angles $\pi/8$ at z_1 , $5\pi/8$ at z_2 and $\pi/4$ at z_3 , the vortex of circulation $-1/2$. On the second arc it is the same triangle with z_1 and z_2 interchanged.

The expression for P is also Gröbli's closed form, given after Lemma 3, at $(m_1, m_2, m_3) = (1, 1, -1/2)$, in another variable [16, § 10, pp. 55–59]. There $\mu_1^G = a_G - 3/2$, $\mu_2^G = a_G + 3/2$ and $\mu_3^G = a_G$, the radicand is $3a_G^2 - 9$, and on the branches $|a_G| > \sqrt{3}$, $P = |\varkappa|/2 = (2a_G^2 - 3)/(2\sqrt{3a_G^2 - 9})$. By our calculation, the substitution $a_G = \sqrt{3}/\cos \chi$, which makes $\mu_1^G : \mu_2^G : \mu_3^G = |w - 1|^2 : |w|^2 : 1$ at the positions above, gives $3a_G^2 - 9 = 9 \tan^2 \chi$ and $2a_G^2 - 3 = 3(3 - \cos 2\chi)/(2 \cos^2 \chi)$, and so turns this into $(3 - \cos 2\chi)/(2 \sin 2\chi)$; in his variable, $P^2 - 2 = (2a_G^2 - 9)^2/(12(a_G^2 - 3))$. The denominator $\mu_1\mu_2\mu_3$ of his Eq. (9) is printed as $\mu_1\mu_3\mu_3$ in Eq. (10.9) of Goodman's English translation cited in [16]; we use the original.

Remark 3. Interchanging the first two vortices and multiplying the circulations by $1/\mu$ maps the family with ratio μ onto the family with ratio $1/\mu$ and exchanges the two orientations, so the arc minima satisfy $P_+(\mu) = P_-(1/\mu)$. For $\mu > 1$ the smaller minimum is therefore attained on $\mathcal{A}_+$.

4 Three vortices in the α -models

In this section the vortices obey (1) with $\alpha > -2$, that is, $\beta > 0$, and we write $B(\alpha) = \sqrt{3 + \alpha}/(2 + \alpha)$. We prove:

Theorem 2. *Let $\alpha > -2$. Every self-similar collapse of three point vortices under (1) has $P > B(\alpha)$, and for every $\delta > 0$ there is one with $P < B(\alpha) + \delta$.*

For $\alpha = 0$, $B(0) = \sqrt{3}/2$ and Theorem 2 is Corollary 1. The proof below works with the side lengths of the triangle and does not use Section 3, and it gives only the bound, not the minima for each ratio of circulations of Theorem 1. It needs two facts proved in Section 4.1: a collapsing triangle is scalene and carries circulations fixed up to a factor by its side lengths (Lemma 5), and P is a symmetric function of the side lengths alone (Lemma 6). First, the motion has the form of Section 2.

Lemma 4. *Let a self-similar motion of three point vortices under (1) change size. Then $\Gamma_1 + \Gamma_2 + \Gamma_3 \neq 0$, and there is $\kappa_0 \in \mathbb{C}$ with $\text{Re } \kappa_0 \neq 0$ such that $\dot{z}_j = \kappa_0(z_j - z_c)$ at $t = 0$ for every j . The motion is given by (3), and it collapses if and only if $\text{Re } \kappa_0 < 0$.*

Proof. The linear impulse $\sum_j \Gamma_j z_j$ is conserved. If $\sum_j \Gamma_j = 0$, then $\sum_j \Gamma_j z_j(t) = b(t) \sum_j \Gamma_j z_j(0)$, and since $|b|$ changes, $\sum_j \Gamma_j z_j(0) = 0$; then $\Gamma_1(z_1 - z_3) + \Gamma_2(z_2 - z_3) = 0$, where the real coefficients Γ_1 and Γ_2 are not both zero, since otherwise $\Gamma_3 = 0$ as well and nothing moves. So the vortices are collinear at $t = 0$, and, the motion being self-similar, at every time. By the formula for $d(r_i^2)/dt$ in Lemma 5 below, whose proof does not use the present lemma, the sides of a triangle that stays collinear keep their lengths, since its signed area vanishes; so the size does not change. Hence $\sum_j \Gamma_j \neq 0$ and z_c is conserved. Since at least two of the $z_j(0)$ differ, $z_j(0) = a(0) + b(0)z_j(0)$ gives $a(0) = 0$ and $b(0) = 1$, and $z_c = a + bz_c$ gives $z_j(t) - z_c = b(t)(z_j(0) - z_c)$, so $\dot{z}_j(0) = \kappa_0(z_j(0) - z_c)$ for every j , with $\kappa_0 = \dot{b}(0)$. By Section 2 the motion is (3); its size changes, so $\text{Re } \kappa_0 \neq 0$, and it tends to 0 in finite time exactly when $\text{Re } \kappa_0 < 0$. $\square$

4.1 Circulations and rates of a collapsing triangle

Label the sides by the opposite vertices: $r_1 = |z_2 - z_3|$, $r_2 = |z_3 - z_1|$, $r_3 = |z_1 - z_2|$, and put $f_i = r_i^{-2\beta}$. Let $A_s = \text{Im}((z_2 - z_1)\overline{(z_3 - z_1)})/2$ be the signed area, positive when z_1, z_2, z_3 are counterclockwise, and $A = |A_s|$. Indices (i, j, k) run over the cyclic permutations of $(1, 2, 3)$.

Lemma 5. *Under (1), $d(r_i^2)/dt = (2A_s/\pi)\Gamma_i(f_j - f_k)$. A motion is self-similar and changes size if and only if the triangle is not collinear, its sides are pairwise different, and*

$$\Gamma_i = c \frac{r_i^2}{f_j - f_k} \quad (i = 1, 2, 3) \quad (12)$$

for some real $c \neq 0$. In that case $\text{Re } \kappa = cA_s/\pi$, and exactly one of a triangle and its mirror image collapses self-similarly.

Proof. For $i = 3$, the terms of $\dot{z}_1 - \dot{z}_2$ proportional to $z_1 - z_2$ do not change $|z_1 - z_2|$, and the rest is $(i\Gamma_3/2\pi)((z_1 - z_3)f_2 - (z_2 - z_3)f_1)$. Since $\text{Im}((z_1 - z_2)\overline{(z_1 - z_3)}) = \text{Im}((z_1 - z_2)\overline{(z_2 - z_3)}) = 2A_s$, $d(r_3^2)/dt = 2\text{Re}((z_1 - z_2)(\dot{z}_1 - \dot{z}_2)) = (2A_s/\pi)\Gamma_3(f_1 - f_2)$, and the other sides follow by cyclic relabeling. In a self-similar motion that changes size, Lemma 4 and Section 2 give $\dot{z}_j = \kappa(z_j - z_c)$ with $\kappa = \kappa_0\lambda^{-2\beta}$ and $\text{Re } \kappa_0 \neq 0$, so $d \ln r_i^2/dt = 2\text{Re } \kappa \neq 0$ for every i and at every time. Hence $(2A_s/\pi)\Gamma_i(f_j - f_k)/r_i^2$ is the same for all i and not 0, so $A_s \neq 0$, $f_j \neq f_k$

for all pairs, and (12) holds with $c = \pi \operatorname{Re} \kappa / A_s$. Conversely, let (12) hold with $c \neq 0$ on a scalene triangle that is not collinear, and write $s_i = r_i^2$. While the triangle is not collinear, $A_s = \sigma \sqrt{2(s_1 s_2 + s_2 s_3 + s_3 s_1) - s_1^2 - s_2^2 - s_3^2} / 4$ by Heron's formula, with a fixed sign σ , so the squared sides obey the closed autonomous system $ds_i/dt = (2A_s/\pi) \Gamma_i (s_j^{-\beta} - s_k^{-\beta})$. Its right side is smooth where $A_s \neq 0$ and all $s_i > 0$, and homogeneous of degree $1 - \beta$ in (s_1, s_2, s_3) . At the initial point it equals $(2cA_s/\pi)(s_1, s_2, s_3)$ by (12), so, by homogeneity, it is a multiple of (s_1, s_2, s_3) at every point of the ray through the initial point; the ray is therefore invariant, and by uniqueness the sides stay proportional to their initial values. With the sign of A_s kept, the triangles at different times are then related by orientation-preserving similarities $z \mapsto a(t) + b(t)z$, so the motion is self-similar, and its size changes because $d \ln r_i^2/dt = 2cA_s/\pi \neq 0$ at $t = 0$; then $\operatorname{Re} \kappa = cA_s/\pi$. Reflecting the triangle changes the sign of A_s and keeps (12), so it changes the sign of $\operatorname{Re} \kappa$. $\square$

In particular, isosceles, equilateral and collinear triangles never collapse self-similarly, and every scalene triangle collapses self-similarly in exactly one orientation, with the circulations (12) for a fixed c . On a scalene triangle, (12) says that the vector $(1/\Gamma_i)$ is proportional to the cross product of (r_i^2) and $(r_i^{2-2\beta})$, that is, orthogonal to both; so it is equivalent to the vanishing of the angular impulse, $\sum_i r_i^2/\Gamma_i = 0$, together with $\sum_i r_i^{-\alpha}/\Gamma_i = 0$, which is the vanishing of $\sum_{j < k} \Gamma_j \Gamma_k r_i^{-\alpha}$ for $\alpha \neq 0$ and the classical condition $\sum_{j < k} \Gamma_j \Gamma_k = 0$ of Lemma 2 for $\alpha = 0$. Conditions of this kind are given in [5, 7, 17, 37, 42]; Lemma 5 is a short self-contained form of them, with the circulations explicit.

Lemma 6. *For a self-similar collapse, with (12),*

$$2\pi \operatorname{Im} \kappa = \frac{1}{2} \sum_{i=1}^3 \Gamma_i (f_j + f_k) = \frac{c}{2} S, \quad S = \sum_{i=1}^3 r_i^2 \coth\left(\beta \ln \frac{r_k}{r_j}\right), \quad P = \frac{|S|}{8A}. \quad (13)$$

Proof. Since $\dot{z}_1 - \dot{z}_2 = \kappa(z_1 - z_2)$, $\operatorname{Im} \kappa = \operatorname{Im}((\dot{z}_1 - \dot{z}_2)/(z_1 - z_2))$. By (1),

$$2\pi \frac{\dot{z}_1 - \dot{z}_2}{z_1 - z_2} = i(\Gamma_1 + \Gamma_2)f_3 + i\Gamma_3 \frac{f_2(z_1 - z_3) - f_1(z_2 - z_3)}{z_1 - z_2},$$

and by the law of cosines $\operatorname{Re}((z_1 - z_3)/(z_1 - z_2)) = (r_2^2 + r_3^2 - r_1^2)/(2r_3^2)$ and $\operatorname{Re}((z_2 - z_3)/(z_1 - z_2)) = (r_2^2 - r_1^2 - r_3^2)/(2r_3^2)$. Hence

$$2\pi \operatorname{Im} \kappa = (\Gamma_1 + \Gamma_2)f_3 + \Gamma_3 \frac{f_1 + f_2}{2} + \Gamma_3 \frac{(f_2 - f_1)(r_2^2 - r_1^2)}{2r_3^2}.$$

By (12) the last term is $c(r_1^2 - r_2^2)/2$, and also $c(r_1^2 - r_2^2) = \Gamma_1(f_2 - f_3) + \Gamma_2(f_1 - f_3)$. Substituting gives $2\pi \operatorname{Im} \kappa = \frac{1}{2} \sum_i \Gamma_i (f_j + f_k)$, and by (12), $\Gamma_i (f_j + f_k) = cr_i^2 (f_j + f_k)/(f_j - f_k) = cr_i^2 \coth(\beta \ln(r_k/r_j))$. With $\operatorname{Re} \kappa = cA_s/\pi$ from Lemma 5 and $\operatorname{Re} \kappa < 0$, $-2\operatorname{Re} \kappa = 2|c|A/\pi$, and (4) gives $P = |S|/(8A)$. The first two identities hold for every self-similar motion that changes size; for an expansion the same computation gives $|\operatorname{Im} \kappa|/(2\operatorname{Re} \kappa) = |S|/(8A)$. $\square$

The circulations drop out: P depends only on the shape of the triangle. Interchanging two labels reverses the cyclic order and sends each term of S to the negative of another term, so $|S|$ is a symmetric function of the side lengths. The first identity says that the rotation rate is half the sum of the rates at which the three pairs would rotate in isolation. Chen and Liu give every self-similar solution of three vortices explicitly, with the rotation angle in closed form in a shape variable and the circulations [7, Theorems 2.1, 2.2], so the rotation of the spiral, and with it P , is explicit there; Lemma 6 writes P through the side lengths alone, in the form in which we minimize it.

4.2 The bound

Only step (iii) of the proof below uses $\beta > 1/2$, and Remark 4 completes that step for $0 < \beta \leq 1/2$. We use the elementary inequality

$$\frac{1}{\ln(1+x)} \geq \frac{1}{x} + \frac{1}{2} - \frac{x}{12} \quad (x > 0). \quad (14)$$

It is trivial where the right side is not positive, that is, for $x \geq 3 + \sqrt{21}$. For $0 < x < 3 + \sqrt{21}$, the function $g(x) = 12x/(12 + 6x - x^2) - \ln(1+x)$ has $g(0) = 0$ and $g'(x) = x^3(24 - x)/((12 + 6x - x^2)^2(1+x)) > 0$, so $\ln(1+x) \leq 12x/(12 + 6x - x^2)$, which is (14).

Proof of Theorem 2. By Lemma 5 a collapsing triangle is scalene, and by Lemma 6, $P = |S|/(8A)$ is symmetric in the sides and invariant under scaling. Order and scale the sides as $r_2 = \rho < r_3 = 1 < r_1$, and write $r_1^2 = 1 + \rho m$. With ψ the angle at z_1 , between the sides of lengths ρ and 1, the law of cosines gives $2 \cos \psi = \rho - m$, so $0 < m < 2 + \rho$, $0 < \rho < 1$, $A = (\rho/2) \sin \psi$ and $4 \sin^2 \psi = 4 - (m - \rho)^2$. Put $X = \rho^{2\beta}$, $Y = (1 + \rho m)^\beta$ and $\ell = (\beta/2) \ln(1 + \rho m)$. Then $0 < X < 1 < Y$ and, from (13) with $(i, j, k) = (1, 2, 3)$ and its cyclic shifts,

$$S = (1 + \rho m) \frac{1 + X}{1 - X} + \rho^2 \coth \ell - \frac{Y + X}{Y - X}.$$

(i) The function $w \mapsto (w + X)/(w - X)$ decreases for $w > X$, so $(Y + X)/(Y - X) < (1 + X)/(1 - X)$ and $S > \rho m(1 + X)/(1 - X) + \rho^2 \coth \ell > \rho m + \rho^2 \coth \ell$. (ii) $\coth \ell > 1/\ell$, and (14) with $x = \rho m$ gives $\rho^2/\ell \geq (\rho/\beta)(2/m + \rho - \rho^2 m/6)$. Hence

$$S > \frac{\rho}{\beta} \Lambda, \quad \Lambda = \beta m + \frac{2}{m} + \rho - \frac{\rho^2 m}{6} > 0,$$

the last because $\rho < 1$ and $m < 2 + \rho$ give $\rho m < 3$, that is, $\rho^2 m/6 < \rho/2$. In particular $S > 0$, and

$$P = \frac{S}{4\rho \sin \psi} > \frac{\Lambda}{2\beta \sqrt{4 - (m - \rho)^2}}.$$

(iii) The identity

$$\Lambda^2 - (1 + 2\beta)(4 - (m - \rho)^2) = \left((1 + \beta)m - \frac{2}{m} - \rho \right)^2 + \frac{\rho^2}{3} \left(1 + 6\beta - \beta m^2 - \rho m + \frac{\rho^2 m^2}{12} \right) \quad (15)$$

holds for all ρ, m, β , and for $\beta > 1/2$ its right side is positive. If $m < 2$, the second bracket exceeds $1 + 6\beta - 4\beta - 2 = 2\beta - 1 > 0$. If $m \geq 2$, then $2/m \leq 1$ and $\rho < 1$, so $(1 + \beta)m - 2/m - \rho > (1 + \beta)m - 2 > 0$, and the second bracket is at least $q_0(m) = 1 + 6\beta - \beta m^2 - m$. If $q_0(m) \geq 0$ the right side is positive; otherwise $\rho^2 q_0(m) > q_0(m)$, and the right side exceeds $D(m) = ((1 + \beta)m - 2)^2 + q_0(m)/3$. Now $D'' = 2\beta^2 + 10\beta/3 + 2 > 0$ and $D'(2) = 4\beta^2 + 8\beta/3 - 1/3 > 0$, so D increases on $[2, \infty)$, and $D(2) = 4\beta^2 + (2\beta - 1)/3 > 0$. Hence, as $\Lambda > 0$, $\Lambda > \sqrt{1 + 2\beta} \sqrt{4 - (m - \rho)^2}$ and $P > \sqrt{1 + 2\beta}/(2\beta) = B(\alpha)$.

Sharpness. Let $\rho \rightarrow 0$ with m fixed in $(0, 2)$. Exactly,

$$S - \rho m - \rho^2 \coth \ell = 2X \left(\frac{1 + \rho m}{1 - X} - \frac{1}{Y - X} \right) = \frac{2X((1 + \rho m)Y - 1 - \rho m X)}{(1 - X)(Y - X)},$$

and for every $\beta > 0$ we have $X = \rho^{2\beta} \rightarrow 0$ and $Y = 1 + O(\rho)$, so the numerator is $O(X\rho)$ and the denominator tends to 1. With $\rho^2 \coth \ell = 2\rho/(\beta m) + O(\rho^2)$ this gives $S/\rho \rightarrow m + 2/(\beta m)$ and

$$P \rightarrow P_0(m) = \frac{\beta m^2 + 2}{2\beta m \sqrt{4 - m^2}},$$

whose only critical point on $(0, 2)$ is $m_* = \sqrt{2/(1 + \beta)}$, since

$$d \ln P_0 / dm = 4((1 + \beta)m^2 - 2)/(m(\beta m^2 + 2)(4 - m^2)),$$

with $P_0(m_*) = \sqrt{1 + 2\beta}/(2\beta)$. By Lemma 5 each of these triangles, or its mirror image, collapses self-similarly with the circulations (12), so P comes arbitrarily close to $B(\alpha)$. $\square$

Remark 4 (the range $-2 < \alpha \leq -1$). Steps (i) and (ii) and the sharpness argument hold for every $\beta > 0$, but for small β the right side of (15) can be negative, for instance at $\alpha = -1.7$, $\rho = 0.99$ and $m = 1.87$: step (i) discards the factor $(1 + X)/(1 - X)$, which is large when $X = \rho^{2\beta}$ is close to 1. For $0 < \beta \leq 1/2$ we complete step (iii) as follows. Let $q = 1 + 6\beta - \beta m^2 - \rho m + \rho^2 m^2/12$ be the second bracket in (15). If $q > 0$, the right side of (15) is positive, and $\Lambda > 0$ by step (ii), since $\rho < 1$ and $m < 2 + \rho$ give $\rho m < 3$, that is, $\rho^2 m/6 < \rho/2$; taking square roots gives $\Lambda > \sqrt{1 + 2\beta} \sqrt{4 - (m - \rho)^2}$, and $P > B(\alpha)$ as in step (iii). If $q \leq 0$, then $s = \rho m \geq 6 - 2\sqrt{6}$. Indeed, for $m^2 \leq 6$, $s - s^2/12 = 1 + \beta(6 - m^2) - q \geq 1$, so $s^2 - 12s + 12 \leq 0$; for $m^2 > 6$, $m < 2 + \rho$ gives $\rho > m - 2$, and since $(m - 2)m$ increases for $m > 1$, $s > (m - 2)m > (\sqrt{6} - 2)\sqrt{6} = 6 - 2\sqrt{6}$. In this case we keep the factor that step (i) discards: exactly

$$S = \rho m \frac{1 + X}{1 - X} + \rho^2 \coth \ell + \frac{2X(Y - 1)}{(1 - X)(Y - X)},$$

where $\rho^2 \coth \ell > 0$ because $\ell > 0$, and the last term is positive because $0 < X < 1 < Y$. With $v = \beta \ln(1/\rho) > 0$ we have $X = e^{-2v}$, so $(1 + X)/(1 - X) = \coth v > 1/v$, and therefore

$$\frac{\beta S}{\rho} > \frac{m}{\ln(1/\rho)} = \frac{s}{\rho \ln(1/\rho)} \geq (6 - 2\sqrt{6})e,$$

because $\rho \ln(1/\rho) \leq 1/e$ for $0 < \rho < 1$. Since $\sin \psi \leq 1$,

$$P = \frac{S}{4\rho \sin \psi} > \frac{(3 - \sqrt{6})e}{2\beta} \geq \frac{\sqrt{1 + 2\beta}}{2\beta} = B(\alpha),$$

where the last inequality holds whenever $1 + 2\beta \leq (15 - 6\sqrt{6})e^2$, that is, for $\beta \leq ((15 - 6\sqrt{6})e^2 - 1)/2 = 0.6196\dots$, and so for every $\beta \leq 1/2$. Hence Theorem 2 holds for every $\alpha > -2$, and so does Corollary 2 below. No step uses a computer; the program `verify_alpha_below.py` checks each of them (Section 8).

In the extremal limit $\rho \rightarrow 0$ the circulations (12) are, up to a factor, $\Gamma_2 = 1$ at the vertex opposite the shortest side and $\Gamma_1 \approx -\Gamma_3$ of order $\rho^{2\beta-1}$ at its ends: by (12), $\Gamma_1/\Gamma_2 = \beta m \rho^{2\beta-1}(1 + o(1)) = -\Gamma_3/\Gamma_2$ as $\rho \rightarrow 0$ with m fixed. Since $2 \cos \psi = \rho - m$, the angle ψ of the triangle at z_1 tends to the obtuse angle $\arccos(-1/\sqrt{2(1 + \beta)})$ at $m = m_*$, so the vector $z_1 - z_2$, from the vortex opposite the shortest side to the vortex of the tight pair that has its sign, and the vector $z_3 - z_1$ across the pair make an angle that tends to $\arccos(1/\sqrt{2(1 + \beta)})$. For $\alpha > -1$ two weak vortices of opposite sign, at distance ρ , spiral into a strong vortex; for $-2 < \alpha < -1$ the tight pair is strong and the third vortex weak, and at $\alpha = -1$ the three circulations stay comparable. For $\alpha = 0$ this is the limit $\mu \rightarrow 0$ of Corollary 1, with the angle 60° of Theorem 3.

Corollary 2. *Let $\alpha > -2$. In every self-similar collapse of three point vortices under (1), each vortex travels more than $(4 + \alpha)/(2 + \alpha)$ times its initial distance from the collision point, the angle between its path and the direction to that point exceeds $\arccos((2 + \alpha)/(4 + \alpha))$, and $|\omega_0|t_c > 2\sqrt{3 + \alpha}/(2 + \alpha)^2$. None of these constants can be improved.*

Proof. No vortex starts at the collision point: if $z_1 = z_c$, then $\Gamma_2(z_2 - z_1) + \Gamma_3(z_3 - z_1) = 0$ with Γ_2 and Γ_3 real and nonzero by (12), so the triangle is collinear, which Lemma 5 excludes; the same holds for z_2 and z_3 . By Lemma 4 and Section 2 the three quantities are $\sqrt{1 + 4P^2}$, $\arctan 2P$ and P/β , which increase with P , and $\sqrt{1 + 4B^2} = (4 + \alpha)/(2 + \alpha)$. $\square$

	$\alpha = 0$ (Euler)	$\alpha = 1/2$	$\alpha = 1$ (SQG)	$\alpha = 3/2$	$\alpha = 2$
$B(\alpha)$	$\sqrt{3}/2 = 0.8660$	$\sqrt{14}/5 = 0.7483$	$2/3$	0.6061	$\sqrt{5}/4 = 0.5590$
path ratio	2	$9/5$	$5/3$	$11/7$	$3/2$
angle	60°	56.25°	53.13°	50.48°	48.19°

Table 1: The bound of Theorem 2 and the two constants of Corollary 2 for some values of α .

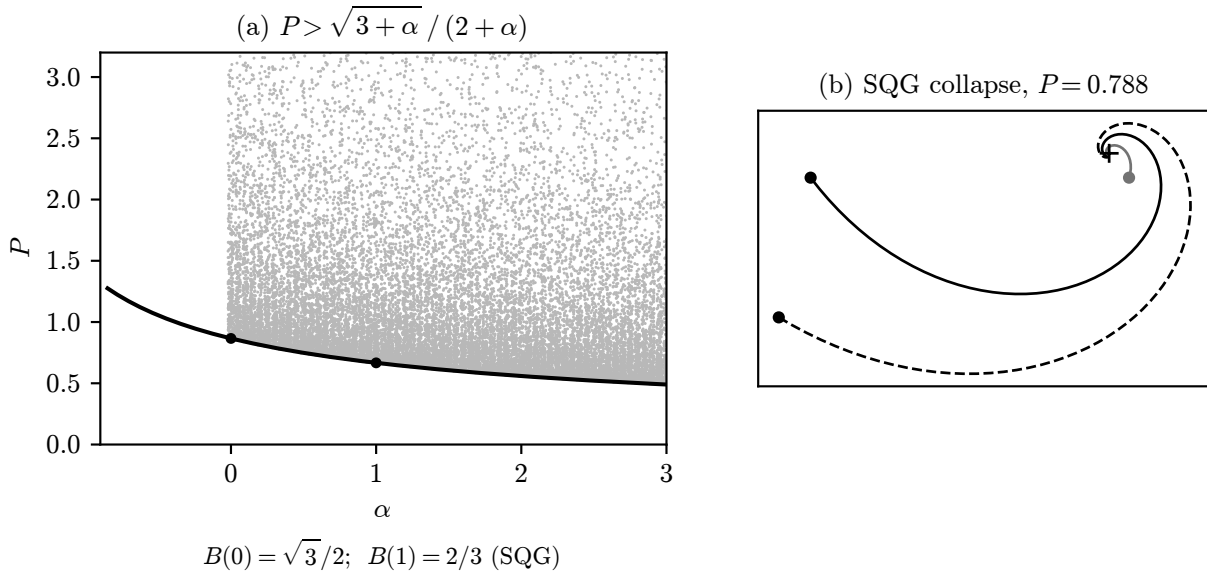


Figure 3: (a) The bound $B(\alpha)$ of Theorem 2 (curve) and $P = |S|/(8A)$, in double precision, for 24395 random collapsing triangles at 61 values of α from 0 to 3, with the shortest side between 10^{-3} and 1 times the middle one (dots, spread slightly in α for visibility); none lies below the curve, which is drawn from $\alpha = -0.85$ and approached as the shortest side tends to 0. (b) The paths of the three vortices of an SQG collapse ($\alpha = 1$) with sides 0.45 , 1 and $(1 + 0.45m_*)^{1/2}$ and the circulations of Lemma 5, integrated from (1) to $10^{-6}t_c$ before the collapse; dots mark the initial positions and the cross the collision point. Panel (a) is dimensionless; panel (b) uses equal spatial scales, divided by the largest initial distance from the collision point. The program is `plot_alpha_winding.py`.

Since $d \ln B/d\alpha = -(4 + \alpha)/(2(3 + \alpha)(2 + \alpha)) < 0$ for $\alpha > -2$, the bound decreases strictly

in α , from ∞ as $\alpha \rightarrow -2$ through $\sqrt{2}$ at $\alpha = -1$, $\sqrt{3}/2$ in the Euler case and $2/3$ for SQG to 0 as $\alpha \rightarrow \infty$ (Table 1 and Figure 3): more local interactions allow tighter spirals, but three vortices never collapse without rotation, which eleven can do at $\alpha = 2$ and sixty can do in the SQG model (Theorem 5). For $\alpha > -1$, that is, $2\beta > 1$, the velocity (1) induced by a vortex decays with distance; the proof uses $2\beta > 1$ only in step (iii), which Remark 4 completes for $-2 < \alpha \leq -1$, where that velocity is constant or grows with distance.

Remark 5 (two equal circulations). For circulations $(1, -\gamma, 1)$ the self-similarly collapsing triangles form a single family. For $\alpha = 2$ it is $r_1 r_3 = r_2^2$, where r_2 joins the two equal vortices, the ratio is $\gamma = r_2^2/(r_1^2 + r_3^2)$, and Lemma 6 gives

$$P^2 = \frac{(1+\gamma)(2\gamma^2 - 2\gamma + 1)^2}{4(1-2\gamma)(1+2\gamma)(3\gamma-1)}, \quad \frac{1}{3} < \gamma < \frac{1}{2},$$

whose minimum, $1.1039451679\dots$, is a root of $27648y^8 - 29952y^6 - 3200y^4 - 1556y^2 - 125$. At $\gamma = 1/3$ the triangle degenerates to the collinear configuration with sides in the ratio $\varphi : 1 : 1/\varphi$, φ the golden ratio, which Yasunaga, Otobe and Iwayama obtain at this end of their family in the limit $\alpha \rightarrow 2$ [42, Sect. 3.1.1]. For SQG the interval $0.387464\dots < \gamma < 1/2$ of [5, Lemma 1] and [7, Theorem 4.1] has as its lower end the root in $(0, 1/2)$ of $4\gamma^4 - 8\gamma^3 + \gamma^2 - 2\gamma + 1$, where the triangle becomes collinear; Iwayama, Yajima and Watanabe give this endpoint in radicals [23, Eq. (4.6)].

5 Two concentric regular polygons

In this section θ denotes the relative rotation of two rings, S , A , ρ and t have meanings of their own, unrelated to those of Section 4, and v is unrelated to $v = \tan \chi$ of Remark 2. Let $n \geq 2$ and $\epsilon = e^{2\pi i/n}$, and place vortices of circulation $x > 0$ at $z\epsilon^k$ and vortices of circulation -1 at $\zeta\epsilon^k$, $k = 0, \dots, n-1$. The motion preserves this symmetry, and the center of vorticity is the origin. Using

$$\sum_{k=1}^{n-1} \frac{1}{1-\epsilon^k} = \frac{n-1}{2}, \quad \sum_{k=0}^{n-1} \frac{1}{z-\zeta\epsilon^k} = \frac{nz^{n-1}}{z^n - \zeta^n},$$

one finds that (2) reduces to [2, Eq. (3)]

$$\bar{z} = \frac{1}{2\pi i} \left(\frac{x(n-1)}{2z} - \frac{nz^{n-1}}{z^n - \zeta^n} \right), \quad \bar{\zeta} = \frac{1}{2\pi i} \left(-\frac{n-1}{2\zeta} + \frac{x n \zeta^{n-1}}{\zeta^n - z^n} \right). \quad (16)$$

A collapse requires the angular impulse $n(x|z|^2 - |\zeta|^2)$ to vanish, and after a rotation and a dilation we take $z = 1$ and $\zeta = \sqrt{x} e^{i\theta}$. The motion is self-similar exactly when $\dot{z}/z = \dot{\zeta}/\zeta$. Writing $v = \zeta^n$, (16) gives

$$\frac{\dot{z}}{z} = \frac{1}{2\pi i} \left(\frac{x(n-1)}{2} - \frac{n}{1-v} \right), \quad \frac{\dot{\zeta}}{\zeta} = \frac{1}{2\pi i} \left(-\frac{n-1}{2x} - \frac{nv}{1-v} \right),$$

and these are equal if and only if

$$(n-1)x^2 - 2nx + (n-1) = 0. \quad (17)$$

This is the circulation condition of Koiller et al. [26, Sect. 11]. Demina and Kudryashov [11, Sect. 3] place circulations Γ_1 and $\Gamma_2 = -\Gamma_1/r^2$ on two such rings, where r is the ratio of the radii, and a vortex of circulation Γ_0 at the center; for $\Gamma_0 = 0$ their equation for r [11, Eq. (37)] is (17) with

$x = r^2$. The condition (17) coincides with the classical condition $\sum_{i < j} \Gamma_i \Gamma_j = 0$ for the $2n$ vortices, whose left side here equals $(n/2)((n-1)x^2 - 2nx + (n-1))$. Its roots are x_n and $1/x_n$, where

$$x_n = \frac{n + \sqrt{2n-1}}{n-1} = e^\eta, \quad \cosh \eta = \frac{n}{n-1}.$$

The root $1/x_n$ gives the same family with the two rings interchanged, as one sees by multiplying the circulations by $-x_n$, reflecting the configuration and rescaling lengths by $x_n^{1/2}$, which does not change P , so we take $x = x_n$. Then every θ gives a self-similar motion. Let $\rho = x_n^{n/2} = e^{n\eta/2} > 1$ and $\tau = n\theta$, so that $v = \rho e^{i\tau}$. The common value of $\bar{\kappa}$ is $S/(2\pi i)$ with $S = x_n(n-1)/2 - n/(1-v)$, so $\kappa = (\text{Im } S + i \text{Re } S)/(2\pi)$, and since $\text{Im}(1-v)^{-1} = \rho \sin \tau / |1-v|^2$,

$$\text{Re } \kappa = -\frac{n\rho \sin \tau}{2\pi|1-v|^2}.$$

The configuration therefore collapses exactly when $\sin n\theta > 0$, that is, for $0 < \theta < \pi/n$ modulo $2\pi/n$. Using $|1-v|^2 = 1 - 2\rho \cos \tau + \rho^2$ one finds

$$\frac{|1-v|^2 \text{Re } S}{\rho} = K_n - ((n-1)x_n - n) \cos \tau, \quad K_n = \frac{(n-1)x_n(\rho + \rho^{-1})}{2} - \frac{n}{\rho}.$$

Here $(n-1)x_n - n = \sqrt{2n-1}$, and writing $n = (n-1) \cosh \eta$ in the last term of K_n gives

$$K_n = (n-1) \sinh \frac{(n+2)\eta}{2}.$$

Moreover

$$K_n - \sqrt{2n-1} = \frac{(n-1)x_n(\rho^{1/2} - \rho^{-1/2})^2}{2} + n(1 - \rho^{-1}) > 0,$$

so $\text{Re } S > 0$, and (4) gives

$$P = \frac{K_n - \sqrt{2n-1} \cos n\theta}{2n \sin n\theta}, \quad 0 < n\theta < \pi. \quad (18)$$

The collapse rate and the rotation rate as functions of the relative angle are given by Koiller et al. [26, Sect. 11]. The constant S is the constant Ω of [11, Eq. (11)], and Demina and Kudryashov write it for these rings, with the central vortex, as an explicit function of $e^{in\theta}$ [11, Eq. (36)]; for $\Gamma_0 = 0$, $\Gamma_1 = x_n$ and radii 1 and $\sqrt{x_n}$ their expression reduces to S by (17). Equation (18) writes the ratio of the two rates in closed form.

Proposition 2. *For $n \geq 2$ the collapsing configurations of two concentric regular n -gons with circulations x_n and -1 satisfy*

$$P \geq F_n = \frac{\sqrt{K_n^2 - (2n-1)}}{2n},$$

with equality if and only if $\cos n\theta = \sqrt{2n-1}/K_n$ and $\sin n\theta > 0$.

Proof. Apply Lemma 1 to (18) with $a = K_n$ and $b = \sqrt{2n-1}$. □

For small n the constants are as follows.

n	x_n	K_n	F_n	$\cos n\theta$ at the minimum
2	$2 + \sqrt{3}$	$4\sqrt{3}$	$3\sqrt{5}/4$	$1/4$
3	$(3 + \sqrt{5})/2$	11	$\sqrt{29}/3$	$\sqrt{5}/11$
4	$(4 + \sqrt{7})/3$	$55\sqrt{7}/9$	$\sqrt{322}/9$	$9/55$
5	2	$127\sqrt{2}/8$	$\sqrt{31682}/80$	$12\sqrt{2}/127$

For $n = 2$ the configuration is a parallelogram of four vortices; Novikov and Sedov [33, Sect. 4] found this collapse and its rates, so its minimum F_2 follows from their formulas, and Gotoda [15] gives its rates in closed form. The ratio x_n is rational exactly when $2n - 1$ is a perfect square, and $n = 5$ is the smallest such n . There $x_5 = 2$ and

$$P = \frac{127\sqrt{2} - 24 \cos 5\theta}{80 \sin 5\theta} \geq \frac{\sqrt{31682}}{80} = 2.2249297741726591 \dots$$

As $n \rightarrow \infty$ the minimum grows without bound:

$$F_n = \frac{1}{4} e^{\sqrt{n/2}} \left(1 + \frac{29}{12\sqrt{2n}} + \frac{265}{576n} + O(n^{-3/2}) \right).$$

Indeed, $\cosh \eta = n/(n-1)$ gives $2 \sinh^2(\eta/2) = 1/(n-1)$, so $E_n = (n+2)\eta/2 = (n+2) \operatorname{arsinh}((2(n-1))^{-1/2})$, and the Taylor series of arsinh at 0 gives $\eta/2 = (2(n-1))^{-1/2} - (2(n-1))^{-3/2}/6 + O(n^{-5/2})$. Since $(2(n-1))^{-1/2} = (2n)^{-1/2}(1 + 1/(2n) + O(n^{-2}))$, we have $(n+2)(2(n-1))^{-1/2} = \sqrt{n/2} + 5/(2\sqrt{2n}) + O(n^{-3/2})$ and $(n+2)(2(n-1))^{-3/2}/6 = 1/(12\sqrt{2n}) + O(n^{-3/2})$, so $E_n = \sqrt{n/2} + 29/(12\sqrt{2n}) + O(n^{-3/2})$. Now $K_n = ((n-1)/2) e^{E_n}(1 - e^{-2E_n})$ and $F_n = (K_n/(2n))(1 - (2n-1)/K_n^2)^{1/2}$, where e^{-2E_n} and $(2n-1)/K_n^2$ are $O(e^{-\sqrt{2n}})$. Hence $F_n = ((n-1)/(4n)) e^{E_n}(1 + O(e^{-\sqrt{2n}}))$. Since $E_n - \sqrt{n/2}$ has no term in $1/n$, $e^{E_n - \sqrt{n/2}} = 1 + 29/(12\sqrt{2n}) + 841/(576n) + O(n^{-3/2})$, where $841/576$ is half the square of $29/(12\sqrt{2})$, and multiplying by $(n-1)/n = 1 - 1/n$ gives the expansion, whose coefficient of $1/n$ is $841/576 - 1 = 265/576$. The ratio of F_n to $e^{\sqrt{n/2}}/4$ is $1.577\dots$, $1.175\dots$ and $1.054\dots$ for $n = 10, 100$ and 1000 .

Now place a vortex of circulation Γ_0 at the center as well. The velocity that each ring induces at the center vanishes, so this vortex stays at rest, and it does not change the angular impulse. It adds Γ_0 to the bracket in the expression for $\dot{z}/z$ above and Γ_0/x to the bracket for $\dot{\zeta}/\zeta$, so the two quotients are equal if and only if

$$(n-1)x^2 - 2(n-\Gamma_0)x + (n-1) - 2\Gamma_0 = 0, \quad \text{that is,} \quad \Gamma_0 = \frac{(n-1)x^2 - 2nx + n-1}{2(1-x)}, \quad (19)$$

and again every θ gives a self-similar motion. On (19) the total circulation does not vanish, since $2(x-1)(n(x-1) + \Gamma_0) = (n+1)x^2 - 2nx + n+1 > 0$, so the center of vorticity is the origin. The left side of (19) equals -2 at $x = 1$ and $n-1-2\Gamma_0$ at $x = 0$, so for every real Γ_0 it has exactly one root $x > 1$, for $\Gamma_0 < (n-1)/2$ also exactly one root in $(0, 1)$, and for $\Gamma_0 \geq (n-1)/2$ none in $(0, 1)$, since it is convex in x and not positive at both ends of that interval. Multiplying the circulations by $-1/x$, reflecting the configuration and rescaling lengths by $x^{-1/2}$, as for $\Gamma_0 = 0$, interchanges the rings and maps the root $x < 1$ for Γ_0 to the root $1/x > 1$ for $-\Gamma_0/x$, with the same P . So we take $x > 1$ and write $x = e^{2t}$ with $t > 0$. With $A = x(n-1)/2 + \Gamma_0$, the common value of $\bar{\kappa}$ is $S/(2\pi i)$ with $S = A - n/(1-v)$.

Proposition 3. Let $n \geq 2$ and $\Gamma_0 \in \mathbb{R}$, and let $x = e^{2t}$, $t > 0$, be the root of (19) greater than 1. The configuration with relative rotation θ collapses exactly when $\sin n\theta > 0$, and then

$$P = \frac{a - b \cos n\theta}{2n \sin n\theta}, \quad a = \coth t \cosh nt + n \sinh nt, \quad b = \coth t. \quad (20)$$

Hence $P \geq \sqrt{D^2 - n^2}/(2n)$, where $D = \coth t \sinh nt + n \cosh nt > 2n$, with equality if and only if $\cos n\theta = b/a$. This minimum over θ exceeds $\sqrt{3}/2$ and decreases strictly from ∞ to $\sqrt{3}/2$ as Γ_0 increases from $-\infty$ to ∞ ; with $h = 1/(\Gamma_0 - 1/2)$ it is

$$\frac{\sqrt{3}}{2} + \frac{\sqrt{3}(1 + 2n^2)}{36} h^2 + O(h^4) \quad \text{as } \Gamma_0 \rightarrow +\infty.$$

In particular $P > \sqrt{3}/2$ for every $n \geq 2$, every Γ_0 and both roots of (19), and no larger constant holds for all of them.

Proof. The central vortex does not change $\text{Im } S = -n\rho \sin \tau/|1 - v|^2$, so the configuration collapses exactly when $\sin n\theta > 0$, and the computation that led to (18) gives $|1 - v|^2 \text{Re } S/\rho = a - b \cos \tau$ with $a = A(\rho + \rho^{-1}) - n/\rho$ and $b = 2A - n$. By (19), $A = (n+1)/2 + 1/(x-1) = (n + \coth t)/2$, and with $\rho = e^{nt}$ this gives the stated a and b . Since $a \mp b = \coth t (\cosh nt \mp 1) + n \sinh nt > 0$, $\text{Re } S > 0$, and (4) gives (20). Lemma 1 gives the minimum $\sqrt{a^2 - b^2}/(2n)$, and $a^2 - b^2 = D^2 - n^2$ because $\cosh^2 nt - \sinh^2 nt = 1$. The difference $\sinh nt - n \sinh t$ vanishes at $t = 0$ and has derivative $n(\cosh nt - \cosh t) \geq 0$, so $\coth t \sinh nt \geq n \cosh t > n$; with $n \cosh nt > n$ this gives $D > 2n$, and hence $D^2 - n^2 > 3n^2$. Moreover $\coth t \sinh nt = \cosh t \sum_{j=0}^{n-1} e^{(n-1-2j)t}$, where the sum pairs into terms $2 \cosh((n-1-2j)t)$, plus 1 when n is odd, so D increases strictly with t , from $2n$ as $t \rightarrow 0$ to ∞ . By (19), $\Gamma_0 = (n + \coth t)/2 - (n-1)e^{2t}/2$, which decreases strictly from ∞ to $-\infty$ as t increases; this gives the monotonicity and the two limits. Finally, D is even in t , with $D = 2n + n(1 + 2n^2)t^2/3 + O(t^4)$, and $\Gamma_0 - 1/2 = 1/(2t) + O(t)$, so $h = 2t + O(t^3)$ and $(D^2 - n^2)/(4n^2) = 3/4 + (1 + 2n^2)h^2/12 + O(h^4)$, which gives the expansion. The root $x < 1$ reduces to a root $x > 1$ by the interchange of the rings described above. $\square$

For $\Gamma_0 = 0$ the root greater than 1 is $x_n = e^\eta$, so $t = \eta/2$. Since $n = (n-1) \cosh \eta$ and $(n-1) \sinh \eta = \sqrt{2n-1}$, we get $b = \coth(\eta/2) = \sqrt{2n-1}$ and $a = (n-1) \sinh((n+2)\eta/2) = K_n$, so (20) is (18) and the minimum is F_n of Proposition 2; in particular $F_n > \sqrt{3}/2$ for every n . The family and its rates are those of Demina and Kudryashov [11, Sect. 3, Eqs. (36)–(37)]: their Eq. (37) with $\Gamma_1 = r^2 = x$ is x times (19), and on (19) their constant Ω is S . For $n = 2$ the configuration is the five-vortex collapse of Novikov and Sedov [33], a parallelogram with a vortex where its diagonals cross, and Gotoda gives its rates in closed form [15], in his Eq. (3.13) in arXiv:2002.09624v1. The parameters of his Fig. 3(b), circulation -1 at $\pm(\sqrt{2}/2)e^{i\theta}$, $1/2$ at ± 1 and $-3/4$ at the origin, are the root $x = 1/2$ of (19) for $n = 2$ and $\Gamma_0 = -3/4$, which the interchange of the rings maps to $x = 2$ and $\Gamma_0 = 3/2$ with the same relative rotation θ . There $\coth t = 3$, $\cosh 2t = 5/4$ and $\sinh 2t = 3/4$, so $a = 21/4$, $b = 3$ and $D = 19/4$, and

$$P = \frac{3}{16} \cdot \frac{7 - 4 \cos 2\theta}{\sin 2\theta} \geq \frac{3\sqrt{33}}{16} = 1.0771054962\dots, \quad 0 < \theta < \frac{\pi}{2},$$

with equality if and only if $\cos 2\theta = 4/7$. For $n = 2$ and $\Gamma_0 = 1/2$ the root is $x = 3$, with $a = 6$, $b = 2$ and $D = 6$, so (20) is the function $(3 - \cos 2\theta)/(2 \sin 2\theta)$ of Remark 2, with the minimum $\sqrt{2}$. The constants κ agree as well: here $A = 2$ and $S = 6/(3 - e^{-2i\theta})$, so $\kappa = 3i/(\pi(3 - e^{2i\theta}))$. For the three vortices of Remark 2, Lemma 3 with $\mu = 1$, whose parameter is $\chi - \pi$, gives $\kappa = 4i/(\pi(3 - e^{2i\chi}))$ at $|z_1 - z_2| = \sqrt{3}/2$, and so $\kappa = 3i/(\pi(3 - e^{2i\chi}))$ at $|z_1 - z_2| = 1$: the three vortices at $\chi = \theta$

have the same κ as these five, for every θ , at these normalizations; if $|z_1 - z_2|$ and the radius of the positive ring are fixed at other values, or the circulations are multiplied by other constants, the two constants differ by a factor independent of θ . We record this as an observation about the two formulas, not as a correspondence between the two motions. Demina and Kudryashov, Novikov and Sedov, and Gotoda give the rates, but none of them minimizes or bounds their ratio. Proposition 3 does not conflict with the seven-vortex collapse with $P = 0.8053\dots$ in Section 3, which has three concentric 2-gons around the central vortex, not two.

6 A strong vortex with weak tight pairs

In Corollary 1 the value $\sqrt{3}/2$ is approached as $\mu \rightarrow 0$, and in Proposition 3 as $\Gamma_0 \rightarrow +\infty$; in both limits a strong vortex carries weak, tight pairs of opposite sign. For three vortices this picture appears in Krishnamurthy and Stremler [29, Sect. 3.6, Fig. 9]: as their parameter g tends to -1 they describe a small vortex dipole moving in the field of a strong, fixed vortex, and on their slice $A = \pi/3$ the self-induced motion of the dipole makes the angle $\pi/3$ with the motion induced by the strong vortex; they compute collapse times and the energy, not the rotation or P . We show that this structure alone forces the constant $\sqrt{3}/2$, for any number of pairs.

Let $m \geq 1$ and $0 < c \leq 1$. We consider $1 + 2m$ point vortices: one of circulation 1 at the origin and, for $j = 1, \dots, m$, one of circulation γa_j at Z_j and one of circulation $-\gamma b_j$ at W_j , where $\gamma > 0$ and, for all j and all $l \neq j$,

$$c \leq a_j, b_j \leq \frac{1}{c}, \quad c \leq |Z_j| \leq \frac{1}{c}, \quad c\gamma|Z_j| \leq |W_j - Z_j| \leq \frac{\gamma|Z_j|}{c}, \quad |Z_j - Z_l| \geq c. \quad (21)$$

We write $d_j = W_j - Z_j$ and

$$u_j = \frac{d_j}{\gamma a_j Z_j} = p_j + iy_j, \quad \nu_j = \frac{a_j - b_j}{\gamma a_j^2},$$

so that $W_j = Z_j(1 + \gamma a_j u_j)$, $b_j = a_j(1 - \gamma a_j \nu_j)$ and, by (21), $c^2 \leq |u_j| \leq c^{-2}$. (In this section u_j , and h_j and D_j below, are unrelated to u of Section 3 and to h and D of Sections 3 and 5.) Pair j has length $\gamma a_j |u_j| |Z_j|$, and $\arg u_j$ is the angle from the direction Z_j , away from the strong vortex, to the direction from the positive to the negative vortex of the pair. The normalization is no loss of generality: a translation puts the strong vortex at the origin, multiplying all circulations by a positive constant leaves P unchanged, and the map $(\Gamma_i, z_i) \mapsto (-\Gamma_i, \bar{z}_i)$ of Section 3 sends κ to $\bar{\kappa}$ and keeps P , so a strong vortex of negative circulation is covered as well, after exchanging the roles of Z_j and W_j (and of a_j and b_j); for $\gamma \leq \gamma_0$ the image satisfies (21) with c replaced by $3c/4$.

Theorem 3 (A strong vortex with weak tight pairs). *Let $m \geq 1$ and $0 < c \leq 1$, and consider configurations as above that collapse self-similarly, that is, $\dot{z}_i = \kappa(z_i - z_c)$ for all $1 + 2m$ vortices, with $\operatorname{Re} \kappa < 0$.*

(a) *For every $\delta > 0$ there is $\gamma_1 > 0$, depending only on m, c and δ , such that every such collapse with $\gamma < \gamma_1$ has $P > \sqrt{3}/2 - \delta$.*

(b) *For every $\delta > 0$ there are $\eta > 0$ and $\gamma_2 > 0$, depending only on m, c and δ , such that every such collapse with $\gamma < \gamma_2$ and $P < \sqrt{3}/2 + \eta$ has $|u_j - e^{i\pi/3}| < \delta$, $|\nu_j - 1| < \delta$ and $||Z_j|/|Z_l| - 1| < \delta$ for all j and l .*

(c) *For every $\delta > 0$ and $\bar{P} > 0$ there is $\gamma_3 > 0$, depending only on m, c, δ and $\bar{P}$, such that every such collapse with $\gamma < \gamma_3$ and $P \leq \bar{P}$ has, for every j , $|p_j - 1/2| < \delta$, $|\nu_j - 1| < \delta$, $y_j > 0$ and*

$$\left| P - \frac{y_j^2 + 3/4}{2y_j} \right| < \delta, \quad \text{where} \quad \frac{y_j^2 + 3/4}{2y_j} = \frac{\sqrt{3}}{2} + \frac{(y_j - \sqrt{3}/2)^2}{2y_j}.$$

For every m , the constant $\sqrt{3}/2$ in (a) cannot be increased when $c \leq 1/2$ for $m = 1$ and $c \leq \min(1/2, 2 \sin(\pi/m))$ for $m \geq 2$.

So, for weak pairs, P is close to $\sqrt{3}/2$ only if every pair has length close to $\gamma a_j |Z_j|$ and makes the angle 60° with the direction away from the strong vortex, all the pairs are at nearly the same distance from the strong vortex, and the negative circulation of each pair, $-\gamma a_j + \gamma^2 a_j^2 \nu_j$, differs from the three-vortex harmonic value $-\gamma a_j / (1 + \gamma a_j)$ by $\gamma^2 a_j^2 (\nu_j - 1) + O(\gamma^3)$, which is small compared with γ^2 .

Proof. It suffices to prove three statements about a sequence of such collapses with $\gamma = \gamma_n \rightarrow 0$, in which we write $P_n, \kappa_n, u_{j,n} = p_{j,n} + i y_{j,n}$, and so on: (S1) $\liminf P_n \geq \sqrt{3}/2$; (S2) if $P_n \rightarrow \sqrt{3}/2$, then $u_{j,n} \rightarrow e^{i\pi/3}$, $\nu_{j,n} \rightarrow 1$ and $|Z_{j,n}|/|Z_{l,n}| \rightarrow 1$ for all j and l ; (S3) if $\sup_n P_n < \infty$, then for every j , $p_{j,n} \rightarrow 1/2$, $\nu_{j,n} \rightarrow 1$, $\liminf_n y_{j,n} > 0$ and $P_n - (y_{j,n}^2 + 3/4)/(2y_{j,n}) \rightarrow 0$. Indeed, if (a) failed for some δ , there would be a sequence with $\gamma_n \rightarrow 0$ and $P_n \leq \sqrt{3}/2 - \delta$, contrary to (S1). If (b) failed for some δ , there would be a sequence with $\gamma_n \rightarrow 0$ and $P_n < \sqrt{3}/2 + 1/n$ for which one of the inequalities fails for every n , and after passing to a subsequence it is always the same inequality, for the same j and l ; by (S1), $P_n \rightarrow \sqrt{3}/2$, contrary to (S2). The same argument with $P_n \leq \bar{P}$ reduces (c) to (S3).

Throughout, $\gamma \leq \gamma_0 = c^3/(8m)$, which along a sequence with $\gamma_n \rightarrow 0$ holds after finitely many terms are discarded, and $O(\gamma^k)$ denotes a quantity bounded by γ^k times a constant that depends only on m and c . Each identity used below is a direct computation, verified in `verify_strong_vortex.py` (Section 8).

Step 1: a priori bounds. By (21), $|d_j| \leq \gamma/c^2 \leq c/8$. Hence $|W_j| \geq c/2$; every vortex of pair j is at distance at least $c - 2\gamma/c^2 \geq c/2$ from every vortex of pair $l \neq j$; and $\sum_i \Gamma_i = 1 + \gamma \sum_j (a_j - b_j)$ lies in $[1/2, 2]$. The identity

$$\sum_i \Gamma_i z_i = \sum_j (\gamma(a_j - b_j)Z_j - \gamma b_j d_j) \quad (22)$$

gives $|\sum_i \Gamma_i z_i| \leq 2m\gamma/c^2$, so that $|z_c| \leq 4m\gamma/c^2 \leq c/2$ and $|Z_j - z_c| \geq c/2$. The contribution of pair l to the sum in (2) at a point z is

$$f_l(z) = \frac{\gamma a_l}{z - Z_l} - \frac{\gamma b_l}{z - W_l} = \frac{\gamma(a_l - b_l)}{z - Z_l} - \frac{\gamma b_l d_l}{(z - Z_l)(z - W_l)}.$$

At the vortices of a pair $j \neq l$ it satisfies $|f_l| \leq 4\gamma/c^2$, and

$$f_l(W_j) - f_l(Z_j) = d_j \left(\frac{\gamma b_l}{(W_j - W_l)(Z_j - W_l)} - \frac{\gamma a_l}{(W_j - Z_l)(Z_j - Z_l)} \right)$$

gives $|f_l(W_j) - f_l(Z_j)| \leq 8\gamma|d_j|/c^3$. Finally, $|\gamma b_j/d_j| \leq c^{-3}$.

Step 2: the equations of a pair. With the strong vortex at the origin, the equation $\dot{z}_i = \kappa(z_i - z_c)$ at Z_j , and the difference of the equations at W_j and Z_j , in which z_c cancels, read by (2)

$$2\pi i \bar{\kappa} (\overline{Z_j - z_c}) = \frac{1}{Z_j} + \frac{\gamma b_j}{d_j} + \sum_{l \neq j} f_l(Z_j), \quad (23)$$

$$2\pi i \bar{\kappa} \overline{d_j} = -\frac{d_j}{Z_j W_j} + \frac{\gamma(a_j - b_j)}{d_j} + \sum_{l \neq j} (f_l(W_j) - f_l(Z_j)). \quad (24)$$

By Step 1 the right side of (23) has absolute value at most $1/c + 1/c^3 + 4m\gamma/c^2 \leq 3/c^3$, and $|Z_j - z_c| \geq c/2$, so $|\kappa| \leq 3/(\pi c^4)$.

Step 3: the relative motion. Multiply (24) by $Z_j/(\gamma a_j)$ and write $-d_j/(Z_j W_j) = -d_j/Z_j^2 + d_j^2/(Z_j^2 W_j)$. Since $d_j = \gamma a_j u_j Z_j$, the multiplication turns $\bar{d}_j$ into $|Z_j|^2 \bar{u}_j$, $\gamma(a_j - b_j)/d_j$ into ν_j/u_j , $-d_j/Z_j^2$ into $-u_j$, and $d_j^2/(Z_j^2 W_j)$ into $\gamma a_j u_j^2/(1 + \gamma a_j u_j)$. Hence, by Step 1 and $|\gamma a_j u_j| = |d_j|/|Z_j| \leq \gamma/c^3 \leq 1/8$,

$$2\pi i \bar{\kappa} |Z_j|^2 \bar{u}_j = -u_j + \frac{\nu_j}{u_j} + \rho_j, \quad |\rho_j| \leq \gamma \left(\frac{2}{c^5} + \frac{8(m-1)}{c^7} \right) \leq \frac{1}{c^4}. \quad (25)$$

Since $|\kappa|$, $|Z_j|$, $|u_j|$ and $1/|u_j|$ are bounded, (25) bounds ν_j , for instance $|\nu_j| \leq 8c^{-10}$; this uses no information on b_j/a_j . So $a_j - b_j = \gamma a_j^2 \nu_j = O(\gamma)$: the total circulation of each pair is $O(\gamma^2)$. Three consequences follow. By (22), $\sum_i \Gamma_i z_i = \sum_j (\gamma^2 a_j^2 \nu_j Z_j - \gamma b_j d_j) = O(\gamma^2)$, so $z_c = O(\gamma^2)$. Next, $b_j/a_j = 1 - \gamma a_j \nu_j = 1 + O(\gamma)$. And by the second form of f_l , $f_l(Z_j) = O(\gamma^2)$ for $l \neq j$, although the bound $O(\gamma)$ of Step 1 suffices below. Multiplying (23) by Z_j and using $\gamma b_j Z_j/d_j = (b_j/a_j)/u_j$ and $(\overline{Z_j - z_c}) Z_j = |Z_j|^2 - \bar{z}_c Z_j$, we obtain

$$2\pi i \bar{\kappa} |Z_j|^2 = 1 + \frac{1}{u_j} + O(\gamma). \quad (26)$$

Step 4: the angular impulse. The angular impulse about z_c vanishes (Section 2). With the strong vortex at the origin, $\sum_i \Gamma_i |z_i - z_c|^2 = \sum_i \Gamma_i |z_i|^2 - |z_c|^2 \sum_i \Gamma_i$, and pair j contributes

$$\gamma a_j |Z_j|^2 - \gamma b_j |W_j|^2 = \gamma^2 |Z_j|^2 (a_j^2 \nu_j - 2a_j b_j p_j - \gamma a_j^2 b_j |u_j|^2)$$

to $\sum_i \Gamma_i |z_i|^2$. Since $z_c = O(\gamma^2)$ and $a_j b_j = a_j^2 - \gamma a_j^3 \nu_j$, dividing by γ^2 gives

$$\sum_j a_j^2 |Z_j|^2 (\nu_j - 2p_j) = O(\gamma). \quad (27)$$

Here $z_c = O(\gamma^2)$ is needed: the a priori bound $z_c = O(\gamma)$ would leave a term of order 1.

Step 5: limits. Along the sequence, a_j , Z_j , u_j , ν_j and κ are bounded, so every subsequence has a further subsequence along which they converge, to a_j^* , Z_j^* , $u_j^* = p_j^* + iy_j^*$, ν_j^* and κ^* . Then $b_j \rightarrow a_j^*$, $a_j^* \geq c$, $|Z_j^*| \geq c$, $|u_j^*| \geq c^2$ and $\text{Re } \kappa^* \leq 0$, and (26), (25) and (27) become

$$2\pi i \bar{\kappa}^* |Z_j^*|^2 = 1 + \frac{1}{u_j^*}, \quad 2\pi i \bar{\kappa}^* |Z_j^*|^2 \bar{u}_j^* = -u_j^* + \frac{\nu_j^*}{u_j^*}, \quad \sum_j a_j^{*2} |Z_j^*|^2 (\nu_j^* - 2p_j^*) = 0. \quad (28)$$

Eliminating κ^* between the first two and multiplying by u_j^* gives

$$2p_j^{*2} + p_j^* - \nu_j^* + iy_j^* (2p_j^* - 1) = 0,$$

so $y_j^* (2p_j^* - 1) = 0$ and $\nu_j^* = 2p_j^{*2} + p_j^*$. By the first equation, $\kappa^* = i(1 + 1/u_j^*)/(2\pi |Z_j^*|^2)$, that is,

$$\text{Re } \kappa^* = -\frac{y_j^*}{2\pi |Z_j^*|^2 |u_j^*|^2}, \quad \text{Im } \kappa^* = \frac{1 + p_j^*/|u_j^*|^2}{2\pi |Z_j^*|^2} \quad (29)$$

for every j .

Case A: some $y_j^ \neq 0$.* Then $\text{Im}(1 + 1/u_j^*) = -y_j^*/|u_j^*|^2 \neq 0$, so $2\pi i \bar{\kappa}^*$ is not real, and the first equation of (28) shows $y_l^* \neq 0$ for every l . Hence $p_l^* = 1/2$ and $\nu_l^* = 1$ for every l , and since $\text{Re } \kappa^* \leq 0$, (29) gives $y_l^* > 0$ for every l and $\text{Re } \kappa^* < 0$. So P , a continuous function of κ on $\text{Re } \kappa < 0$, converges along the subsequence to

$$P^* = \frac{\text{Im } \kappa^*}{-2 \text{Re } \kappa^*} = \frac{|u_l^*|^2 + p_l^*}{2y_l^*} = \frac{y_l^{*2} + 3/4}{2y_l^*} = \frac{\sqrt{3}}{2} + \frac{(y_l^* - \sqrt{3}/2)^2}{2y_l^*} \geq \frac{\sqrt{3}}{2},$$

the same number for every l . Thus every y_l^* is one of the two positive roots, y^* and $3/(4y^*)$, of $y^2 - 2P^*y + 3/4$; different pairs may take different roots, and they then lie at different distances from the strong vortex, by the first equation of (28). If $P^* = \sqrt{3}/2$, the two roots coincide, every $y_l^* = \sqrt{3}/2$, so $u_l^* = e^{i\pi/3}$, and the first equation of (28) gives the same $|Z_l^*|$ for every l .

Case B: every $y_j^ = 0$.* Then $1 + 1/u_j^*$ is real, so κ^* is purely imaginary. If $\kappa^* \neq 0$, then $\text{Re } \kappa_n \rightarrow 0^-$ while $|\text{Im } \kappa_n| \rightarrow |\kappa^*| > 0$, so $P_n \rightarrow \infty$ along the subsequence. If $\kappa^* = 0$, then $1 + 1/p_j^* = 0$, so $p_j^* = -1$ and $\nu_j^* = 1$ for every j , and the last equation of (28) reads $3 \sum_j a_j^{*2} |Z_j^*|^2 = 0$, which is impossible. This is the only place where the angular impulse enters.

Conclusion. Every subsequence has a further subsequence along which either $P_n \rightarrow \infty$, or Case A holds and $P_n \rightarrow P^* \geq \sqrt{3}/2$; this proves (S1). If $P_n \rightarrow \sqrt{3}/2$, only Case A with $P^* = \sqrt{3}/2$ can occur, so every subsequence has a further subsequence along which $u_{j,n} \rightarrow e^{i\pi/3}$, $\nu_{j,n} \rightarrow 1$ and $|Z_{j,n}|/|Z_{l,n}| \rightarrow 1$; since these limits do not depend on the subsequence, the whole sequence converges to them, which proves (S2). If $\sup_n P_n < \infty$, only Case A can occur, and the same argument gives $p_{j,n} \rightarrow 1/2$, $\nu_{j,n} \rightarrow 1$ and $P_n - (y_{j,n}^2 + 3/4)/(2y_{j,n}) \rightarrow 0$, while $\liminf_n y_{j,n} > 0$ because every subsequence has a further subsequence along which $y_{j,n}$ has a positive limit; this proves (S3). Finally, the families of Corollary 1 and Proposition 3 contain collapses that satisfy (21), for $m = 1$ and for every $m = n \geq 2$, and have $P \rightarrow \sqrt{3}/2$, as shown next; so $\sqrt{3}/2$ in (a) cannot be increased. $\square$

The two families lie in the class (21) near their minima. For three vortices, translate vortex 1 to the origin and take $m = 1$, $\gamma = \mu$, $a_1 = 1$, $b_1 = 1/(1 + \mu)$, $Z_1 = z_2 - z_1$ and $W_1 = z_3 - z_1$ in (7); then $\nu_1 = 1/(1 + \mu)$ and $|Z_1| = \sqrt{R}/(1 + \mu)$, and along the configurations with $\sin \theta = -\sqrt{3}\mu/2$ used for sharpness in the direct proof of Corollary 1, $u_1 \rightarrow e^{i\pi/3}$ as $\mu \rightarrow 0$. For the rings of Proposition 3 with $\Gamma_0 > 0$, divide the circulations by Γ_0 and take $m = n$, $\gamma = 1/\Gamma_0$, $a_j = x$, $b_j = 1$, $Z_j = e^j$ and $W_j = \sqrt{x} e^{i\theta} e^j$; then $\nu_j = (x - 1)\Gamma_0/x^2 \rightarrow 1$ as $\Gamma_0 \rightarrow +\infty$, and at the minimizing θ , where $\cos n\theta = b/a$ in (20), $\theta/t \rightarrow \sqrt{3}$ and $u_j = (\sqrt{x} e^{i\theta} - 1)\Gamma_0/x \rightarrow e^{i\pi/3}$. In both cases (21) holds for all small γ , with $c = 1/2$ for three vortices and $c = \min(1/2, 2\sin(\pi/n))$ for the rings, and $P \rightarrow \sqrt{3}/2$ by Corollary 1 and Proposition 3. This is why the same constant appears in both results. Theorem 3 does not reprove them: it is asymptotic, Proposition 4 below gives a bound only for γ below a threshold that is not explicit, and for three vortices the fact that a bounded P forces a tight pair as $\mu \rightarrow 0$ follows from (9) and Theorem 1(c) (on $\mathcal{A}_+$, $P \geq P_+(\mu) \rightarrow \infty$; on $\mathcal{A}_-$, a bounded P forces $\theta - \pi$ to be of order μ), not from the theorem. Nor does it say anything about collapses outside its hypotheses, and these cannot simply be dropped: the seven-vortex collapse of Section 3, with $P = 0.8053\dots < \sqrt{3}/2$, has no dominant vortex, since its other circulations are 0.33, 0.35 and 0.49 times the central one; its six other vortices are four positive and two negative, so they do not form opposite-signed pairs; and the nearest vortices of opposite sign in it are separated by 0.38 times their distance from the center.

Proposition 4 (The bound at a fixed circulation). *Let $m \geq 1$ and $0 < c \leq 1/2$, and consider collapses as in Theorem 3. There are $\gamma_4 > 0$, $\eta_4 > 0$ and $K_4 > 0$, depending only on m and c , such that every such collapse with $\gamma < \gamma_4$ has*

$$P \geq \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{8} c^2 \gamma^2 > \frac{\sqrt{3}}{2}. \quad (30)$$

If moreover $P < \sqrt{3}/2 + \eta_4$, then the directions $\phi_j = \arg Z_j$ satisfy $|e^{i\phi_j} - e^{i\phi_l}| \geq 3c^2/4$ for $j \neq l$, and

$$P \geq \frac{\sqrt{3}}{2} + (C_* - K_4\gamma)\gamma^2, \quad C_* = \max_j C_j, \quad C_j = \frac{3\sqrt{3}}{4} a_j^2 + \sum_{l \neq j} a_l^2 \chi(\phi_l - \phi_j), \quad (31)$$

where

$$\chi(\theta) = \frac{2\sqrt{3}}{3} \cos \theta + \frac{\sqrt{3}}{6 \sin^2(\theta/2)} + \frac{\cos(\theta/2)(1 + 2 \sin^2(\theta/2))}{6 \sin^3(\theta/2)}; \quad (32)$$

and $C_* \geq (\sqrt{3}/4) \min_j a_j^2 \geq (\sqrt{3}/4) c^2$.

Since the class (21) grows as c decreases, the proposition applies for $1/2 < c \leq 1$ with c replaced by $1/2$. The constants γ_4 and η_4 come from Theorem 3(b), which is proved by compactness, and are not explicit; K could be made explicit from the proof. Unlike Theorem 3(a), the proposition gives $P > \sqrt{3}/2$ at every fixed $\gamma < \gamma_4$.

Proof. Let $\gamma_* = \min(\gamma_0, c^3/32)$ and $\delta_0 = c^2/4 \leq 1/16$. Throughout, $\gamma \leq \gamma_*$, so that $h_j = \gamma a_j \leq \gamma/c \leq c^2/32$, and K denotes a positive constant that depends only on m and c , not always the same one; $O(X)$ denotes a quantity bounded by KX . We use Steps 1–3 of the proof of Theorem 3, write $\Lambda = 2\pi i \bar{\kappa}$, $R_j = \Lambda |Z_j|^2$, $\varrho_j = b_j/a_j = 1 - h_j \nu_j$, $t_{jl} = Z_l/Z_j$, $\tau_{jl} = t_{jl}/|t_{jl}| = e^{i(\phi_l - \phi_j)}$, $\xi_j = p_j - 1/2$ and $\Delta_j = y_j - \sqrt{3}/2$, and set

$$\delta_* = \max_{j,l} \max \left(|u_j - e^{i\pi/3}|, |\nu_j - 1|, \left| \frac{|Z_j|}{|Z_l|} - 1 \right| \right).$$

Since $\operatorname{Re} \Lambda = 2\pi \operatorname{Im} \kappa$ and $\operatorname{Im} \Lambda = 2\pi \operatorname{Re} \kappa < 0$, (4) reads $P = |\operatorname{Re} \Lambda|/(2|\operatorname{Im} \Lambda|)$. Steps 1–5 below assume $\delta_* \leq \delta_0$, and the constants in them do not depend on δ_* . Then $|u_j| \leq 17/16$, $y_j \geq \sqrt{3}/2 - 1/16 > 1/2$, $|\nu_j| \leq 2$ and $|t_{jl}| \leq 17/16$. By (21), $|1 - t_{jl}| = |Z_j - Z_l|/|Z_j| \geq c^2$, and $|t_{jl} - \tau_{jl}| = ||t_{jl}| - 1| \leq \delta_*$, so $|e^{i\phi_j} - e^{i\phi_l}| = |1 - \tau_{jl}| \geq 3c^2/4$ and $|\sin((\phi_l - \phi_j)/2)| \geq 3c^2/8$; hence $|C_j| \leq K$.

Step 1: the exact equations of a pair. Multiply (23) by Z_j and (24) by $Z_j/(\gamma a_j)$, as in Step 3 of the proof of Theorem 3, but keep every term. Since $2\pi i \bar{\kappa} (Z_j - z_c) Z_j = R_j - \Lambda \bar{z}_c Z_j$, $\gamma b_j Z_j/d_j = \varrho_j/u_j$, $2\pi i \bar{\kappa} \bar{d}_j Z_j/(\gamma a_j) = R_j \bar{u}_j$, $-d_j/(Z_j W_j) \cdot Z_j/(\gamma a_j) = -u_j/(1 + h_j u_j)$ and $\gamma(a_j - b_j)/d_j \cdot Z_j/(\gamma a_j) = \nu_j/u_j$,

$$R_j = 1 + \frac{\varrho_j}{u_j} + \varepsilon_j, \quad \varepsilon_j = \sum_{l \neq j} Z_j f_l(Z_j) + \Lambda \bar{z}_c Z_j, \quad (33)$$

$$R_j \bar{u}_j = -\frac{u_j}{1 + h_j u_j} + \frac{\nu_j}{u_j} + \sigma_j, \quad \sigma_j = \frac{Z_j}{\gamma a_j} \sum_{l \neq j} (f_l(W_j) - f_l(Z_j)). \quad (34)$$

Multiplying (34) by u_j and inserting (33) gives

$$(2p_j^2 + p_j - \nu_j) + iy_j(2p_j - 1) = |u_j|^2 + \bar{u}_j + u_j^2 - \nu_j = G_j := h_j \nu_j \bar{u}_j + \frac{h_j u_j^3}{1 + h_j u_j} + \sigma_j u_j - \varepsilon_j |u_j|^2. \quad (35)$$

Step 2: the sources. Inserting $b_l = a_l \varrho_l$ and $W_l = Z_l(1 + h_l u_l)$ into f_l and into (22), where $\sum_i \Gamma_i = 1 + \gamma^2 \sum_l a_l^2 \nu_l$, gives the exact forms

$$Z_j f_l(Z_j) = \gamma^2 a_l^2 A(t_{jl}, u_l, \nu_l, h_l), \quad \frac{Z_j}{\gamma a_j} (f_l(W_j) - f_l(Z_j)) = \gamma^2 a_l^2 u_j B(t_{jl}, u_l, \nu_l, h_l, u_j, h_j),$$

$$\Lambda \bar{z}_c Z_j = R_j \bar{z}_c / Z_j = \gamma^2 R_j \frac{\overline{\sum_l a_l^2 (\nu_l - \varrho_l u_l) t_{jl}}}{1 + \gamma^2 \sum_l a_l^2 \nu_l} \quad (t_{jj} = 1),$$

where

$$A(t, u, \nu, h) = \frac{\nu}{1-t} - \frac{(1-h\nu)ut}{(1-t)(1-t-hut)},$$

$$B(t, u', \nu', h', u, h) = -\frac{\nu'}{(1-t)(1-t+hu)} + \frac{(1-h'\nu')u't(2-2t-h'u't+hu)}{(1-t)(1-t-h'u't)(1-t+hu)(1-t-h'u't+hu)}.$$

At $u = u' = e^{i\pi/3}$, $\nu' = \nu = 1$ and $h = h' = 0$ they become $A_0(t) = 1/(1-t) - e^{i\pi/3}t/(1-t)^2$ and $B_0(t) = -1/(1-t)^2 + 2e^{i\pi/3}t/(1-t)^3$. Let $\mathcal{K}$ be the compact set of the (t, u', ν', h', u, h) with $|t| \leq 2$, $|1-t| \geq c^2/2$, $|u'|, |\nu'|, |u| \leq 2$ and $0 \leq h, h' \leq c^2/32$. On $\mathcal{K}$ the factors $1-t-h'u't$, $1-t+hu$ and $1-t-h'u't+hu$ have modulus at least $3c^2/8$, $7c^2/16$ and $5c^2/16$, so A and B are smooth on a neighbourhood of $\mathcal{K}$, and they and their first derivatives are bounded on $\mathcal{K}$ by constants that depend only on c . The point $(t_{jl}, u_l, \nu_l, h_l, u_j, h_j)$ lies in $\mathcal{K}$, and so does the segment from it to $(\tau_{jl}, e^{i\pi/3}, 1, 0, e^{i\pi/3}, 0)$, since along it $|t| \leq 17/16$ and $|1-t| \geq |1-t_{jl}| - \delta_* \geq c^2/2$. The mean value inequality along this segment gives $A(t_{jl}, u_l, \nu_l, h_l) = A_0(\tau_{jl}) + O(\delta_* + \gamma)$ and $u_j B(\dots) = e^{i\pi/3} B_0(\tau_{jl}) + O(\delta_* + \gamma)$. In the last term, $|R_j| = |\Lambda||Z_j|^2 \leq 6/c^6$ by Step 2 of the proof of Theorem 3, $\nu_l - \varrho_l u_l = 1 - e^{i\pi/3} + O(\delta_* + \gamma)$, $t_{jl} = \tau_{jl} + O(\delta_*)$ and the denominator is $1 + O(\gamma^2)$. Hence first $|\varepsilon_j| + |\sigma_j| = O(\gamma^2)$; then, by (33), $|R_j - 1 - e^{-i\pi/3}| \leq |1/u_j - e^{-i\pi/3}| + h_j |\nu_j/u_j| + |\varepsilon_j| = O(|u_j - e^{i\pi/3}| + \gamma)$; and, since $(1 + e^{-i\pi/3})(1 - e^{i\pi/3}) = \sqrt{3}e^{i\pi/6}$,

$$\begin{aligned} \varepsilon_j &= \gamma^2 E_j + O(\gamma^2(\delta_* + \gamma)), & \sigma_j &= \gamma^2 S_j + O(\gamma^2(\delta_* + \gamma)), \\ E_j &= \sum_{l \neq j} a_l^2 A_0(\tau_{jl}) + \sqrt{3}e^{i\pi/6} \sum_l a_l^2 \bar{\tau}_{jl}, & S_j &= e^{i\pi/3} \sum_{l \neq j} a_l^2 B_0(\tau_{jl}). \end{aligned} \quad (36)$$

Step 3: one pair. Fix j and let $F_j = \text{Re}(e^{-i\pi/3} R_j)$. By (33), $2|u_j|^2 F_j = |u_j|^2 + \varrho_j(p_j - \sqrt{3}y_j) + 2|u_j|^2 \text{Re}(e^{-i\pi/3} \varepsilon_j)$. Insert $|u_j|^2 + p_j - \sqrt{3}y_j = \Delta_j^2 + 2\xi_j + \xi_j^2$, $\varrho_j = 1 - h_j \nu_j$, $2\xi_j = \text{Im } G_j/y_j$ from (35), and $\text{Im}(u_j^3)/y_j = 3p_j^2 - y_j^2 = 3\xi_j + 3\xi_j^2 - \sqrt{3}\Delta_j - \Delta_j^2$; this gives the identity (dropping the index j)

$$\begin{aligned} 2|u|^2 F &= (1-h)\Delta^2 + \xi^2 + h(3-\nu)\xi + 3h\xi^2 + \sqrt{3}h(\nu-1)\Delta - \frac{h^2}{y} \text{Im} \frac{u^4}{1+hu} \\ &\quad + \frac{\text{Im}(\sigma u)}{y} + |u|^2 \left(2 \text{Re}(e^{-i\pi/3} \varepsilon) - \frac{\text{Im} \varepsilon}{y} \right). \end{aligned} \quad (37)$$

By Step 2, $G = O(\gamma)$, so $\xi = \text{Im } G/(2y) = O(\gamma)$, and the real part of (35), $\nu-1 = 2\xi(3/2+\xi) - \text{Re } G$, gives $\nu-1 = O(\gamma)$. Next, $y(2\xi+h) = \text{Im } G + hy = -h(\nu-1)y + h \text{Im}(u^3/(1+hu)) + \text{Im}(\sigma u) - |u|^2 \text{Im} \varepsilon$, and $\text{Im}(u^3/(1+hu)) = \text{Im}(u^3 - e^{i\pi}) + O(h) = O(\delta_* + \gamma)$, so $\zeta = \xi + h/2 = O(\gamma(\delta_* + \gamma))$. Then

$$\xi^2 + h(3-\nu)\xi = -\frac{3}{4}h^2 + h\zeta + \zeta^2 + \frac{1}{2}h^2(\nu-1) - h(\nu-1)\zeta = -\frac{3}{4}h^2 + O(\gamma^2(\delta_* + \gamma)),$$

$3h\xi^2 = O(\gamma^3)$, $\sqrt{3}h(\nu-1)\Delta = O(\gamma^2\delta_*)$, and, since $\text{Im}(e^{4i\pi/3})/(\sqrt{3}/2) = -1$ and $(u, h) \mapsto \text{Im}(u^4/(1+hu))/\text{Im } u$ is Lipschitz on $|u - e^{i\pi/3}| \leq 1/16$, $0 \leq h \leq 1/128$, the sixth term is $h^2 + O(\gamma^2(\delta_* + \gamma))$. Replacing u by $e^{i\pi/3}$ and y by $\sqrt{3}/2$ in the last two terms costs $O((|\varepsilon| + |\sigma|)\delta_*) = O(\gamma^2\delta_*)$, and $\text{Im}(\sigma e^{i\pi/3})/(\sqrt{3}/2) = (2/\sqrt{3}) \text{Re}(e^{-i\pi/6}\sigma)$, $2 \text{Re}(e^{-i\pi/3}\varepsilon) - (2/\sqrt{3}) \text{Im} \varepsilon = (2/\sqrt{3}) \text{Re}(e^{-i\pi/6}\varepsilon)$. With $-\frac{3}{4}h^2 + h^2 = \frac{1}{4}h^2$, $h_j = \gamma a_j$ and (36), we obtain for every j

$$2|u_j|^2 F_j = (1-h_j)\Delta_j^2 + \sqrt{3}\hat{C}_j\gamma^2 + O(\gamma^2(\delta_* + \gamma)), \quad \sqrt{3}\hat{C}_j = \frac{a_j^2}{4} + \frac{2}{\sqrt{3}} \text{Re}(e^{-i\pi/6}(E_j + S_j)). \quad (38)$$

Step 4: the coefficients. For $\tau = e^{i\theta}$, $\theta \notin 2\pi\mathbb{Z}$, a direct computation gives

$$\begin{aligned} \frac{2}{\sqrt{3}} \operatorname{Re}(e^{-i\pi/6}(A_0(\tau) + e^{i\pi/3}B_0(\tau))) &= \frac{1}{2\sin^2(\theta/2)} + \frac{\cos(\theta/2)(1 + 2\sin^2(\theta/2))}{2\sqrt{3}\sin^3(\theta/2)}, \\ \frac{2}{\sqrt{3}} \operatorname{Re}(e^{-i\pi/6}\sqrt{3}e^{i\pi/6}\bar{\tau}) &= 2\cos\theta, \end{aligned}$$

and the sum of the two right sides is $\sqrt{3}\chi(\theta)$. The term $l = j$ of the second sum in E_j contributes $2a_j^2$, and $a_j^2/4 + 2a_j^2 = \sqrt{3} \cdot 3\sqrt{3}a_j^2/4$. Hence $\hat{C}_j = C_j$. The first two terms of χ are even and the last is odd, so in $\sum_j a_j^2 C_j$ the odd part cancels between (j, l) and (l, j) , and with $\sum_{j \neq l} a_j^2 a_l^2 \cos(\phi_l - \phi_j) = |\sum_j a_j^2 e^{i\phi_j}|^2 - \sum_j a_j^4$,

$$\sum_j a_j^2 C_j = \frac{\sqrt{3}}{12} \sum_j a_j^4 + \frac{2\sqrt{3}}{3} \left| \sum_j a_j^2 e^{i\phi_j} \right|^2 + \frac{\sqrt{3}}{6} \sum_{j \neq l} \frac{a_j^2 a_l^2}{\sin^2((\phi_j - \phi_l)/2)}. \quad (39)$$

Put $x_j = a_j^2$, $x_1 = \min_j x_j$ after relabeling, $\Sigma_1 = \sum_j x_j$ and $\Sigma_2 = \sum_j x_j^2 \leq \Sigma_1^2$. For $m = 1$, $C_* = C_1 = (3\sqrt{3}/4)x_1$. For $m \geq 2$, $C_* \geq \sum_j x_j C_j / \Sigma_1$, and (39) with $\sin^2 \leq 1$ gives $\sum_j x_j C_j \geq (\sqrt{3}/12)\Sigma_2 + (\sqrt{3}/6)(\Sigma_1^2 - \Sigma_2)$. For $m = 2$ this exceeds $(\sqrt{3}/4)x_1\Sigma_1$ by $(\sqrt{3}/12)(x_2 - x_1)(x_2 + 2x_1) \geq 0$; for $m \geq 3$ it is at least $(\sqrt{3}/12)\Sigma_1^2 \geq (\sqrt{3}/12)m x_1 \Sigma_1 \geq (\sqrt{3}/4)x_1\Sigma_1$. In both cases $C_* \geq (\sqrt{3}/4)\min_j a_j^2$.

Step 5: from the pairs to P . By Step 2, $\operatorname{Re} R_j \geq 3/2 - O(\delta_* + \gamma) \geq 1$ and $|\operatorname{Im} R_j| = \sqrt{3}/2 + O(\delta_* + \gamma)$ once $\delta_* + \gamma$ is at most a constant $\delta_2 > 0$ that depends only on m and c , which we assume from now on, decreasing δ_2 where needed. Then $\operatorname{Re} \Lambda = \operatorname{Re} R_j / |Z_j|^2 > 0$, and for every j

$$P - \frac{\sqrt{3}}{2} = \frac{\operatorname{Re} \Lambda + \sqrt{3} \operatorname{Im} \Lambda}{2|\operatorname{Im} \Lambda|} = \frac{\operatorname{Re}(e^{-i\pi/3}\Lambda)}{|\operatorname{Im} \Lambda|} = \frac{F_j}{|\operatorname{Im} R_j|} = \frac{2|u_j|^2 F_j}{D_j},$$

where $D_j = 2|u_j|^2 |\operatorname{Im} R_j| = \sqrt{3} + O(\delta_* + \gamma)$, so that $\sqrt{3}/2 \leq D_j \leq 2\sqrt{3}$. By (38) and $h_j < 1$, $2|u_j|^2 F_j \geq L_j = \sqrt{3} C_j \gamma^2 - O(\gamma^2(\delta_* + \gamma))$, and $|L_j| = O(\gamma^2)$ because $|C_j| \leq K$. Since $|L_j/D_j - L_j/\sqrt{3}| = |L_j||D_j - \sqrt{3}|/(\sqrt{3} D_j) = O(\gamma^2(\delta_* + \gamma))$, there is K_1 , depending only on m and c , such that

$$P - \frac{\sqrt{3}}{2} \geq (C_* - K_1(\delta_* + \gamma))\gamma^2. \quad (40)$$

Step 6: conclusion. Let $\delta_1 = \min(\delta_0, \delta_2/2, \sqrt{3}c^2/(16K_1))$, and let $\eta > 0$ and $\gamma_2 > 0$ be given by Theorem 3(b) for $\delta = \delta_1$; they depend only on m and c . Put $\eta_4 = \eta$ and

$$\gamma_4 = \min\left(\gamma_*, \gamma_2, \frac{\delta_2}{2}, \frac{\sqrt{3}c^2}{16K_1}, \left(\frac{8\eta}{\sqrt{3}c^2}\right)^{1/2}\right).$$

Let $\gamma < \gamma_4$. If $P \geq \sqrt{3}/2 + \eta$, then $P \geq \sqrt{3}/2 + (\sqrt{3}/8)c^2\gamma^2$ because $\gamma^2 < 8\eta/(\sqrt{3}c^2)$. Otherwise Theorem 3(b) gives $\delta_* < \delta_1$, so Steps 1–5 apply, the directions are separated as shown before Step 1, and (40) with $K_1(\delta_* + \gamma) < \sqrt{3}c^2/8$ and $C_* \geq (\sqrt{3}/4)c^2$ gives (30).

Now let also $P < \sqrt{3}/2 + \eta_4$, so that $\delta_* < \delta_1$. If $P - \sqrt{3}/2 \geq C_*\gamma^2$, (31) holds. Otherwise $0 < P - \sqrt{3}/2 < C_*\gamma^2 \leq K\gamma^2$, so $2|u_j|^2 F_j = D_j(P - \sqrt{3}/2) = O(\gamma^2)$, and (38) with $1 - h_j \geq 1/2$ gives $\Delta_j^2 = O(\gamma^2)$. With $\xi_j = O(\gamma)$ and $\nu_j - 1 = O(\gamma)$ from Step 3, $u_j - e^{i\pi/3} = O(\gamma)$; then $R_j = 1 + e^{-i\pi/3} + O(\gamma)$ by Step 2, and since $|Z_j|^2/|Z_l|^2 = |R_j|/|R_l|$, $|1 + e^{-i\pi/3}| = \sqrt{3}$ and $|\sqrt{x} - 1| \leq |x - 1|$ for $x > 0$, $||Z_j|/|Z_l| - 1| = O(\gamma)$. So $\delta_* \leq K'\gamma$ with a constant K' that depends only on m and c , and (40) gives (31) with $K_4 = K_1(1 + K')$. $\square$

Remark 6. The exponent 2 in (30) cannot be lowered, and (31) is attained up to $O(\gamma^3)$ by the rings of Proposition 3. As shown after the proof of Theorem 3, near their minima they lie in the class with $m = n \geq 2$, $\gamma = 1/\Gamma_0$, $a_j = x = 1 + O(\gamma)$ and the directions of a regular n -gon. Since $\sum_{l=1}^{n-1} \cos(2\pi l/n) = -1$, $\sum_{l=1}^{n-1} 1/\sin^2(\pi l/n) = (n^2 - 1)/3$ and the odd part of χ cancels between l and $n - l$, every C_j equals $x^2(3\sqrt{3}/4 - 2\sqrt{3}/3 + \sqrt{3}(n^2 - 1)/18) = x^2\sqrt{3}(1 + 2n^2)/36$, and Proposition 3 gives the least P of the family as $\sqrt{3}/2 + \sqrt{3}(1 + 2n^2)/(36(\Gamma_0 - 1/2)^2) + O(\Gamma_0^{-4})$, that is, $\sqrt{3}/2 + C_*\gamma^2 + O(\gamma^3)$. For $n = 2$ this is the case $C_1 = C_2 = (\sqrt{3}/4)x^2$ of equality in $C_* \geq (\sqrt{3}/4)\min_j a_j^2$. For $m = 1$ every collapse in the class is a three-vortex collapse with $\mu = \gamma a_1$, and $C_* = (3\sqrt{3}/4)a_1^2$ is also the coefficient in $P_-(\mu) = \sqrt{3}/2 + (3\sqrt{3}/4)\mu^2 + O(\mu^3)$: by Theorem 1, $P_-(\mu)^2$ is the root of $Q(\mu, \cdot)$ that tends to $3/4$, which is analytic at $\mu = 0$ because $Q(0, y) = -16(4y - 3)$ has a simple root there, and its Taylor coefficients follow from (10). For other arrangements we have checked numerically, but not proved, that for fixed a_j and ϕ_j the least value of P over the collapses is $\sqrt{3}/2 + C_*\gamma^2 + O(\gamma^3)$, attained with $y_J - \sqrt{3}/2 = O(\gamma^2)$ for a pair J with $C_J = C_*$ and with $y_j - \sqrt{3}/2 \approx \pm\gamma(\sqrt{3}(C_* - C_j))^{1/2}$ for the others. By (39) the weighted mean of the C_j is positive for every arrangement of distinct directions; the separation $|Z_j - Z_l| \geq c$ enters the proof only through uniform bounds (Theorem 3(b), the set $\mathcal{K}$, $|C_j| \leq K$), not through the sign of C_* .

7 More than three vortices

Self-similar collapse of any number of Euler vortices is described by the same equations $\dot{z}_j = \kappa(z_j - z_c)$ [11, 15, 25, 34], and by Section 2 so is self-similar collapse in the α -models, with P given by (4). Once translations, rotations and the scalings of lengths and of circulations are divided out, a self-similar collapse of N vortices is a solution of the $2N$ real equations $\dot{z}_j = \kappa(z_j - z_c)$ in $3N - 3$ unknowns. Two of them follow from the others, since $\sum_j \Gamma_j(\dot{z}_j - \kappa(z_j - z_c)) = 0$ identically: $\sum_j \Gamma_j \dot{z}_j = 0$ because the interactions cancel in pairs, and $\sum_j \Gamma_j(z_j - z_c) = 0$ by the definition of z_c . Where the Jacobian of the remaining $2N - 2$ equations has full rank, the collapses form a manifold of dimension $N - 1$, the collapse set; for $N = 3$ its coordinates are μ and θ of Section 3. The bound of Corollary 1 does not extend to more vortices: the seven-vortex collapse of Section 3 has $P = 0.8053\dots$. We show that already four vortices go below $\sqrt{3}/2$, that P has strict local minima on the collapse sets of four, five and six vortices, and that in the α -models four vortices go below the three-vortex bounds of Theorem 2, eleven can collapse without rotating at $\alpha = 2$, and sixty can do so in the SQG model. The next two theorems are computer-assisted: their proofs, described in Section 8, are carried out in interval arithmetic by the program `certify_collapses.py`, and, for part (b) of Theorem 5, by `certify_sqg60.py`.

Theorem 4. (a) *For $N = 4, 5$ and 6 there is a self-similar collapse of N point vortices under (2) with $P < \sqrt{3}/2$, in which all circulations and their sum are nonzero, no two vortices coincide and the angular impulse vanishes.*

(b) *For $N = 4, 5$ and 6 there is such a collapse at which P has a strict local minimum P_N on the collapse set of N vortices, a manifold of dimension $N - 1$ near it, with*

$$P_4 = 0.79789678387986348076760587182\dots,$$

$$P_5 = 0.74481445696249246102875218111\dots,$$

$$P_6 = 0.71368014852939247497732477960\dots$$

(c) *Under (1) with $\alpha = 1$ and with $\alpha = 2$ there is a self-similar collapse of four vortices at which P has a strict local minimum on the collapse set, with the values $0.5499151189\dots < B(1) = 2/3$ and $0.4116738600\dots < B(2) = \sqrt{5}/4$.*

The program encloses each of these minima in an interval of width 10^{-50} . In the normalization $\Gamma_1 = 1$, $z_c = 0$, $\kappa = -1 + ib$ the four-vortex minimizer of part (b) has $b = 2P_4 = 1.5957935677597269615\dots$ and

$$\begin{aligned}\Gamma &= (1, 0.1673914318538, 0.1041331880362, -0.2272513003829), \\ z &= (0.0166495669172, -0.2191226402232 - 0.3290676220068i, \\ &\quad -0.1579874300122 - 0.2864759934272i, -0.1605333843048 - 0.3736601672808i),\end{aligned}$$

and the Hessian of the Lagrangian of the minimization of P , restricted to the three-dimensional tangent space of the constraints and written in an orthonormal basis of that space for the Euclidean metric of the unknowns $(x_1, x_2, y_2, x_3, y_3, x_4, y_4, \Gamma_2, \Gamma_3, \Gamma_4, b)$, where $z_j = x_j + iy_j$ and $y_1 = 0$, has the eigenvalues 1.106, 6.289 and 9.377; the eigenvalues depend on these coordinates, their signs do not. Like the minimizers of part (c), it has one strong and three weak vortices. The minima are local: we have not proved that P_4 , P_5 and P_6 are the least values of P over all collapses of four, five and six vortices. For four vortices a numerical minimization from many random starts ends at the configuration above, up to relabeling, and for five and six vortices P_5 and P_6 are the least values found by multistart searches.

Theorem 5. (a) *Under (1) with $\alpha = 2$ there is a self-similar collapse of eleven point vortices with κ real, so that $P = 0$: every vortex moves on a straight line into the collision point, and $|z_j(t) - z_c|^4 = (1 - t/t_c) |z_j(0) - z_c|^4$. In it all circulations and their sum are nonzero, no two vortices coincide and the angular impulse vanishes. Near it, the self-similar collapses of eleven vortices with κ real form, modulo translations, rotations and the scalings of lengths and of circulations, a manifold of dimension nine.*

(b) *Under (1) with $\alpha = 1$ (SQG) there is a self-similar collapse of sixty point vortices with κ real, so that $P = 0$: every vortex moves on a straight line into the collision point, and $|z_j(t) - z_c|^3 = (1 - t/t_c) |z_j(0) - z_c|^3$. In it all circulations and their sum are nonzero, no two vortices coincide and the angular impulse vanishes. Near it, the self-similar collapses of sixty vortices with κ real form, modulo the same symmetries, a manifold of dimension fifty-eight.*

In the normalization of the program, $2\pi\kappa = -1$, so that $t_c = \pi/2$ for part (a), the collision point is the origin, one circulation is 1 and nine others are fixed rational numbers. The configuration has one strong vortex of circulation 27.6..., the one closest to the collision point, one of circulation $-8.94\dots$ and nine positive vortices with circulations between 0.37 and 2.3; the total circulation is 27.74..., the distances from the collision point lie between 0.21 and 5.08, and no two vortices are closer than 0.3487.

For part (b), a search that adds one vortex at a time to a family of minimizers reached this self-similar collapse of sixty SQG vortices without rotation; Newton's method with κ held real refines it at 60 digits to a residual of 10^{-58} , and in it the angular impulse and $\sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^{-1}$, which every self-similar collapse makes vanish, vanish to the same precision; its circulations span a factor of 83, and its closest pair is 0.0094 times its size. In the normalization of the certifying program, $t_c = 2\pi/3$, the collision point is the origin, one circulation is 1, fifty-eight others are fixed rational numbers and one, Γ_7 , is the certified enclosure 1.0484485069.... The configuration has one strong vortex of circulation 41.00..., the one closest to the collision point, one of circulation $-37.44\dots$ and fifty-eight positive vortices with circulations between 0.496 and 1.717; the total circulation is 43.48..., the distances from the collision point lie between 0.656 and 9.499, and no two vortices are closer than 0.0893.

By Theorem 2 three vortices cannot collapse without rotation in any α -model with $\alpha > -2$, and no Euler collapse without rotation is known for any number of vortices (Section 10).

As controls, the same pipeline certifies the three-vortex minimum of Proposition 1, $P_-(1/2) = 1.06470597627120433735498624890810207179\dots$, which it encloses within 10^{-94} of the closed form, and the minimum $P_-(1/20) = 0.869112874060804665568\dots$; numerical searches with the same equations find no three-vortex collapse with $P < \sqrt{3}/2$ and none for $\alpha = 2$ with κ real, as Corollary 1 and Theorem 2 require, while the same search with $P = 11/10$ finds one, which is then certified.

The following results are numerical, not proofs. Multistart searches for $N = 7, \dots, 12$ Euler vortices found no value of P below $0.6936617908\dots$, $0.6797704208\dots$, $0.6592598630\dots$, $0.6377055919\dots$, $0.6179510627\dots$ and $0.6044746933\dots$ respectively. Up to $N = 8$ these minimizers have one strong vortex, one negative vortex and a chain of weak positive vortices; from $N = 9$ on a two-arm family is lower, with a vortex at the center of vorticity, two negative vortices and two arms of weak positive vortices. Continued in N with no symmetry imposed, the values of this family decrease: $P = 0.5172291322\dots$ for 33 vortices, $0.4981844392\dots$ for 61 and $0.4793959201\dots$ for 603, so a self-similar collapse can turn by less than half a radian per unit decrease of $\ln \lambda^2$. For $N \geq 301$ the values fit

$$P = 0.477364 + 1.2206/N + 2.93/N^2$$

with a root-mean-square residual of $7 \cdot 10^{-10}$, and cubic Richardson extrapolation in $1/N$ through $N = 101, 203, 403, 603$ gives 0.4773635 . Whether a different structure takes over at much larger N , as the two-arm family took over at $N = 9$, is open.

8 Numerical verification and computer-assisted proofs

The results were checked numerically as summarized in Table 2. For the Euler sections, except where the table says otherwise, the checks evaluate the Biot–Savart velocities (2) directly for configurations built independently of (7) (two vortices fixed and the third moved around the zero-impulse circle), test self-similarity by comparing $\dot{z}_j/(z_j - z_c)$ across all vortices, and obtain P from (4). The algebraic steps in the proofs were verified in exact arithmetic (SymPy): Lemma 3 for all three vortices, (11), the resultant in the proof of Theorem 1(b), the discriminant and the resultant in the proof of Theorem 1(c), the irreducibility of Q and of each of the 277 sextics in Remark 1, the constants in Proposition 1 and its closed forms c_m , the forms of P , the equality condition, the angles of the minimizing triangle and the reduction to Gröbli’s coefficient in Remark 2, Gröbli’s closed form of P for every triple of circulations that can collapse, after Lemma 3, the triple of Chen, Walsh and Wheeler as a member of the family with $\mu = 1/2$, and, for general n , the identities of Section 5 from (17) to (18), together with the table of constants. The program `verify_general_mu.py` carries out these exact computations with SymPy 1.14.0 and the high-precision checks with mpmath 1.3.0; `verify_floors_independent.py` repeats the minimizations independently and contains the exact sign changes of G and the interval enclosures in the proof of Proposition 1, computed with the interval arithmetic of mpmath at 30 digits. The program `verify_direct_proof.py` checks every identity in the direct proof of Corollary 1 exactly, with SymPy. The program `verify_central_vortex.py` checks Proposition 3: exactly, with SymPy, the derivation of (19) and (20) for symbolic n , every identity in the proof, the case $\Gamma_0 = 0$ and the expansion, the two examples with $n = 2$ after its proof, the quartet of Chen, Walsh and Wheeler as a member of the family of Proposition 2, and the expansion of F_n as $n \rightarrow \infty$, and also the leading-order relations used in Step 5 of the proof of Theorem 3, with the limits of u_j and P along the two families; numerically, the Biot–Savart velocities and the minima over θ at 50 and 30 digits, the configuration of Gotoda’s Fig. 3(b), that quartet and F_n up to $n = 10^{12}$ at 50 digits; and negative controls, checks that must fail and do. The program `verify_strong_vortex.py` makes 143 checks of Theorem 3: exactly, with SymPy, every identity its proof uses; the explicit constants of Steps 1–3 ($\gamma_0 = c^3/(8m)$, $|z_c| \leq 4m\gamma/c^2$,

$|\kappa| \leq 3/(\pi c^4)$, the bound on $|\rho_j|$ and $|\nu_j| \leq 8c^{-10}$, exactly where they are algebra and on 960 random configurations in the class (21) where they are bounds; self-similar collapses computed at 50 digits for $m = 1$ and $m = 3$ weak pairs, symmetric and asymmetric, at $\gamma = 10^{-2}$, 10^{-3} and 10^{-4} , and for $m = 2$ antipodal pairs at $\gamma = 10^{-4}$, each accepted only after a Biot–Savart residual below 10^{-40} , along which P stays above $\sqrt{3}/2$ and approaches it; that the two families lie in the class, with $c = 1/2$ for three vortices and $c = \min(1/2, 2\sin(\pi/n))$ for the rings; and negative controls: the seven-vortex collapse of Section 3 lies outside the class, unconverged and perturbed points are rejected, $y_j \leq 0$ does not collapse, and without the angular impulse the case $\kappa^* = 0$ is not excluded. The program `verify_pairs_bound.py` makes 69 checks of Proposition 4 and Remark 6: exactly, with SymPy, every identity of the proof, among them (33), (34), (35), the exact forms of the sources in Step 2, (37), the coefficients of Step 4 and (39), the constants of the set $\mathcal{K}$, and the coefficients of Remark 6 for the rings and for three vortices; numerically, the remainders $O(\gamma^2(\delta_* + \gamma))$ of Steps 2 and 3 on 1500 random configurations of the class and 3000 random solutions of (35), with $\delta_* + \gamma$ down to 10^{-7} , where they do not grow as $\delta_* + \gamma \rightarrow 0$, while in two negative controls, each with a wrong leading term, they grow like $1/(\delta_* + \gamma)$; and 90 self-similar collapses computed at 50 digits, $m = 1$ to 4, $\gamma = 10^{-2}$, 10^{-3} and 10^{-4} , each accepted only after a Biot–Savart residual below 10^{-40} , which satisfy (33), (34) and (37) to 10^{-45} and have $P \geq \sqrt{3}/2 + (\sqrt{3}/8)c^2\gamma^2$. At these fixed values of γ , which may exceed the threshold γ_4 , $P > \sqrt{3}/2$ is a numerical observation.

For Section 4, the program `verify_alpha_winding.py` checks each step in high-precision arithmetic. On 700 random scalene triangles with $\alpha \in \{0, 0.3, 1, 1.7, 3\}$, the circulations (12) make the velocities (1) self-similar to $2 \cdot 10^{-38}$, and P computed from the velocities agrees with (13) to $3 \cdot 10^{-38}$. The derivative of g and the identity (15) are verified symbolically, and the critical point and value of P_0 as well. The chain of inequalities in the proof holds at 20000 random points with β from just above $1/2$ to 20 and ρ down to 10^{-8} , computed with precision adapted to $\rho^{2\beta}$; in double precision the first step appears to fail at small ρ and large β , where its margin, of order $\rho^{2\beta}$, is below the rounding error. For α from -1.8 to -1 the minimum of $P - B(\alpha)$ over the ordered triangles is positive in every run, as Remark 4 requires, again only with precision adapted to ρ . For $\alpha = 1$, the scalene triangles on which (12) gives two equal circulations carry the ratios $-\Gamma_2/\Gamma_1$ from 0.3876 to 0.49999 on a grid, in agreement with the interval of [5, Lemma 1], and at $\Gamma = 0.49$, solved exactly at 30 digits, the ratio of sides is 0.7514840918..., which agrees with the value 0.751484 of [5] in every digit given there. Near-extremal triangles at $\rho = 10^{-6}$ and $m = m_*$ collapse, and $P - B(\alpha)$ is $1.3 \cdot 10^{-12}$, $1.8 \cdot 10^{-13}$, $1.6 \cdot 10^{-13}$ and $1.6 \cdot 10^{-13}$ for $\alpha = 0, 1/2, 1, 2$. The program `verify_alpha_equal_circulations.py` checks Remark 5, exactly with SymPy and against the velocities at 40 digits.

The program `verify_alpha_below.py` checks Remark 4 and the sharpness argument for $0 < \beta \leq 1/2$. With SymPy it verifies exactly every identity they use, the roots $6 \pm 2\sqrt{6}$ and the maximum of $\rho \ln(1/\rho)$; in the ball arithmetic of FLINT/Arb it confirms $(6 - 2\sqrt{6})e = 2.9928... > 2\sqrt{2}$ and $((15 - 6\sqrt{6})e^2 - 1)/2 > 1/2$; and, independently of the algebra, an Arb bisection of $0 \leq \beta \leq 1/2$, $0 \leq \rho \leq 1$, $0 \leq m \leq 3$, which visits 733 boxes and ends in 336 leaves that meet the domain $m < 2 + \rho$, shows that every such leaf has $q > 0$ or $m/\ln(1/\rho) > 2\sqrt{1+2\beta}$. On 1500 random collapsing triangles with $-2 < \alpha < -1$ and ρ down to 10^{-300} , the velocities (1) at 50 to 950 digits are self-similar to $2 \cdot 10^{-49}$, P agrees with (13) to $8 \cdot 10^{-47}$, and P exceeds the lower bound of its case, which exceeds $B(\alpha)$. The case $q \leq 0$ is far from the bound: on a grid of 2732 such collapses $P > 4.2 B(\alpha)$. Near-extremal collapses at $\alpha = -1.5, -1.8$ and -1.99 , with ρ down to 10^{-4800} , have $P - B(\alpha) > 0$, decreasing to 0 like $\rho^{2\beta}$. Six negative controls fail, as they must: among them steps (i) to (iii) alone at $\alpha = -1.7$, $\rho = 0.99$, $m = 1.87$, the bound of the case $q \leq 0$ near the extremal limit, and the subdivision with its threshold raised by ten per cent.

The program `certify_collapses.py`, with its modules `certify_*.py`, proves Theorem 4 and

part (a) of Theorem 5 in the ball arithmetic of Arb [24], through python-flint 0.9.0, at 320 bits. The equations are those of a self-similar collapse with the collision point at the origin, $2\pi v_j = 2\pi\kappa z_j$ for the velocities v_j , with $2\pi\kappa = -1 + 2iP$, and their first and second derivatives are computed by automatic differentiation in ball arithmetic, so that the Jacobian F' of the system $F = 0$ is enclosed over the whole of a box X by a set $F'(X)$ of matrices. The proofs rest on the theorem of Krawczyk and Moore [27, 31], in the form [38, Theorem 13.3]: if $\tilde{x} \in X$, Y is a real matrix and the Krawczyk operator $K(X) = \tilde{x} - YF(\tilde{x}) + (I - YF'(X))(X - \tilde{x})$ lies in the interior of X , then Y and every matrix in $F'(X)$, among them the Jacobian at every point of X , are nonsingular, and F has exactly one zero in X . The program takes $\tilde{x}$ from a Newton iteration and Y from a floating-point inverse; floating point chooses only these, the charts and the brackets of the eigenvalue enclosures, which can make a certification fail but not succeed wrongly. The rational values at which coordinates or circulations are fixed, and 2π , enter as balls, so each conclusion holds for their exact values. What must be trusted is the correctness of Arb, of python-flint and of the functions of `certify*.py` that the docstring of `certify_collapses.py` lists.

For Theorem 4(a) the unknowns are P and $2N - 1$ of the coordinates and circulations, the other $N - 1$ being fixed at rational values, and on the box the program proves that every circulation and their sum are nonzero, that the vortices are distinct and bounded away from each other, and that $P < \sqrt{3}/2$. The angular impulse vanishes by Section 2, as in every self-similar collapse; the program checks, as a consistency test, that its enclosure contains 0. For Theorem 4(b) and (c) the Krawczyk operator is applied to the Lagrange system of the minimization of P on the collapse set. By the same theorem the Jacobian block of the $2N$ coordinates that are not chart coordinates is invertible on the box, so there the collapse set is the graph of a smooth function of the $N - 1$ chart coordinates s , and P on it is a smooth function $p(s)$. At the Lagrange point $\nabla p = 0$, and differentiating the constraints twice shows that the Hessian of p is $Z^T(\nabla^2 L)Z$, where L is the Lagrangian and the columns of Z span the tangent space of the constraints. The program encloses this matrix over a box that contains the Lagrange point and proves that all its leading principal minors are positive; by Sylvester's criterion the Hessian of p is positive definite at that point, so p , and with it P on the collapse set, has a strict local minimum there. The program also encloses the eigenvalues quoted after Theorem 4, by inertia counts, checks that the objective has the sign that makes the certified point a minimum of P and not a maximum, and checks that the four-vortex configurations of part (c) have one vortex whose circulation exceeds four times the absolute value of each of the others. For Theorem 5(a), $2\pi\kappa = -1$ is imposed exactly and nine circulations are fixed at rational values; the Krawczyk operator proves a unique solution of the 22 real equations in a box of radius 10^{-6} , where, by the same theorem, their Jacobian is invertible, which gives the nine-dimensional family by the implicit function theorem. The rational values of the fixed chart coordinates and circulations, and the enclosures of the certified points, are listed in the output of the program, `data/certify_collapses-2026-09-25.txt`. The program also carries out the controls of Section 7, tests its ball arithmetic derivatives against mpmath, and re-checks every certified point with separate Biot-Savart code at 80 digits. It makes 94 checks in one to two minutes.

The program `certify_sqg60.py` proves Theorem 5(b) the same way, reusing the modules `certify_ball_ad.py`, `certify_alpha_model.py`, `certify_pipeline.py` and `certify_crosschecks.py` unmodified. It refines the stored configuration at 100 digits (mpmath, non-rigorous), fixes fifty-eight circulations at 12-decimal rationals and, for the remaining 120 real unknowns (119 position coordinates and Γ_7), the Krawczyk operator proves a unique solution of the collapse equations $2\pi v_j = -z_j$ ($2\pi\kappa = -1$ imposed exactly) in a box of radius 10^{-9} ; on the box every Γ_j and their sum are nonzero, the vortices are pairwise distinct, and the enclosures of the angular impulse and of $\sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^{-1}$, which vanish in every self-similar collapse, contain 0; the 120×120 Jacobian, invertible on the box by the same theorem, gives the fifty-eight-dimensional

family by the implicit function theorem. The fixed circulations and the certified enclosures are listed in `data/sqg60-certificate.json`. The program re-checks the certified point with separate Biot–Savart code at 80 digits and runs four negative controls, among them a point shifted by 10^{-10} in every unknown, which fails the test in a box of radius 10^{-12} that does not contain the zero, and the mirror image, which expands and fails the collapse equations; as a positive control, the same shifted point passes in the box of radius 10^{-9} , which contains the zero. It makes 11 checks in well under a minute.

The program `verify_many_vortices.py` checks the numerical results of Section 7 from the stored configurations: the minimizers for $N = 7, \dots, 12, 33$ and 61 Euler vortices, stored to 60 digits, satisfy the self-similarity equations to $4 \cdot 10^{-47}$ relative, and the 603-vortex one, stored in double precision, to 10^{-13} , with $\sum_{j < k} \Gamma_j \Gamma_k$ and the angular impulse vanishing to the same precision; the program reproduces the fit and the extrapolation of the two-arm family from its stored values, and it refines the SQG collapse of sixty vortices with κ held real, with the residuals of Section 7. Each program prints every check with its result and stops with an error if one fails; under Python 3.11 each runs in about a minute or less, and their output is in `data/`.

Table 2: Numerical checks and computer-assisted proofs of the results.

Check	Result
Lemma 3 against the Biot–Savart velocities of all three vortices, at the positions (7) for 36 pairs (μ, θ) on a grid with $0.1 \leq \mu \leq 7$, and for 30 triangles built independently of (7), 50 digits	relative difference $\leq 2 \times 10^{-49}$
Arc minima by direct minimization of P , computed from the quotient of the first vortex, for 26 values of $\mu \in (0, 1]$ at 30 digits and 13 rational values at 50 digits	agree with the roots of $Q(\mu, \cdot)$ to 5×10^{-21} and 5×10^{-45}
$\mu = 1/2$: the three quotients over 4000 shapes on the zero-impulse circle, 60 digits	equal to 60-digit working precision; the collapsing shapes fill half of the circle, the arcs $\mathcal{A}_\pm$
$\mu = 1/2$: local minima of P over the collapsing shapes, located numerically	two, equal to the values of Proposition 1 to 60-digit working precision
$\mu = 1/2$: formula for $P(\theta)$ against direct evaluation at the positions (7), 399 interior points of each arc	relative difference $\leq 10^{-59}$
$\mu = 1/2$: integration of (2) from the minimizing configuration on $\mathcal{A}_-$ (Taylor method, 30 digits)	every $ z_j - z_c ^2$ follows $1 - t/t_c$, and every rotation angle follows $-\omega_0 t_c \ln(1 - t/t_c)$, to 30-digit working precision up to $t = 0.9 t_c$
$\mu = 1/2$: the closed forms c_0, c_1, c_2 of Proposition 1, the reduction of $G = 0$ to the depressed cubic (exact, SymPy), and P at $\cos \theta = c_0$ and c_1 , 60 digits	residual of the cubic $\leq 10^{-59}$; $c_0 \in (0.67, 0.68)$, $c_1 \in (-0.93, -0.92)$, $c_2 < -1$; $P = P_+$ and P_- there to 10^{-60}
Five configurations at the positions (7) with $\mu \in \{0.3, 0.5, 2\}$, on both arcs: integration of (2) to $t = 0.9 t_c$ (fourth-order Runge–Kutta, 30 digits)	distances to the collision point, rotation angles, the angle $\arctan 2P$ between each velocity and the direction to the collision point, and path lengths $r_0 \sqrt{1 + 4P^2} (1 - \sqrt{1 - t/t_c})$ agree to 2×10^{-10}
$\mu = 1$ (Remark 2): P from (2) against $(3 - \cos 2\chi)/(2 \sin 2\chi)$ on both collapsing arcs, and its minimum, 50 digits	difference $\leq 2 \times 10^{-47}$; minimum $\sqrt{2}$ at $\cos 2\chi = 1/3$ on both arcs

Check	Result
$\mu = 1$: the configuration with the fastest collapse at $ z_1 - z_2 = 1$, 50 digits	$\cos 2\chi = 3/5$, $t_c = 4\pi/3$ and $P = 3/2$ to 10^{-40}
$\mu = 1$: Kimura's rates A , B [25, Eq. (4.4)] at ten angles in $0 < \chi < \pi/2$, 50 digits	$\kappa = (A + iB)/(4\pi)$ for $\Gamma = (1, 1, -1/2)$, to 10^{-45}
$\mu = 1$ (Remark 2): $P = v + 1/(2v)$, $P - \sqrt{2} = (\sqrt{2}v - 1)^2/(2v)$, the invariance under $v \mapsto 1/(2v)$ and the angles of the minimizing triangle (exact, SymPy); P from (2) at five angles and the angles of the minimizers located numerically on both arcs, 50 digits	identities exact; difference $\leq 3 \times 10^{-50}$; angles $(\pi/8, 5\pi/8, \pi/4)$ and $(5\pi/8, \pi/8, \pi/4)$ at (z_1, z_2, z_3) to 2×10^{-27}
Gröbli's Eqs. (8), (9), (11) and (12) of § 10: at $(m_1, m_2, m_3) = (1, 1, -1/2)$, symbolic in a_G , with the substitution $a_G = \sqrt{3}/\cos \chi$ and the ratios of the squared sides, and the translation's denominator $\mu_1\mu_3\mu_3$ run through the same substitution as a negative control; for every triple with $1/m_1 + 1/m_2 + 1/m_3 = 0$, symbolic in m_1 , m_2 , a_G and λ , against Lemma 6 at $\beta = 1$ with $r_i^2 = \lambda\mu_i^G$ and Heron's formula factored in the side lengths, and under a transposition of two circulations and a change of the signs of all three (exact, SymPy); against Biot–Savart for $\mu \in \{0.1, 0.3, 0.5, 0.8, 1, 2\}$, two angles on each arc, on 188 triangles of 47 triples in random order and sign, among them $(3, -2, 6)$ and $(1, 2, -2/3)$, with a_G random on both branches and $16A^2$ from the coordinates, and on his example of Fig. 6, 50 digits; $-K$ in place of K as a negative control	$P = (2a_G^2 - 3)/(2\sqrt{3a_G^2 - 9})$, which becomes $(3 - \cos 2\chi)/(2 \sin 2\chi)$; $P^2 - 2 = (2a_G^2 - 9)^2/(12(a_G^2 - 3))$; the misprinted denominator does not give $v + 1/(2v)$; S is $-\lambda$ times the numerator of $\varkappa$ and $16A^2$ is λ^2 times the radicand; a transposition exchanges the two branches and keeps $ \varkappa $; $ \varkappa /2 = P$ to 2×10^{-48} , relative to 3×10^{-48} on the random triangles; with $-K$, P is missed in each of the 184 with $ K > 0.01$; Fig. 6: $\varkappa = 5/\sqrt{3}$, $P = 5\sqrt{3}/6$
Two rings, $n = 2, \dots, 10$: the reduced equations (16) and the two sums over roots of unity against the full Biot–Savart sum, 50 digits	difference $\leq 10^{-49}$
Two rings, $n = 2, \dots, 8$: P from (2) against (18) over the collapsing range, and its numerical minimum against F_n , 60 digits	relative difference $\leq 3 \times 10^{-58}$; minima agree to 2×10^{-60}
Two rings: the equations of Demina and Kudryashov [11, Eqs. (36)–(37)] at $\Gamma_0 = 0$ (exact, SymPy), and their constant against the Biot–Savart sum, $n = 2, \dots, 8$, 50 digits	(37) is (17) with $x = r^2$; (36) minus S is a multiple of (17); difference $\leq 10^{-45}$
The collapsing configuration of seven vortices in [11, Table 1, Fig. 1a], Biot–Savart at 50 digits	self-similar, the printed Ω to 10^{-45} , and $P = 12433/(1240\sqrt{155}) < \sqrt{3}/2$
Two rings with a central vortex, $n \in \{2, 3, 4, 5, 7\}$, $-1999.5 \leq \Gamma_0 \leq 250$, each positive root of (19): Biot–Savart for all $2n + 1$ vortices, 295 configurations, 50 digits; minima over θ , 13 cases, 30 digits	quotients equal to κ to 3×10^{-48} , P to (20) to 6×10^{-41} , [11, Eq. (36)] holds to 3×10^{-48} ; minima $\sqrt{D^2 - n^2}/(2n)$ to 4×10^{-27}
Two rings with a central vortex, $n = 2$: $\Gamma_0 = 3/2$ and $\Gamma_0 = 1/2$ (exact, SymPy); the configuration of Gotoda's Fig. 3(b) at five angles, and κ for $\Gamma_0 = 1/2$ against the three vortices of Remark 2 at six angles, two of them expanding, Biot–Savart at 50 digits	$(a, b, D) = (21/4, 3, 19/4)$ and $(6, 2, 6)$; self-similar and collapsing, P to the formula to 2×10^{-50} , minimum $3\sqrt{33}/16$; the two κ agree to 5×10^{-51}

Check	Result
The triple and the quartet of Chen, Walsh and Wheeler [8, Eqs. (4.5), (4.6)]: the triple against (7) with $\mu = 1/2$ at $\theta = \pi/2$, its center of vorticity and κ , and $\Omega - i/(2\kappa)$ formed from their printed Ω and collapse time κ , and the quartet, turned by $\pi/6$, against Proposition 2 with $n = 2$ at $\theta = \pi/12$ (exact, SymPy); Biot–Savart for both at 50 digits; their map V [8, Eq. (4.4)] at the triple shifted to its center of vorticity, with that Ω , at the quartet with its printed Ω , and along $\mathcal{A}_+$ for $\mu = 1/2$, and its behaviour under rotations, dilations, a factor in the circulations and relabeling; the triple as printed, the printed real Ω alone, and the quartet with its third position as printed in arXiv:2506.04093v1 and in the journal version as negative controls; the rank of $D_\Lambda V$ at 199 points of $\mathcal{A}_+$ for $\mu = 1/2$ and of $0 < \theta < \pi/2$ for $n = 2$, numerical, 30 digits	shape ratio $1/3 - (\sqrt{7}/3)i$ and $P = 5\sqrt{7}/2$; $z_c = -2i/\sqrt{7}$, $\kappa = (-\sqrt{7} + 35i)/(264\pi)$ and $\Omega - i/(2\kappa) = (35 - \sqrt{7}i)/(264\pi) = i\bar{\kappa}$; circulations $(4 - \sqrt{3})\pi(x_2, -1)$ and $P = 2\sqrt{3} - 3/4$; self-similar to 10^{-50} , the similarity to (7) to 2×10^{-51} ; $V = 0$ to 10^{-50} with $\Omega = i\bar{\kappa}$, and $V \circ L = A \circ V$ to 10^{-50} ; unshifted, $\min_\Omega V = 0.049$, and with the real Ω alone $\max_k V_k = 0.011$; the printed position gives $z_c \neq 0$ and no self-similar motion; the $2N$ -th singular value is at least 7.9×10^{-4} and 2.9×10^{-4} times the largest
F_n as $n \rightarrow \infty$: the series of $E_n = (n+2) \operatorname{arsinh}((2(n-1))^{-1/2})$ and of $((n-1)/n) e^{E_n - \sqrt{n/2}}$ in $n^{-1/2}$ (SymPy), and F_n at 50 digits for $n = 10^k$, $k = 1, \dots, 6$, and $n = 10^8, 10^{12}$; the coefficient 30 in place of 29 as a negative control	$E_n = \sqrt{n/2} + 29/(12\sqrt{2n}) + O(n^{-3/2})$, coefficients $29/(12\sqrt{2})$ and $265/576$; $n(4e^{-\sqrt{n/2}}F_n - 1 - 29/(12\sqrt{2n}))$ increases from 0.374 to 0.45997 for $k = 1, \dots, 6$ and to 0.4600693 at $n = 10^{12}$, toward $265/576 = 0.4600694\dots$, and $n^{3/2}$ times the remainder of the expansion lies in $[-0.28, -0.07]$; with 30 it grows like $\sqrt{n}$
Theorem 3: self-similar collapses computed at 50 digits, $m = 1$ and $m = 3$ weak pairs at $\gamma = 10^{-2}, 10^{-3}, 10^{-4}$ and $m = 2$ at $\gamma = 10^{-4}$, each accepted after a Biot–Savart residual below 10^{-40}	P stays above $\sqrt{3}/2$ and approaches it
Proposition 4 (<code>verify_pairs_bound.py</code>): its identities (exact, SymPy); the remainders of Steps 2 and 3 on 1500 random configurations and 3000 random pair solutions; 90 collapses computed at 50 digits, $m = 1$ to 4, $\gamma = 10^{-2}, 10^{-3}, 10^{-4}$, Biot–Savart residual below 10^{-40}	remainders bounded by $20\gamma^2(\delta_* + \gamma)$ on the collapses; $P \geq \sqrt{3}/2 + (\sqrt{3}/8)c^2\gamma^2$ on all 90
Lemmas 5 and 6: the velocities (1) of 700 random scalene triangles with $\alpha \in \{0, 0.3, 1, 1.7, 3\}$ and the circulations (12), 40 digits	self-similar to 2×10^{-38} ; $P = S /(8A)$ to 3×10^{-38}
Proof of Theorem 2: the chain of inequalities at 20000 random points, β from just above $1/2$ to 20, ρ down to 10^{-8} , precision adapted to $\rho^{2\beta}$	no violation; least $(P - B(\alpha))/B(\alpha)$ is 9.3×10^{-9}
Sharpness in Theorem 2: triangles with $\rho = 10^{-6}$ and $m = m_*$ for $\alpha = 0, 1/2, 1, 2$, velocities (1) at 60 digits	collapses; $P - B(\alpha) = 1.3 \times 10^{-12}$, 1.8×10^{-13} , 1.6×10^{-13} , 1.6×10^{-13}
Theorem 2 for $-1.8 \leq \alpha \leq -1$: minimization of $P - B(\alpha)$ with precision growing with $ \ln \rho $	the minimum is positive at $\alpha = -1, -1.2, -1.5, -1.8$
Remark 4 (<code>verify_alpha_below.py</code>): its identities (exact, SymPy); its constants and a subdivision of $0 \leq \beta \leq 1/2$, $0 \leq \rho \leq 1$, $0 \leq m \leq 3$ in Arb ball arithmetic; Biot–Savart at 50 to 950 digits for 1500 random collapsing triangles with $-2 < \alpha < -1$, and near-extremal collapses at $\alpha = -1.5, -1.8, -1.99$	a bisection of 733 boxes, all 336 leaves resolved; P above the bound of its case, and so above $B(\alpha)$, in every triangle; $P - B(\alpha) \rightarrow 0^+$

Check	Result
SQG, circulations $(1, -\gamma, 1)$: the collapsing triangles of Lemma 5 on a grid	γ from 0.3876 to 0.49999, as in [5, Lemma 1]; side ratio 0.7514840918... at $\gamma = 0.49$, solved exactly
Remark 5: the $\alpha = 2$ family and its minimum (exact, SymPy), the formula against the velocities at five values of γ , 40 digits, and the SQG endpoint (exact)	the octic and the quartic; difference $\leq 4 \times 10^{-40}$
Theorem 4: Krawczyk operator in ball arithmetic at 320 bits, on the collapse equations and on the Lagrange systems	unique solutions in boxes of radius 10^{-4} to 10^{-8} ; the minima enclosed to width 10^{-50} ; reduced Hessians positive definite
Theorem 5(a): Krawczyk operator with $2\pi\kappa = -1$ and nine circulations fixed	unique solution in a box of radius 10^{-6} ; Jacobian invertible
Theorem 5(b): Krawczyk operator (<code>certify_sqg60.py</code>) with $2\pi\kappa = -1$ and fifty-eight circulations fixed	unique solution in a box of radius 10^{-9} ; Jacobian invertible; independent recheck at 80 digits
Controls: Proposition 1 and $\mu = 1/20$ by the same pipeline; searches for three-vortex collapses with $P < \sqrt{3}/2$, or with $\alpha = 2$ and κ real	$P_-(1/2)$ within 10^{-94} of the closed form; no such three-vortex collapse found
Every point certified by <code>certify_collapses.py</code> , with separate Biot–Savart code at 80 digits	self-similar to 4×10^{-79} ; P agrees with the enclosures
Numerical minimizers for $N = 7, \dots, 12, 33, 61$ (60 digits) and 603 (double precision) Euler vortices, Biot–Savart at 65 and 30 digits	self-similar to 4×10^{-47} and 10^{-13} relative; P as stated
The two-arm family, $N = 9, \dots, 603$: fit over $N \geq 301$ and extrapolation in $1/N$	$0.477364 + 1.2206/N + 2.93/N^2$, rms 7×10^{-10} ; extrapolated limit 0.4773635
Independent recheck of the certified SQG collapse of sixty vortices, Newton’s method with κ held real at 60 digits	residual 6×10^{-59} ; angular impulse and $\sum_{j < k} \Gamma_j \Gamma_k z_j - z_k ^{-1}$ below 10^{-61} relative

9 Discussion

Lemma 3 gives, in a form suited to minimization, the collapse and rotation rates that Aref [3, Eqs. (25a), (25d)] expresses through the side lengths of the triangle (see also [15], [14, Eq. (E.1)], and, for equal circulations, [25, Eq. (4.4)]); Aref also notes that the rotation rate times the collapse time, that is, the ratio of the two rates, fixes the logarithmic spiral [3, Eq. (29c)]. Theorem 1 concerns the minimization of that ratio, which none of Gröbli [16, § 10], Aref [3], Conte and de Seze [9] and Kimura [25] considers; Gröbli gives the ratio in closed form for every triple of circulations that can collapse, $1/\Gamma_1 + 1/\Gamma_2 + 1/\Gamma_3 = 0$ (Section 3, after Lemma 3), but does not discuss its extremum. When the two circulations of the same sign differ, $\mu < 1$, the two orientations of the triangle are not equivalent and the minima on the two arcs differ, so a minimization restricted to one orientation finds only one of them. The infimum $\sqrt{3}/2$ of Corollary 1 is approached only as $\mu \rightarrow 0$ and is not attained.

Leoncini, Kuznetsov and Zaslavsky [30] analyze the motion near collapse when two of the vortices are identical; the configuration that collapses fastest at a fixed distance between the identical vortices, found by Kimura [25, Eq. (4.6)] and shown in [30, Fig. 18], has $P = 3/2$, the value at $\chi = \pi/4$ by the symmetry $v \mapsto 1/(2v)$ of Remark 2, and the value $\sqrt{3}/2$ in that caption is a value of their energy parameter $\Lambda = e^{4\pi H}$, not of P . For $\sum_{j < k} \Gamma_j \Gamma_k = 0$, Tavantzis and Ting [40, Sect. II], who work with the side lengths of the triangle, show that the contracting family of similar

configurations is unstable and the expanding family asymptotically stable, stability referring to the ratios of the sides. Reinaud, Dritschel and Scott [37] extend the collapse conditions to generalized Euler and quasi-geostrophic models and map the collapse time over the collapsing configurations, with the distance between the like-signed vortices fixed, finding a minimum of that time along a curve in one of their parameter maps [37, Fig. 3]; P , unlike the collapse time, does not depend on the choice of length and time scales. Krishnamurthy and Stremmer [29, Sects. 3.3–3.5] relate the interior angles of the triangle, the circulation ratios, the energy, the collapse time and the distance traveled before collapse by the circumcenter of the triangle. When the angular impulse about the center of vorticity vanishes, the circumcircle passes through the center of vorticity at all times [28, Eq. (40)], so in a collapse the circumcenter stays one circumradius from the collision point; since it is carried by the self-similar motion, it travels $\sqrt{1 + 4P^2}$ times the initial circumradius before the collapse. In their notation $P = \tilde{\tau}|K_1|$ [29, Eq. (3.28a)], and they observe numerically that this normalized distance, their $\tilde{s}(1)$, exceeds 2 [29, Sect. 3.5]; Corollary 1 proves this observation and shows that the constant 2 cannot be improved.

Hiraoka [20] blows up the triple collision to McGehee’s collision manifold and shows that it is topologically regularizable in Easton’s sense when the two vortices of the same sign have equal circulations, $\mu = 1$ in our normalization, once those two vortices are identified [20, Theorem 1], and not when their circulations differ by a small nonzero amount [20, Theorem 2]; his variables are the side lengths and the signed area of the triangle, so the rotation does not enter. Gallay and Šverák [14] ask when a collision of three point vortices is regularizable, that is, a limit of nearby motions without collision. Whether it is depends on the perturbation [14, Theorems 5.8, 5.10, 5.15], and when it is, the limit is determined only up to a rotation [14, Sect. 1], because the triangle turns through the angle $s \ln(1 - t/T)$ and so makes infinitely many turns before the collision [14, Sect. 5.2]. Grotto, Romito and Viviani [18] select a continuation after collapse by adding a vanishing stochastic diffusion, which yields a probability distribution over continuations rather than a single one. Gallay and Šverák note that $s \neq 0$ [14, Remark E.2]; since $|s| = P$, Corollary 1 sharpens this to $|s| > \sqrt{3}/2$ for every collision, and no larger constant holds for all of them.

For two rings, Aref [2] derived the reduced equations (16) for arbitrary circulations and analyzed equal and opposite circulations, $x = 1$, which do not satisfy (17). Koiller et al. [26] found the collapsing configurations and their rates. Demina and Kudryashov [11] study relative equilibria, collapse and scattering of point vortices with arbitrary circulations by a polynomial method. For two regular polygons with a vortex of circulation Γ_0 , possibly zero, at the center, they give the equation for the ratio of the radii, which is (19) with $x = r^2$, and the constant Ω , which on (19) is S , as an explicit function of the relative rotation [11, Eqs. (36)–(37)], and they state that every relative rotation with $e^{in\theta} \neq \pm 1$ gives a collapse or a scattering. For this family they do not say which of the two occurs, and they do not write out the ratio of the two rates, although their general solution contains it [11, Eq. (10)]. Equations (18) and (20) write this ratio in closed form, for $\Gamma_0 = 0$ and for every Γ_0 , and Propositions 2 and 3 give its minimum over the relative rotation and, for every n , its infimum $\sqrt{3}/2$ over the whole family. Theorem 3 explains why this infimum is the same as that of Corollary 1: both are approached by a strong vortex carrying weak, tight pairs of opposite sign, and every such configuration has $P > \sqrt{3}/2$ once the pairs are weak enough (Proposition 4), with near-equality only at the shape $u_j = e^{i\pi/3}$. O’Neil [35] proves that for generic circulations three concentric rings have finitely many relative equilibria and, for each fixed complex rate, finitely many collapse configurations. We have not found the minimal values of Theorem 1 and Propositions 1–3 stated in the literature, from Gröbli’s § 10 [16] on, nor a proof of the bounds of Corollary 1, Theorem 2, Proposition 3 and Theorem 3 or of their sharpness.

Self-similar collapse of three vortices in the α -models has been studied from several directions. Badin and Barry [5] derive the necessary conditions for collapse in Nambu form and show that

in the SQG case collapse is self-similar when the energy vanishes and not self-similar otherwise; for circulations $(1, -\Gamma, 1)$ they find self-similar SQG collapse exactly for $0.387464\dots < \Gamma < 1/2$ [5, Lemma 1], an interval that Lemma 5 reproduces. Yasunaga, Otake and Iwayama [42] give necessary conditions, the orbits of self-similar motion and its spiral, and Iwayama and Yajima [22] show that the collapsing motions are linearly unstable and the expanding ones stable. Reinaud [36] treats the self-similar collapse of three geophysical vortices; as the Correction to that article records, it predicts the collapse time analytically, confirms it with simulated collapses and maps it over a family of collapsing configurations. Chen and Liu [7] prove that the existence of self-similar collapse and expansion depends only on the circulations, give explicit expressions for all nontrivial self-similar solutions [7, Theorems 2.1, 2.2], and obtain the interval of Badin and Barry analytically [7, Theorem 4.1], a description from the side of the circulations that Lemma 5 complements from the side of the shape. Iwayama, Yajima and Watanabe [23] prove that collapse that is not self-similar cannot occur for $-1 \leq \alpha < 0$ and show its possible existence for $0 < \alpha < 2$. Grotto and Pappalettera [17, Prop. 2.1] characterize the self-similar motions and write the collapsing spiral, as a step toward bursts out of arbitrary configurations. We have not found a bound on the winding, the spiral angle or the distance traveled in this literature. Theorem 2 concerns self-similar collapse only; it says nothing about other singular motions of the α -models, such as the collapses at nonzero energy studied in [5, 23].

For more than three vortices, self-similar collapse has mostly been studied at fixed circulations. O’Neil [34] defines collapse configurations through a complex rate [34, Definition 1.1.3] and computes one-parameter families of them for four vortices [34, Sect. 8.5]; Gotoda [15] continues such families numerically for up to ten vortices, and Novikov and Sedov [33] and Demina and Kudryashov [11] give explicit examples. None of these works bounds or minimizes P , and we have not found a self-similar collapse without rotation reported for any α or number of vortices. The minima of Theorem 4 are over the circulations and the shape together, and the collapse of Theorem 5 shows that in the α -models the winding of a self-similar collapse, bounded below for three vortices by Theorem 2, can vanish for more.

Meaning and limits. While the size of a collapsing configuration shrinks from r_0 to r , it turns through the angle $P \ln(r_0^2/r^2)$. By Corollary 1, a triangle of three vortices that shrinks by the factor 10 therefore turns through more than $(\sqrt{3}/2) \ln 100 = 3.988\dots$ radians, about 0.63 of a turn, whatever the circulations, and by Proposition 3 so do the ring configurations; by Proposition 4 the same holds for a strong vortex with sufficiently weak, tight pairs; and the same constant bounds the spin $|s| = P$ of Gallay and Šverák, as noted above. For three vortices in the α -models the angle exceeds $B(\alpha) \ln 100$ by Theorem 2. Four vortices can turn less than three (Theorem 4). The comparison with gravitation is instructive. A homographic solution of the Newtonian n -body problem, one that keeps its shape, ends in a collision of all the bodies only if its angular momentum vanishes, and it is then homothetic: it shrinks without turning [41, §§ 370 bis, 378]. A collapsing vortex configuration also has zero angular impulse, the invariant associated with rotations, but that invariant depends on the positions alone, and three vortices, like the ring configurations, cannot collapse without turning; in the $\alpha = 2$ model eleven vortices can, and in the SQG model sixty can (Theorem 5). The exact minima of Theorem 1 and Propositions 1–3, the certified minima of Theorem 4, and the configurations that attain them, can serve as benchmarks for point-vortex codes near a collapse. Since the contracting three-vortex configurations are unstable in shape [40, Sect. II], [22], a perturbed configuration follows a self-similar collapse only transiently. The results concern point vortices. Vortex patches exist for all time in the Euler equations, since bounded, compactly supported vorticity has a unique global weak solution, by Yudovich’s theorem [43], proved there

for bounded domains (for the whole plane see [10, Theorems 1.6 and 3.3], and the discussion in [8, Sect. 1]); so finite cores of that kind regularize the collision, and we expect, but do not prove, that the bounds describe the approach only while the cores are small compared with the separations. Hollow vortices, whose cores are regions of constant pressure, can collapse: Chen, Walsh and Wheeler prove that every collapsing configuration of point vortices that is non-degenerate in their sense [8, Definition 4.1] can be desingularized into a family of hollow vortices that implode self-similarly in finite time [8, Theorems 1.3 and 4.4], and they verify the hypothesis for an explicit collapsing triple and quartet [8, Examples 4.2 and 4.3, Corollary 4.6]. Both lie in our families. Their triple [8, Eq. (4.5)], circulations $(1, 2, -2/3)$ at $-2, 1$ and $\sqrt{7}i$, becomes the configuration $\theta = \pi/2$ of (7) with $\mu = 1/2$, which lies on $\mathcal{A}_+$, when its first two vortices are interchanged and its circulations halved: its shape ratio is then $w = 1/3 - (\sqrt{7}/3)i$, and $P = 5\sqrt{7}/2$ by (9). Their quartet [8, Eq. (4.6)], turned by $\pi/6$, is the configuration of Proposition 2 with $n = 2$ and $\theta = \pi/12$, with the circulations multiplied by $(4 - \sqrt{3})\pi$, and $P = 2\sqrt{3} - 3/4$ by (18). (In arXiv:2506.04093v1 and in the published version the third position of the quartet is printed as $-\sqrt{3}/2 - 1 - i/2$; it has to be read as $-\sqrt{3}/2 - 1 + i/2$, for only then are the four points a parallelogram centered at the origin, as stated there.) Their map is $V_k(\Lambda) = \sum_{j \neq k} \gamma_j / (2\pi i(z_k - z_j)) + i\Omega \bar{z}_k$, $k = 1, \dots, N$, on the parameters $\Lambda = (z_1, \dots, z_N, \gamma_1, \dots, \gamma_N, \Omega) \in \mathbb{C}^N \times \mathbb{R}^N \times \mathbb{C}$ [8, Eq. (4.4)], where they write M for our N , and non-degeneracy is the full rank of its derivative $D_\Lambda V$ at a zero [8, Definition 4.1]. By (2), $V(\Lambda) = 0$ says that the vortices with the circulations γ_j move self-similarly about the origin with $\Omega = i\bar{\kappa}$. Since the double sum cancels in pairs, $\sum_k \gamma_k V_k = i\Omega \overline{\sum_k \gamma_k z_k}$. For their triple as quoted above, whose center of vorticity is $-2i/\sqrt{7}$, this vanishes only for $\Omega = 0$, where V is the conjugate of the velocities, which do not vanish; so no Ω makes that triple a zero of V . Shifted to its center of vorticity, as they do, the triple is a zero of V with $\Omega = i\bar{\kappa} = (35 - \sqrt{7}i)/(264\pi)$; the argument below starts from this shifted triple, since its maps L do not include translations. For the triple they print, from Aref's formulas, a real constant $\Omega = 35/(264\pi)$, in regular type while the Ω of V is in bold, and $1/\kappa = \sqrt{7}/(132\pi)$, where their κ is the collapse time; by the definition at the start of their Sect. 4.1 the Ω of V is the real constant minus $i/(2\kappa)$, which is the value above. For the quartet, printed with its center of vorticity at the origin, the printed Ω is that of V , and it is $i\bar{\kappa}$. The argument uses two properties of V , both read off from the formula for V . First, V is a rational function of the real coordinates of Λ whose denominators $|z_k - z_j|^2$ do not vanish while the vortices are distinct, so it is real-analytic there. Second, a rotation $(z, \gamma, \Omega) \mapsto (e^{i\phi}z, \gamma, \Omega)$, a dilation $(z, \gamma, \Omega) \mapsto (\lambda z, \gamma, \lambda^{-2}\Omega)$ with $\lambda > 0$, a factor $(z, \gamma, \Omega) \mapsto (z, c\gamma, c\Omega)$ with $c > 0$ and a relabeling are real-linear isomorphisms L of the parameter space with $V \circ L = A \circ V$, where A is multiplication by $e^{-i\phi}$, λ^{-1} or c , or the same permutation; so $D_\Lambda V(L\Lambda) L = A D_\Lambda V(\Lambda)$, and L maps zeros to zeros and keeps the rank. Hence one member of each of these two families is non-degenerate. Along each arc the positions are real-analytic in θ , by (7) and by the construction of Section 5, the circulations are fixed, and $\Omega = i\bar{\kappa}$ is real-analytic in θ , by Lemma 3 and by the formula for S in Section 5; since V fixes neither the size nor Ω , the members need no renormalization along the arc. So the maximal minors of $D_\Lambda V$ along each arc are real-analytic in θ ; a minor that does not vanish at one point of an interval vanishes only at isolated points of it, so the configurations of $\mathcal{A}_+$ for $\mu = 1/2$, and those with $0 < \theta < \pi/2$ for $n = 2$, are non-degenerate except at isolated values of θ . Numerically, at 199 equally spaced points of each of these two arcs the $2N$ -th singular value of $D_\Lambda V$ is at least 2.9×10^{-4} times the largest (Table 2); this observation is not used in the argument. We have not checked the minimizers, or the other families, and we do not study the winding of hollow or other cored vortices.

10 Open problems

1. Is there a positive lower bound for P over self-similar collapses of any number of Euler vortices? The certified local minima decrease with the number of vortices, $P_4 > P_5 > P_6$ (Theorem 4), and the smallest values found numerically decrease further, to 0.4793959201... for 603 vortices. The two-arm family that gives them tends to 0.47736; is there a structure that goes lower at larger N , and can P be made arbitrarily small?
2. Can Euler vortices collapse self-similarly without rotation, for some number of vortices? Since the Euler law is holomorphic, such a collapse, with κ real, is the same as a relative equilibrium, with angular velocity $-\kappa$, of the imaginary circulations $i\Gamma_j$. For $\alpha \neq 0$ there is no such correspondence, and Theorem 5 gives collapses without rotation for $\alpha = 2$ and for SQG.
3. Are the local minima P_4 , P_5 and P_6 of Theorem 4 the least values of P over all self-similar collapses of four, five and six Euler vortices, and are they algebraic numbers?
4. Are the four-vortex minima 0.5499151189... (SQG) and 0.4116738600... ($\alpha = 2$) of Theorem 4(c) the least values for four vortices? What is the least number of vortices that can collapse self-similarly without rotation, for $\alpha = 2$, where eleven suffice, and for SQG, where sixty suffice (Theorem 5)?
5. For $\alpha = 2$ the value of P on the collapsing triangles is an explicit algebraic function of the circulations, and on the family with two equal circulations its minimum is the number 1.1039451679... of Remark 5. For SQG with two equal circulations the collapsing shape solves a quartic, which Iwayama, Yajima and Watanabe solve in radicals [23, Eqs. (4.2), (4.3)], so P is again algebraic along that family. What is its minimum there, and what is the minimal polynomial of that minimum?

Data availability. The programs `verify_general_mu.py`, `verify_floors_independent.py`, `verify_direct_proof.py`, `verify_central_vortex.py`, `verify_strong_vortex.py`, `verify_pairs_bound.py`, `verify_alpha_winding.py`, `verify_alpha_below.py`, `verify_alpha_equal_circulations.py`, `verify_many_vortices.py`, `certify_collapses.py` with its modules and `certify_sqg60.py`, used for Section 8, and `plot_minimal_winding.py` and `plot_alpha_winding.py`, used for Figures 1–3, and their output are in the repository <https://github.com/ChaseHendrick/minimal-winding>: the programs in `code/`, the output of the twelve verification programs and the stored configurations of Section 7 in `data/`, and the figures in `paper/figures/`. Release 2.2.0 of that repository holds the programs and data of this version of the paper (<https://doi.org/10.5281/zenodo.22994932>). Release 2.2.2 archives the same proof programs and data with supplementary checks at <https://doi.org/10.5281/zenodo.23047024>. The previous release 2.1.0 remains at <https://doi.org/10.5281/zenodo.22966989>.

Funding. This research received no external funding.

This work was prepared with AI assistance. The author takes full responsibility for its content.

Rights and licenses. Copyright (c) 2026 Chase Hendrick. The manuscript and figures are all rights reserved. Code and data are licensed under Apache License 2.0, subject to their component notices. These notices do not revoke licenses previously granted for earlier material.

References

- [1] H. Aref, Motion of three vortices, *Phys. Fluids* **22** (1979) 393–400.
- [2] H. Aref, Point vortex motions with a center of symmetry, *Phys. Fluids* **25** (1982) 2183–2187.
- [3] H. Aref, Self-similar motion of three point vortices, *Phys. Fluids* **22** (2010) 057104.
- [4] H. Aref, N. Rott and H. Thomann, Gröbli’s solution of the three-vortex problem, *Annu. Rev. Fluid Mech.* **24** (1992) 1–21.
- [5] G. Badin and A. M. Barry, Collapse of generalized Euler and surface quasigeostrophic point vortices, *Phys. Rev. E* **98** (2018) 023110; arXiv:1805.10127.
- [6] A. V. Borisov and V. G. Lebedev, Dynamics of three vortices on a plane and a sphere — III. Noncompact case. Problems of collapse and scattering, *Regul. Chaotic Dyn.* **3** (1998), no. 4, 74–86; arXiv:nlin/0503057.
- [7] J. Chen and Q. Liu, Sufficient and necessary conditions for self-similar motions of three point vortices in generalized fluid systems, *Physica D* **470** (2024) 134392.
- [8] R. M. Chen, S. Walsh and M. H. Wheeler, Finite-time self-similar implosion of hollow vortices, *Math. Ann.* **396** (2026) 5; doi:10.1007/s00208-026-03541-2; arXiv:2506.04093v1, whose numbering is cited; the numbers cited are the same in the journal version.
- [9] R. Conte and L. de Seze, Exact solution of the planar motion of three arbitrary point vortices, report DPhG/PSRM/1697/80, CEN Saclay (1980); also *Mod. Phys. Lett. B* **29** (2015) 1530017; arXiv:1511.00069v1, whose page numbers are cited.
- [10] G. Crippa and G. Stefani, An elementary proof of existence and uniqueness for the Euler flow in localized Yudovich spaces, *Calc. Var. Partial Differential Equations* **63** (2024), no. 7, Paper No. 168; arXiv:2110.15648.
- [11] M. V. Demina and N. A. Kudryashov, Rotation, collapse, and scattering of point vortices, *Theor. Comput. Fluid Dyn.* **28** (2014) 357–368; doi:10.1007/s00162-014-0319-4.
- [12] M. Donati and L. Godard-Cadillac, Hölder regularity for collapses of point-vortices, arXiv:2111.14230v4 (2023), whose appendix numbering is cited.
- [13] T. D. Drivas, B. A. Khanikati and V. A. Khanikati, On the collapse of three point vortices on surfaces, preprint, arXiv:2607.16490 (2026).
- [14] T. Gallay and V. Šverák, The three-vortex system: Hopf fibration, symplectic reduction, and near-collisions, preprint, arXiv:2609.10847 (2026).
- [15] T. Gotoda, Self-similar motions and related relative equilibria in the N -point vortex system, *J. Dyn. Differ. Equ.* **33** (2021) 1759–1777; arXiv:2002.09624.
- [16] W. Gröbli, *Specielle Probleme über die Bewegung geradliniger paralleler Wirbelfäden*, Inaugural-Dissertation, Göttingen; Zürcher und Furrer, Zürich, 1877. English translation by R. Goodman, arXiv:2404.01305 (2024).
- [17] F. Grotto and U. Pappalètera, Collapse and burst of generalized surface quasi-geostrophic point vortices, preprint, arXiv:2505.19782 (2025).
- [18] F. Grotto, M. Romito and M. Viviani, Zero-noise dynamics after collapse for three point vortices, *Physica D* **457** (2024) 133947; arXiv:2307.05133.

- [19] A. Hernández-Garduño and E. A. Lacomba, Collisions and regularization for the 3-vortex problem, *J. Math. Fluid Mech.* **9** (2007) 75–86; arXiv:math-ph/0412024, whose theorem numbering is cited.
- [20] Y. Hiraoka, Topological regularizations of the triple collision singularity in the 3-vortex problem, *Non-linearity* **21** (2008) 361–379.
- [21] Y. Hiraoka, Remarks on collision manifolds and nonexistence of non self-similar collision solutions in the 3-vortex problem, *RIMS Kôkyûroku Bessatsu* **B13** (2009) 35–43.
- [22] T. Iwayama and T. Yajima, Linear stability of self-similar motions of three point vortices in a generalized two-dimensional fluid system, *J. Phys. Soc. Jpn.* **92** (2023) 084401.
- [23] T. Iwayama, T. Yajima and T. Watanabe, Non-self-similar collapse of three point vortices in a generalized two-dimensional fluid system, *J. Phys. A* **58** (2025) 075701; doi:10.1088/1751-8121/adaef8.
- [24] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66** (2017) 1281–1292; doi:10.1109/TC.2017.2690633.
- [25] Y. Kimura, Similarity solution of two-dimensional point vortices, *J. Phys. Soc. Jpn.* **56** (1987) 2024–2030.
- [26] J. Koiller, S. Pinto de Carvalho, R. Rodrigues da Silva and L. C. Gonçalves de Oliveira, On Aref’s vortex motions with a symmetry center, *Physica D* **16** (1985) 27–61.
- [27] R. Krawczyk, Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken, *Computing* **4** (1969) 187–201; doi:10.1007/BF02234767.
- [28] V. S. Krishnamurthy, H. Aref and M. A. Stremler, Evolving geometry of a vortex triangle, *Phys. Rev. Fluids* **3** (2018) 024702; arXiv:1706.00731v2, whose equation numbers are cited.
- [29] V. S. Krishnamurthy and M. A. Stremler, Finite-time collapse of three point vortices in the plane, *Regul. Chaotic Dyn.* **23** (2018) 530–550.
- [30] X. Leoncini, L. Kuznetsov and G. M. Zaslavsky, Motion of three vortices near collapse, *Phys. Fluids* **12** (2000) 1911–1927.
- [31] R. E. Moore, A test for existence of solutions to nonlinear systems, *SIAM J. Numer. Anal.* **14** (1977) 611–615.
- [32] P. K. Newton, *The N-Vortex Problem: Analytical Techniques*, Applied Mathematical Sciences 145, Springer, 2001.
- [33] E. A. Novikov and Yu. B. Sedov, Vortex collapse, *Sov. Phys. JETP* **50** (1979) 297–301 [*Zh. Eksp. Teor. Fiz.* **77** (1979) 588–597].
- [34] K. A. O’Neil, Stationary configurations of point vortices, *Trans. Amer. Math. Soc.* **302** (1987) 383–425.
- [35] K. A. O’Neil, Relative equilibrium and collapse configurations of heterogeneous vortex triple rings, *Physica D* **236** (2007) 123–130.
- [36] J. N. Reinaud, Self-similar collapse of three geophysical vortices, *Geophys. Astrophys. Fluid Dyn.* **115** (2021) 369–392, doi:10.1080/03091929.2020.1828402; Correction, *ibid.* **115** (2021) I–III, doi:10.1080/03091929.2020.1848127.
- [37] J. N. Reinaud, D. G. Dritschel and R. K. Scott, Self-similar collapse of three vortices in the generalised Euler and quasi-geostrophic equations, *Physica D* **434** (2022) 133226.
- [38] S. M. Rump, Verification methods: rigorous results using floating-point arithmetic, *Acta Numer.* **19** (2010) 287–449; doi:10.1017/S096249291000005X.

- [39] J. L. Synge, On the motion of three vortices, *Canad. J. Math.* **1** (1949) 257–270.
- [40] J. Tavantzis and L. Ting, The dynamics of three vortices revisited, *Phys. Fluids* **31** (1988) 1392–1409.
- [41] A. Wintner, *The Analytical Foundations of Celestial Mechanics*, Princeton Mathematical Series 5, Princeton University Press, Princeton, 1941.
- [42] T. Yasunaga, N. Otake and T. Iwayama, Self-similar motion of three point vortices for a generalized two-dimensional fluid system, *J. Phys. Soc. Jpn.* **90** (2021) 124401; doi:10.7566/JPSJ.90.124401.
- [43] V. I. Yudovich, Non-stationary flows of an ideal incompressible fluid, *Zh. Vychisl. Mat. Mat. Fiz.* **3** (1963) 1032–1066; English translation: Non-stationary flow of an ideal incompressible liquid, *USSR Comput. Math. Math. Phys.* **3** (1963) 1407–1456; doi:10.1016/0041-5553(63)90247-7.